MONITORING REPORT: IPOÁ REDD+ PROJECT

Document Prepared by Carbonext

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Project Title	IPOÁ REDD+ PROJECT
Project ID	3324
Version	2.2
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Date of Issue	16/02/2024
Project Location	Novo Aripuanã municipality, State of Amazonas, Brazil
Project Proponent(s)	- Primary contact: Carbonext Tecnologia em Soluções Ambientais Ltda: + 55 (11) 3168-8521 | redd.ipoa@carbonext.com.br - Sílvio Antonio Balen: silviobalen@hotmail.com - Carina Marcia Klein Balen
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GHG Accounting/ Crediting Period	17 March 2022 – 16 March 2052; 30 - year total period The CCB and VCS periods are the same
Monitoring Period of this Report	17 March 2022 – 31 December 2023 The CCB and VCS periods are the same
History of CCB Status	First Verification, validation is being audited concurrently with this MR1.
Gold Level Criteria	**Community**: The project meets CCB Gold Level concept GL2.1b with activities targeted at a low-income community: Improving communication and electricity infrastructure for communities living below the poverty line, ensuring access to technology and information.

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1 SUMMARY OF PROJECT BENEFITS

1.1 Unique Project Benefits

Outcome or Impact	Achievements during the Monitoring Period	Section Reference	Achievements during the Project Lifetime
Increase in community members with access to the internet and electricity, broadening their life opportunities.	10 people with access to electricity: Boys: 08; Girls: 02 29 people with access to communication technology: Men: 09; Women: 08; Boys: 8; Girls: 4	4.1.1 6	10 people with access to electricity: Boys: 08; Girls: 02 29 people with access to communication technology: Men: 09; Women: 08; Boys: 8; Girls: 4[1]
Contribute to quality education for communities. School facilities in PZ adequate and conducive to quality education.	19 electricity and telecommunication equipment items installed within education facilities. 38 educational equipment and material items purchased or transported 2 enrolments in training and educational courses BRL 20,900 spent on teacher and student training and courses	4.1.1 6	19 electricity and telecommunication equipment items installed within education facilities. 38 educational equipment and material items purchased or transported 2 enrolments in training and educational courses BRL 20,900 spent on teacher and student training and courses
Recognition of areas of community land-use and natural resources in the Project Zone by community members	Participatory Mapping workshop, mapping areas fundamental for the livelihoods of communities, including: community cultivation areas;	4.1.1 4.1.4 6	Participatory Mapping workshop, mapping areas fundamental for the livelihoods of communities, including:

[1] Achievements during the project lifetime are the same as the "achievements during the monitoring period" column, as this is the first MR.

	forest resource areas; and areas of cultural importance		community cultivation areas; forest resource areas; and areas of cultural importance
Reduction in deforestation within the PZ. Reduction in illegal trespassing and land-squatting attempts within the PZ.	5 deforestation events detected remotely 3 deforestation events visited in the field 8 meetings carried out to combat deforestation	2.5.4 3.1.3 6	5 deforestation events detected remotely 3 deforestation events visited in the field 8 meetings carried out to combat deforestation
Increase in people following occupational health and safety guidelines.	12 people (men: 9; women: 3) trained in best practices for occupational health and safety	4.1.1 6	12 people (men: 9; women: 3) trained in best practices for occupational health and safety

1.2 Standardized Benefit Metrics

Category	Metric	Achievements during Monitoring Period	Section Reference	Achievements during the Project Lifetime
GHG emission reductions & removals	Net estimated emission removals in the project area, measured against the without-project scenario	N/A	N/A	N/A
	Net estimated emission reductions in the project area, measured against the without-project scenario	692,909 tCO_2	3.2.4	692,909 tCO2[2]
Forest[3] cover	For REDD[4] projects: Number of hectares of reduced forest loss in the project area measured against the without-project scenario	1,420.32 ha	3.2.1	1,420.32 ha
	For ARR[5] projects: Number of hectares of forest cover increased in the project area measured against the without-project scenario	N/A	N/A	N/A
Improved land management	Number of hectares of existing production forest land in which IFM[6] practices have occurred	N/A	N/A	N/A

[2] Achievements during the project lifetime are the same as the "achievements during the monitoring period" column, as this is the first MR.

[3] Land with woody vegetation that meets an internationally accepted definition (e.g., UNFCCC, FAO or IPCC) of what constitutes a forest, which includes threshold parameters, such as minimum forest area, tree height and level of crown cover, and may include mature, secondary, degraded and wetland forests (*VCS Program Definitions*)

[4] Reduced emissions from deforestation and forest degradation (REDD) - Activities that reduce GHG emissions by slowing or stopping conversion of forests to non-forest land and/or reduce the degradation of forest land where forest biomass is lost (*VCS Program Definitions*)

[5] Afforestation, reforestation and revegetation (ARR) - Activities that increase carbon stocks in woody biomass (and in some cases soils) by establishing, increasing and/or restoring vegetative cover through the planting, sowing and/or human-assisted natural regeneration of woody vegetation (*VCS Program Definitions*)

[6] Improved forest management (IFM) - Activities that change forest management practices and increase carbon stock on forest lands managed for wood products such as saw timber, pulpwood and fuelwood (*VCS Program Definitions*)

Category	Metric	Achievements during Monitoring Period	Section Reference	Achievements during the Project Lifetime
	as a result of the project's activities, measured against the without-project scenario			
	Number of hectares of non-forest land in which improved land management has occurred as a result of the project's activities, measured against the without-project scenario	N/A	N/A	N/A
Training	Total number of community members who have improved skills and/or knowledge resulting from training provided as part of project activities	38 participants in association training events 12 participants trained in best practices for occupational health and safety	4.1.1 4.4.6	38 participants in association training events 12 participants trained in best practices for occupational health and safety
	Number of female community members who have improved skills and/or knowledge resulting from training provided as part of project activities of project activities	18	4.1.1 4.1.2 4.1.3	18
Employment	Total number of people employed in of project activities,[7] expressed as number of full-time employees[8]	N/A	N/A	N/A

[7] Employed in project activities means people directly working on project activities in return for compensation (financial or otherwise), including employees, contracted workers, sub-contracted workers and community members that are paid to carry out project-related work.

[8] Full time equivalency is calculated as the total number of hours worked (by full-time, part-time, temporary and/or seasonal staff) divided by the average number of hours worked in full-time jobs within the country, region or economic territory (adapted from the UN System of National Accounts (1993) paragraphs 17.14[15.102];[17.28])

Category	Metric	Achievements during Monitoring Period	Section Reference	Achievements during the Project Lifetime
	Number of women employed in project activities, expressed as number of full time employees	N/A	N/A	N/A
Livelihoods	Total number of people with improved livelihoods[9] or income generated as a result of project activities	N/A	N/A	N/A
	Number of women with improved livelihoods or income generated as a result of project activities	N/A	N/A	N/A
Health	Total number of people for whom health services were improved as a result of project activities, measured against the without-project scenario	N/A	N/A	N/A
	Number of women for whom health services were improved as a result of project activities, measured against the without-project scenario	N/A	N/A	N/A
Education	Total number of people for whom access to, or quality of, education was improved as a result of project activities, measured against the without-project scenario	10	4.1.1 6	10
	Number of women and girls for whom access to,	3[10]	4.1.1	3

[9] Livelihoods are the capabilities, assets (including material and social resources) and activities required for a means of living (Krantz, Lasse, 2001. *The Sustainable Livelihood Approach to Poverty Reduction*. SIDA). Livelihood benefits may include benefits reported in the Employment metrics of this table.

[10] This includes 2 girls studying at the PZ school, and the teacher who is female, and is benefitting from improved schooling as part of MR1 actions (see sections 4.4.3 and 6)

Category	Metric	Achievements during Monitoring Period	Section Reference	Achievements during the Project Lifetime
	or quality of, education was improved as a result of project activities, measured against the without-project scenario		4.4.3 6	
Water	Total number of people who experienced increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario	N/A	N/A	N/A
	Number of women who experienced increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario	N/A	N/A	N/A
Well-being	Total number of community members whose well-being[11] was improved as a result of project activities	29 community members[12]	4.1.1	29 community members
	Number of women whose well-being was improved as a result of project activities	18 women	4.1.1 4.4.3	18 women
Biodiversity conservation	Change in the number of hectares significantly better managed by the	1,420.32 ha	3.2.1	1,420.32 ha

[11] Well-being is people's experience of the quality of their lives. Well-being benefits may include benefits reported in other metrics of this table (e.g. Training, Employment, Livelihoods, Health, Education and Water), and may also include other benefits such as strengthened legal rights to resources, increased food security, conservation of access to areas of cultural significance, etc.
[12] Considered as the whole population of the Taracuá (Boa Esperança) community

Category	Metric	Achievements during Monitoring Period	Section Reference	Achievements during the Project Lifetime
	project for biodiversity conservation,[13] measured against the without-project scenario			
	Number of globally Critically Endangered or Endangered species[14] benefiting from reduced threats as a result of project activities,[15] measured against the without-project scenario	32[16]	5.1.1	32

[13] Biodiversity conservation in this context means areas where specific management measures are being implemented as a part of project activities with an objective of enhancing biodiversity conservation.
[14] Per IUCN's Red List of Threatened Species
[15] In the absence of direct population or occupancy measures, measurement of reduced threats may be used as evidence of benefit
[16] Number predicted by literature (see section 5.1.1), to be confirmed by biodiversity contractor

2 GENERAL

2.1 Project Description

2.1.1 Implementation Description

During monitoring period n°1 (henceforth “MR1"), the project achieved total GHG emission reductions of 692,909 tCO2, due to activities implemented to preserve its forests.
Various activities in the areas of community, climate and biodiversity were conducted by the project proponent and together with communities from the Project Zone, as summarized below:

- Forest monitoring via CarbonEye, with weekly frequency;
- Deforestation prevention and mitigation actions, against both illegal invasive deforestation as well as negotiated commitments to deal with minor community deforestation events;
- Installation of solar panels and internet access via StarLink [17];
- Community land-use participatory mapping workshop;
- Donation of hygiene products;
- Community training on associations and cooperatives;
- Community women’s workshop;
- Occupational health and safety training including distribution of Personal Protective Equipment;
- Professional training for community school teacher;
- Donation of school supplies and equipment;
- Initial meetings towards health clinic activities.

During the Monitoring Period five occurrences of deforestation within the Project Area (PA) were dectected, totaling 11.61 hectares, indicating the strong deforestation pressure it is subjected to.

The non-permanence risk for this MR period was assessed and managed using the AFOLU “Non-Permanence Risk Tool” VCS Version 4, Procedural Document, 19 September 2019, v4.0, fully described in the accompanying Non-Permanence Risk Report. Leakage is duly monitored and managed through remote sensing technicques applying the methodologies specified by the leakage modules listed in section 2.1.8. In the present MR, the leakage displaced from the Project Area to the Leakage Belt was of 17.06 ha, equating to 10,802 tCO2 in emissions and 581 tCO2 in biomass burning, as reported in section 3.2.3.

[17] A promotional video of these activities is available via the link: https://www.instagram.com/reel/CpDMrP0MdnN/?utm_source=ig_web_copy_link&igshid=MzRlODBiNWFlZA==

2.1.2 Project Category and Activity Type

Sectorial Scope: 14 Agriculture, Forestry and Other Land Use (AFOLU);

Project Category: Reducing Emissions from Deforestation and Degradation (REDD)

Project Activity: this project comprises Avoided Planned Deforestation (APD) and Avoided Unplanned Deforestation (AUD) areas;

The Project corresponds to the VCS Scope VM0007 REDD+ Methodology Framework (REDD+MF), v1.6.

This is a grouped project (more details in sector 2.1.21 Grouped Projects).

Project Scale	
Project	X
Large project	

2.1.3 Project Proponent(s)

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2.1.4 Other Entities Involved in the Project

No other entities are involved in the VCS/ CCB development of the present project.

2.1.5 Project Start Date (G1.9)

The project start date of the Ipoá REDD+ project is March 17, 2022. Explanations per component follow below.

AUD component:

The activities that led to the generation of GHG emission reductions began on March 17, 2022.

The project proponent (landowner) acquired the Campos property on August 21, 2019, and the Maanaim property on October 13, 2019, with the objective, as stated in the purchase and sale agreements, of carbon credit activities in the component of the legal reserve, being 80% of forests on the property (corresponding to the AUD component), and timber production activities, which includes a component of deforestation, to maximize timber production capacity, in the remaining 20% of the property (corresponding to the APD component).

In December 2021, the landowner and Carbonext (Project Developer) signed an agreement for the development and implementation of the REDD+ project.

After the recognition and analysis step, Carbonext started the implementation of project activities in the field on March 17, 2022. This event is considered as the Project Start Date.

On this occasion, an experienced social technician from Carbonext, accompanied by a local consultant, visited the Project Area, carrying out public consultations with the communities in the Project Zone, presenting the principles and benefits generated by the REDD+ project. The presentation lecture given to local inhabitants is considered the first activity to generate reductions in the project's GHG emissions, as it made communities aware of the forest preservation activities and encouraged them to monitor possible illegal deforestation.

APD component:

Regarding APD activities, May 3rd, 2022 was the date that the project owner submitted a request to the official organ IPAAM, to obtain permission to deforest the legal proportion (the aforementioned 20%) of the Maanaim and Campos properties in the Project Area (PA). The documents required and submitted to IPAAM for the request were made available at audit.

The planned deforestation license was considered necessary by the project owner as a form of backup plan, in case the REDD+ activities are unsuccessful, in which case he would opt for the regional business-as-usual of deforestation with timber extraction, and possible installation of permanent agricultural activity, following deforestation. This business-as-usual activity is also what the proponents personal background is in, and would provide the financial returns he requires, in the scenario of non-success of the present REDD+ project. To summarize, the proprietor’s stated plan, since the purchase of the land, was: REDD carbon activities as the primary plan, with planned deforestation combined with timber production activities, as the backup alternative.

Regarding planned deforestation documentation, the VM0007 methodology, VMD0006-BL-PL_v1.3 module, states that: “if government approval is required for deforestation to occur, the intention to deforest within the project area must be demonstrated by evidence:

- Recent approval from relevant government department (local to national) for conversion of forest to an alternative land use; or
- Documentation that a request for approval has been filed with the relevant government department for permission to deforest and convert to alternative land use;”

In accordance with these methodology requirements, the filed deforestation request mentioned above is the documentation corresponding to the proof and start of the project's APD activities.

2.1.6 Project Crediting Period (G1.9)

The crediting period is from March 17, 2022, to March 16, 2052, therefore comprising a duration of 30 years.

2.1.7 Project Location

The IPOÁ REDD+ Project project is in Northern Brazil's Amazon region, in the state of Amazonas, municipality of Novo Aripuanã, as shown in the figure below. Novo Aripuanã falls within the Mesoregion of the South Amazon and Microregion of the Madeira watershed.

The Project Area (PA) is of 17,329.73 ha in area and is composed of two properties: the Maanaim property, geodetic coordinates: 05°31'04,33" S 59°46'00,53" W. And the Campos property, geodetic coordinates: 05°50'15,93" S 59°48'26,53" W (see Figure below). The PA has also been submitted as a KML file to Verra.

The primary means of accessing the region is by water, on large boats or speedboats that leave the state capital, Manaus, crossing parts of the Amazon River and Madeira River. The duration of the route is 24 hours in larger boats and 10 hours in speedboats, as there are 450 kilometers that separate the two cities. However, by hydroplane from Manaus the journey takes under two hours. The main bodies of water in the vicinity of the project area are the Sucunduri River and the Acari River.

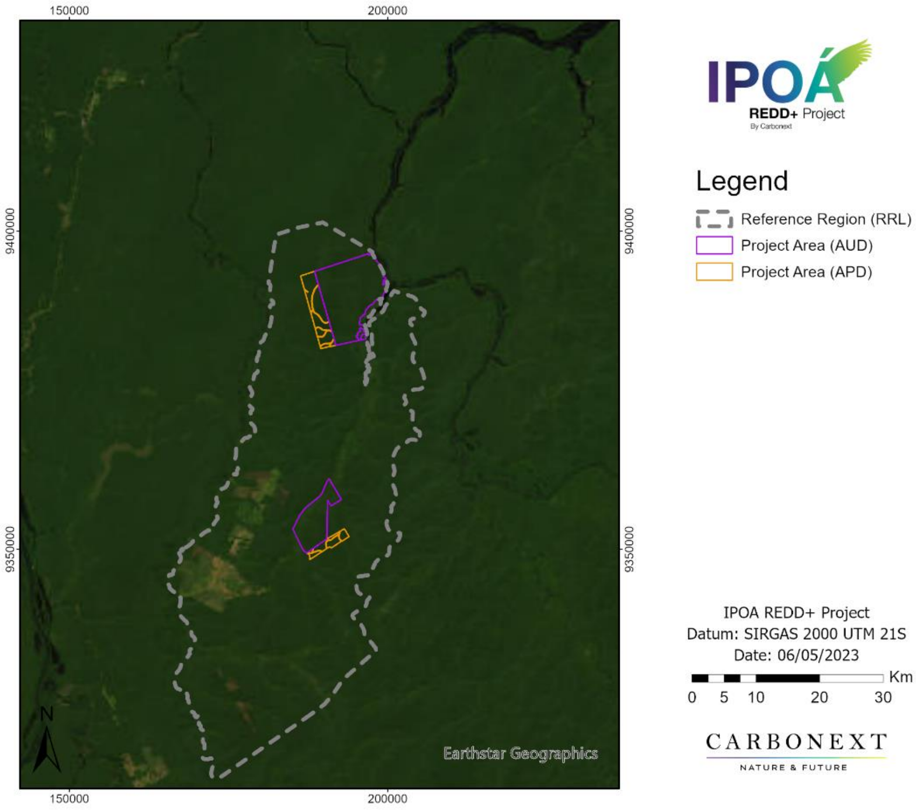

Figure 1. Ipoá REDD+ Project PA and Reference Region

2.1.8 Title and Reference of Methodology

Methodology applied:

Approved VCS Methodology VM0007 "REDD+ Methodology Framework (REDD+ MF)", Version 1.6, 08 September 2020[18].

Carbon pool modules:

- VMD0001 "Estimation of carbon stocks in the above- and belowground biomass in live tree and non-tree pools" (CP-AB), v1.1
- VMD0005 "Estimation of carbon stocks in the long-term wood products pool" (CP-W), v1.1

[18] Available at: https://verra.org/wp-content/uploads/2020/09/VM0007-REDDMF_v1.6.pdf

Baseline module:

- VMD0006 Estimation of baseline carbon stock changes and greenhouse gas emissions from planned deforestation/forest degradation and planned wetland degradation (BL-PL), v1.3
- VMD0007 "Estimation of baseline carbon stock changes and greenhouse gas emissions from unplanned deforestation and unplanned wetland degradation" (BL-UP), v3.3

Leakage modules:

- VMD0009 Estimation of emissions from activity shifting for avoided planned deforestation/forest degradation and avoided planned wetland degradation (LK-ASP), v1.3
- VMD0010 Estimation of emissions from activity shifting for avoiding unplanned deforestation and avoiding unplanned wetland degradation (LK-ASU), v1.2
- VMD0011 "Estimation of emissions from market-effects" (LK-ME), v1.1

Miscellaneous Modules:

- VMD0013 Estimation of greenhouse gas emissions from biomass and peat burning (E–BPB), v1.2
- VMD0016 Methods for stratification of the project area (X-STR), v1.2
- VMD0017 Estimation of uncertainty for REDD+ project activities (X-UNC), v2.2

Tools:

- VT0001 Tool for The Demonstration and Assessment of Additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities – Version 3.0
- AFOLU "Non-Permanence Risk Tool" VCS Version 4, Procedural Document, 19 September 2019, v4.0

2.1.9 Other Programs (G5.9)

The Ipoá Project has not been registered under any emissions trading programs. The project will not claim credit for the same GHG emission reduction under any other programs. The project has not sought or received another form of GHG-related environmental credit, including renewable energy certificates, during this monitoring period.

2.1.10 Sustainable Development

Priority Areas for Conservation

The IPOÁ REDD+ project area is within two priority areas for conservation, sustainable use and benefit sharing of biodiversity, according to the Brazilian environmental ministry, which are: AMZ-359 and AMZ-669. The areas designated in the 2nd update[19] of the priority areas for conservation in the Amazon, specify the biological importance, biodiversity conservation targets, priority level, recommended actions (main and secondary), opportunities for the area, general characteristics and conflicting activities in the region.

Despite the project area being located in a priority conservation location, which underscores the importance of its preservation, this has not prevented the advancement of deforestation and changes in land-use in the region.

Such information and data are fundamental in validating the importance of the present REDD+ project in this region. The IPOÁ REDD+ project has provided significant climate, community and biodiversity benefits in an integrated and sustainable way, cooperating with the state to preserve these areas with the expansion of tools and investments in forest conservation and biodiversity, the inspection and control of illegal activities, integrated and participatory management, recovery of degraded areas and restoration of ecosystem services (see 4.3 Community Impact Monitoring)[20].

[19] https://www.gov.br/mma/pt-br/assuntos/servicosambientais/ecossistemas-1/conservacao-1/areas-prioritarias/2a-atualizacao-das-areas-prioritarias-para-conservacao-da-biodiversidade-2018. Published 09/12/2021.

[20] https://www.gov.br/mma/pt-br/assuntos/ecossistemas-1/conservacao-1/areas-prioritarias/arquivos/fichas_amz_.pdf

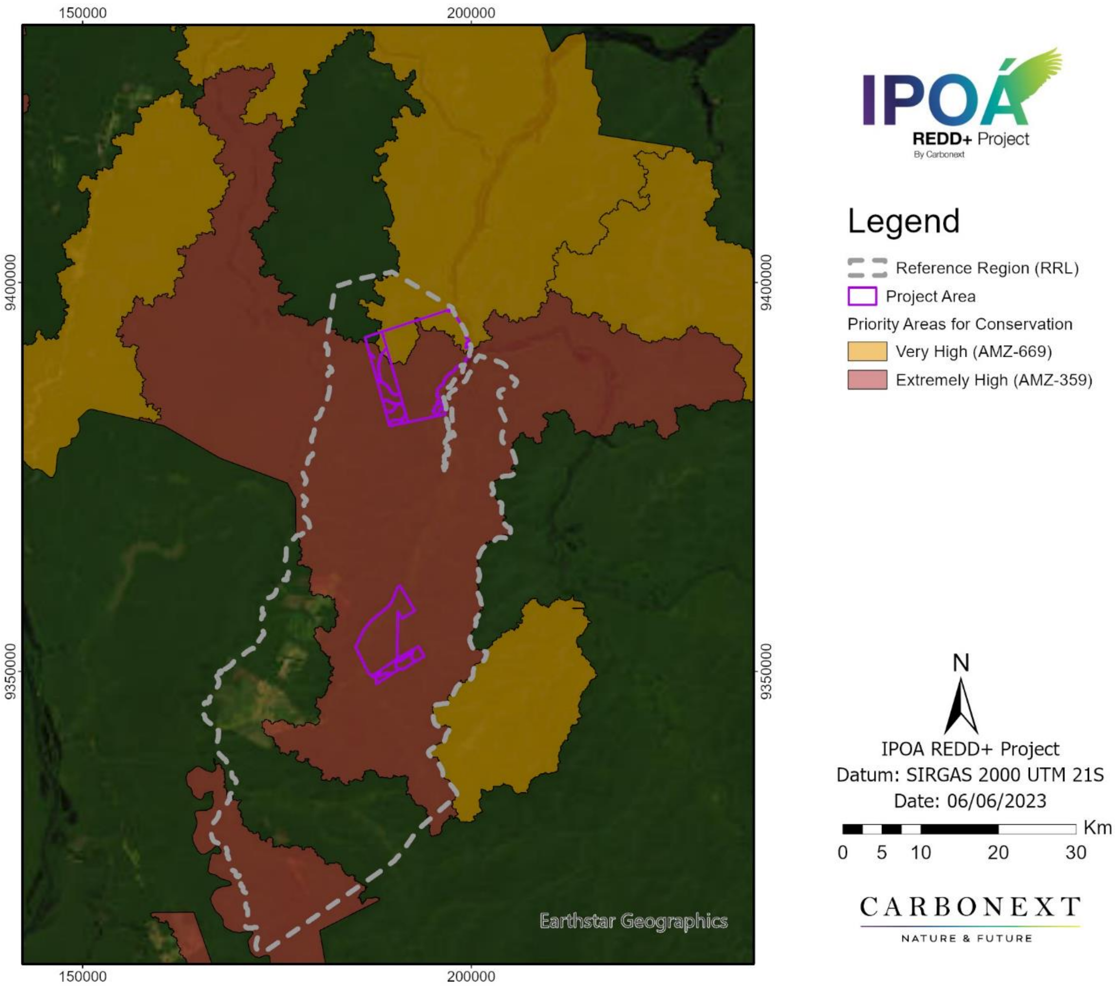

Figure 2 - Priority Areas for Conservation surrounding and overlapping with the project

Territorial Management in Brazil

The IPOÁ REDD+ project is aligned with the main premises of the Action Plan for the Prevention and Control of Deforestation in the Amazon PPCDAm[21]. The PPCDAm carries out mitigation actions and

[21] http://redd.mma.gov.br/pt/acompanhamento-e-a-analise-de-impacto-das-politicas-publicas/ppcdam#:~:text=O%20Plano%20de%20A%C3%A7%C3%A3o%20para,desenvolvimento%20sustent%C3%A1vel%20na%20Amaz%C3%B4nia%20Legal. Access 28/03/2023.

continuous actions to reduce deforestation and is linked to the Legal Ecological-Economic Macrozoning of the Amazon (MacroZEE)[22] of the State of Amazonas established by Decree n. º 7.378/2010[23].

In order for the Action Plan for the Prevention and Control of Deforestation in the Amazon - PPCDAm to include the objective, it is necessary to adhere to initiatives adopted with independent actions, such as those proposed within the scope of this project.

Brazil's Nationally Determined Contribution (NDC)

At the international level, the IPOÁ REDD+ Project is aligned with the environmental priorities set by the Brazilian Federal Administration at COP 14 in December 2008, where a 70% deforestation decrease by the year 2018 was declared, and then further targets of zero illegal deforestation by 2030, and offsetting greenhouse gas emissions from illegal sources of vegetation removal were declared. The latter are elements of the Brazilian NDC, which the country intends to adopt more firmly in the framework of the Paris Climate Accords (COP-21)[24]. Recently Brazil made official at COP 26, in November 2021, that the country's goal is to eliminate deforestation by 2028, increase to 50% the goal of reducing greenhouse gas emissions by 2030, and reduce by 30% its methane emissions in the atmosphere[25]. At this COP, the Glasgow Climate Pact was signed, which reinforced the parties' commitments to keep warming below the 1.5°C limit[26]. Another important decision taken was the authorization of the global carbon market, which means that countries will be able to exchange carbon credits among themselves, favoring Brazil because of the Amazon Forest (IDESAM, 2021).

Sustainable Development Goals

The SDGs - Sustainable Development Goals are a campaign of the UN, the United Nations Organization, to promote positive changes in the world of the future. In 2015, the conclusion of the negotiations on the 2030 Agenda culminated in an ambitious result for Brazil, which was the proposal of 17 Sustainable Development Goals; 231 Sustainable Development Global Indicators to monitor and measure progress in the implementation of the SDGs; and 169 targets, representing the central axis of the 2030 Agenda, which

[22] https://antigo.mma.gov.br/component/k2/item/8204. Access 28/03/2023.
[23] https://www2.camara.leg.br/legin/fed/decret/2010/decreto-7378-1-dezembro-2010-609606-publicacaooriginal-130911-pe.html#:~:text=Aprova%20o%20Macrozoneamento%20Ecol%C3%B3gico%2DEcon%C3%B4mico,que%20lhe%20confere%20o%20art. Access 28/03/2023.
[24] Source MMA: http://redd.mma.gov.br/pt/redd-e-a-indc-brasileira
[25] Source FGV: https://portal.fgv.br/artigos/quem-e-brasil-cop-26. Access 12/05/2022.
[26] Source IDESAM (2021): https://idesam.org/opiniao/cop26-entre-erros-e-acertos/.

will guide actions in the three dimensions of sustainable development[27]. Brazil has highlighted this as a great opportunity to eradicate poverty that may be within the mandate of this new Agenda.

Considering the national sustainable development goals and indicators, the IPOÁ REDD+ Project has aligned its activities so that, at the local level, its activities contribute to some of the national goals[28]. The goals and targets related to the activities executed in this present Monitoring Report are presented in the table below.

Table 1. Sustainable development goals contributed to in this monitoring period by specific project actions.

Sustainable Development Goal	Corresponding CCB Action in MR1
1- No poverty	Community association creation
4 - Quality education	School infrastructure and equipment and Teacher and student education and training
5 - Gender Equality	Women's workshop
7 - Affordable and Clean Energy	Installation and maintenance in communities of infrastructure and equipment for electricity, telecommunication, and internet access
8 - Decent work and Economic Growth	Training and workshop: Occupational health and safety and Personal protective equipment (PPE) distribution for worker health and safety
9 - Industry, inovation and infrastructure	Installation and maintenance of infrastructure and equipment for electricity, telecommunication, and internet access

[27] Source: ODS – Metas Nacionais dos Objetivos de Desenvolvimento Sustentável (https://portalods.com.br/publicacoes/ods-metas-nacionais-dos-objetivos-de-desenvolvimento-sustentavel/?msclkid=ae9f0bf1cfa611ecb5630a6edc181d31)
[28] Source: https://www.pactoglobal.org.br/ods

	15 - Life on Land	Deforestation prevention and mitigation actions
	11- Sustainable Cities and Communities	Community association creation

Cancun Safeguards

The IPOA Project was developed in accordance with the Climate, Community and Biodiversity (CCB) Standards and as described in the CCB Standards[29], projects developed with such standards are aligned with the Cancun Safeguards.

The Cancun Safeguards were agreed upon in 2010 in Cancun, Mexico, at the United Nations Framework Convention on Climate Change (UNFCCC) with the objective of implementing activities that reduce emissions from deforestation and forest degradation so that they contribute to conservation, sustainable management of forests, and enhancement of forest carbon stocks (REDD+)[30]. These clauses have also been adapted by Brazil, as reported by the Ministry of Environment (MMA)[31]:

> *"When undertaking [REDD+] activities, the following safeguards should be promoted and supported: (a) That actions complement or are consistent with the objectives of national forest programmers and relevant international conventions and agreements; (b)Transparent and effective national forest governance structures, taking into account national legislation and sovereignty; (c)Respect for the knowledge and rights of indigenous peoples and members of local communities, taking into account relevant international obligations, circumstances and national laws, and noting that the UN General Assembly has adopted the UN Declaration on the Rights of Indigenous Peoples; (d) The full and effective participation of relevant stakeholders, in particular indigenous peoples and local communities, in [REDD+] actions; (e) That actions are consistent with the conservation of natural forests and biological diversity, ensuring that [REDD+] actions are not used for the conversion of natural forests, but rather to encourage the protection and conservation of natural forests and their ecosystem services, and to enhance other social and*

[29] https://verra.org/wp-content/uploads/2017/12/CCB-Standards-v3.1_ENG.pdf
[30] Source: "The Cancun Agreements: Outcome of the work of the Ad Hoc Working Group on Long-term Cooperative Action under the Convention", FCCC/CP/2010/7/Add.1. https://unfccc.int/documents/6527. Access: 28/03/2023.
[31] Source: http://redd.mma.gov.br/images/publicacoes/salvaguardas_1sumario.pdf

environmental benefits; (f) Actions to address reversal risks; (g) Actions to reduce emissions displacement[32]."

2.2 Project Implementation Status

2.2.1 Implementation Schedule (G1.9)

The activities and diagnoses implemented during the present MR1 period are presented in the table below. Full details regarding them are present in sections 2.3.2 – 2.3.3 for the socioeconomic diagnoses, 3.1.2 for implemented project activities in the climate pillar, and 4.1.1 and 4.3.1. for implemented project activities in the social pillar.

Table 2. All project activities and diagnoses conducted in the MR1 period

Date	Milestone(s) in the project's development and implementation
March/2022	Socioeconomic diagnosis
April/2022	Socioeconomic diagnosis
April/2022	Discovery workshop for women
April/2022	Distribution of ecological women's hygiene kits
April/2022	Distribution of oral hygiene kits
May/2022	CarbonEye Alert
June/2022	Telephone frequency test
July/2022	Reporting of deforestation event to command and control organs
September/2022	Sign installation – Maanaim Farm
October/2022	Antenna installation test
October/2022	Sign installation – Campos Farm
October/2022	Socioeconomic diagnosis – Part II
February/2023	Solar panel and Starlink internet installation
February/2023	Participatory Mapping Workshop
February/2023	Personal Protective Equipment (PPE) diagnosis

[32] Source: UNFCCC Decision 1/CP.16, Annex I, paragraph 2, available at: https://www.un-redd.org/glossary/cancun-safeguards

March-December/2023	Starlink monthly payments
July/2023	Delivery of school equipment and materials to the community school
July/2023	Training on associations and cooperatives
July/2023	Community Public Consultation
July/2023	Participatory Mapping Workshop - Validation
July/2023	Survey of results after implemented actions
July/2023	Women's workshop
July/2023	Community occupational health and safety training
July-August/2023	Delivery of PPE equipment to community members
October 2023	Meeting with a prominent Amazonas State health foundation
August-December/2023	Weekly Meetings with the Alicerce (Education) team

2.2.2 Methodology Deviations

The methodology deviation cited in the PD applies to the present Monitoring Period. It is as follows.

The methodology's VMD0007-BL-UP_v3.3 module requires that at least one land-tenure variable be included as an input variable, as described in the module's section 3.1.2. However, in the present project variables related to actual land tenure and management – specifically public and protected areas – were disregarded in the model, because only private areas are included in the project's Reference Region (RR). This is in order to respect the following similarity requirement of the aforementioned module, section 1.1.1.1: "Policies and regulations having an impact on land-use change patterns within the RRD and the project area must be of the same type or have an equivalent effect at the start of the historical reference period, taking into account the current level of enforcement". Following this requirement, Conservation Units and Indigenous Territories, which have specific policies and regulations, were excluded from the RR.

To this extent, in order to correctly follow the methodological requirement in section 1.1.1.1, it was considered necessary to disregard the requirement in section 3.1.2.

A second deviation emerged in the context of MR1, as follows. Regarding estimation of emissions from market effects (LK-ME), it was noted that the methodology for calculating *ex-post* emissions from market leakage through decreased timber harvest is not specified in the VMD0011 module. For this reason, a procedure for ex-post calculation was created, with the following steps:

1. subtract unavoided deforestation measured in the PA from APD and AUD baseline values (which are used in the baseline LK-ME calculation)
2. If hectare result is <0, consider it as zero to avoid positive leakage.
3. If hectare result is >0, proceed to calculate MR values for $LK_{MarketEffects,timber}$, applying the rest of the procedure as described in VMD0011, however replacing the A_i in equation (6) with the new baseline minus measured deforestation value (in hectares). We will call this variable MR_1A_i (ha).

No further methodology deviations were identified for the Ipoá REDD+ Project.

2.2.3 Minor Changes to Project Description (*Rules* 3.5.6)

No minor changes to the Project Description were identified in the monitoring period.

2.2.4 Project Description Deviations (*Rules* 3.5.7 – 3.5.10)

New impacts (long-term) have been included in the Further Leisure, Cultural and Sporting Activities scope of the Theory of Change in section 4.4.1.

2.2.5 Grouped Projects

This is a grouped project. During MR1 no new instances were added to the Project Area.

1) **New Project Areas and Communities (G1.13)**
   No new Project Areas were added during MR1.

2) **Removed Project Areas and Communities (G1.13)**
   No new Project Areas were added during MR1.

3) **Eligibility Criteria for Grouped Projects (G1.14)**
   No new Project Areas were added during MR1.

4) **Scalability Limits for Grouped Projects (G1.15)**
   No new Project Areas were added during MR1.

5) **Risk Mitigation for Grouped Projects (G1.15)**
   No new Project Areas were added during MR1.

6) **Project Zone Map (G1.13)**
   No new Project Areas were added during MR1.

7) **Changes to Management (G4.1)**

No new Project Areas were added during MR1.

2.2.6 Risks to the Project (G1.10)

Some natural and human-induced risks to the project's climate, community, and biodiversity benefits are expected during the project lifetime. Each risk type has corresponding mitigation measures, as presented in the table below. In accordance with previsions in the PD, this item has been updated to reflect the renewed risk assessments for the present monitoring period.

It is further noted that section 2.3.4 below presents project risks focused on community aspects, and the Adaptive Management Plan 2.3.8 identifies, assesses, and creates a mitigation plan to address them. And remaining risk types were duly analyzed using the guidelines of the Non-Permanence Risk Tool, version 4.0, as presented in the IPOÁ Non-Permanence Risk Report.

Table 3. Potential project risks on community, biodiversity, and climate aspects.

Identify Risk	Potential impact of risk on community, biodiversity, and climate	Actions needed and designed to mitigate the risk
Disputes over access/use rights (or overlapping rights)	Disputes may negatively impact communities. Or indeed compromise the continuity of the project itself. And consequently, the benefits to the climate, biodiversity and communities may be compromised.	Application of Deforestation Event Protocol relating to deforestation event which occurred between May 12 and 22, 2022. This was an invasive[33]e deforestation event suspected to have been carried out by neighbouring properties, involving a road being cut through the project's Maanaim property.
Changes in policies and laws governing external actors	Changes can compromise the continuity of the project and consequently the	No instances of this project risk type occurred during MR1.

[33] See evidence folder "08. Deforestation"

	benefits to the climate, biodiversity, and communities.	
Increase of fires	Carbon emissions, biodiversity loss and risks for the communities.	Although project field and GIS analyses assessed this risk as low in the PZ, the possibility of fire from anthropogenic causes, natural causes, or an interaction between the two was continually monitored over the MR1 period, on a weekly basis by the CarbonEye system. As presented in section 3.1.2, based on an analysis of heat points over the period, the fire risk in MR1 remains classified as low.
Deforestation/Degradation	Deforestation or degradation of forest cover.	The road built by the invaders in the project's Maanaim property led to the deforestation of 8 hectares within the PA.[34]

2.2.7 Benefit Permanence (G1.11)

This MR is the first in the project's life cycle, so the considerations relating to "measures required and implemented since the last CCB validation or verification to maintain and improve the climate, the community, and biodiversity" will not apply to this first period. For this reason, the following explanation focuses on how each of the CCB project actions was designed, from the start, to provide permanence beyond the project lifetime.

Regarding the Social Pillar of the Theory of change, communication and electrification activity scope, the installed communication infrastructure is designed to be long-lasting to serve the community beyond the project period. Regarding the solar and antenna equipment's longevity the Social Pillar of the Theory of change, communication and electrification activity scope, the installed communication infrastructure is

[34] See evidence folder "08. Deforestation"

designed to be long-lasting to serve the community beyond the project period. Regarding the solar and antenna equipment's longevity Regarding the longevity of solar panels, according to information from the solar energy market in Brazil[35], on average, standard solar panels have an efficiency guarantee for 25 years, operating at 80% of its original performance. After this period, there is a reduction in efficiency, but they remain fully operational. The lithium battery, according to information provided by the supplier, has an efficiency of 10 years and must then be replaced. Regarding the antenna, according to information from the supplier, it will continue to operate, with only occasional software updates being necessary.

During the training given in February 2023, the specialist in solar panels trained[36] the communities in maintenance and good practice of the installed equipment, capacitating them directly with a view to ensuring that the benefits of the project continue beyond its lifetime, by making the beneficiaries themselves responsible and trained to conduct replacement when required.

Furthermore the structures being implemented by the project, as described in section 2.3.10 of the present MR and 2.4.1. of the PD create a governance framework, intrinsically involving the community in decision-making and implementation processes, which go beyond just the relationship with the project proponents, to other partners and the public sector, which aim to be long-lasting and transformative. For example the community association creation activity, planned to initiate as of MR2, will be indispensable in this regard providing community members with ever increasing agency over project activity design and implementation. These structures will affect all three of the elements, "C,C, and B" as described in the Theory of Change (see section 6 of the present MR).

Turning to the climate pillar, the Ipoá REDD+ Project has developed a set of strategies to maintain forest integrity within the PA and PZ, through forest monitoring that combats deforestation, burning, and illegal invasions, and provision of a framework to monitor, manage and communicate with communities regarding deforestation events. This strategy framework is comprised primarily of Operational Procedures (OPs), as fully presented in section 4.3.1 and 3.1.2. These procedures are designed to standardize procedures, establish the flows of actions, management, reporting, and records storage. The OPs, together with remote monitoring of the area, help to reduce emissions from deforestation and degradation, thereby benefiting the climate.

2.3 Stakeholder Engagement

2.3.1 Stakeholder Access to Project Documents (G3.1)

[35] https://www.portalsolar.com.br/quanto-tempo-duram-os-paineis-solares
[36] See evidence folder "05. Communication and Electrification Member_trained"

The IPOÁ project is committed to ensuring that communities and stakeholders are fully informed about the project and its implications. Information regarding the project was disseminated orally to local communities, through meetings that took place in, April 23 and 25/2022 and February 2023 (see Figure 4). This guaranteed access to information and answered questions that the community had.

Both communities and institutional stakeholders have been informed about the project's development and verification through public consultations between March and April 2022, and in meetings during activities with communities in the Project Zone on March 16, April 23 – 25 2022 and February 2023.

Ipoá's Project Description and other documentation is publicly available online on the VERRA[37] website. Carbonext also provides, on its website[38], a summary of the project and its most recent developments in accessible, direct language, in both English and Portuguese.

Figure 3 - Meetings with the community in the Project Zone.

2.3.2 Dissemination of Summary Project Documents (G3.1)

[37] https://registry.verra.org/app/projectDetail/VCS/3324
[38] https://www.carbonext.com.br/projects/ipoa

In February 2023, the summary documentation of the project, including the minimum points describing the overall project profile described in G1.1-9, was sent by email to institutional stakeholders. Also, it was disseminated to the communities in the field during a technical visit, with illustrative material in Portuguese including leaflets and banners to facilitate comprehension (figures 5 and 6), and direct explanation to community members, (figure 6), all in accessible language.

The summaries of the Monitoring Reports will be delivered after each verification period to the institutional stakeholders by e-mail, and a printed copy in Portuguese will be handed out to communities during the next site visit, which is planned during the MR2 period.

In addition, the summaries of the project documents will be made publicly available through the VERRA website.

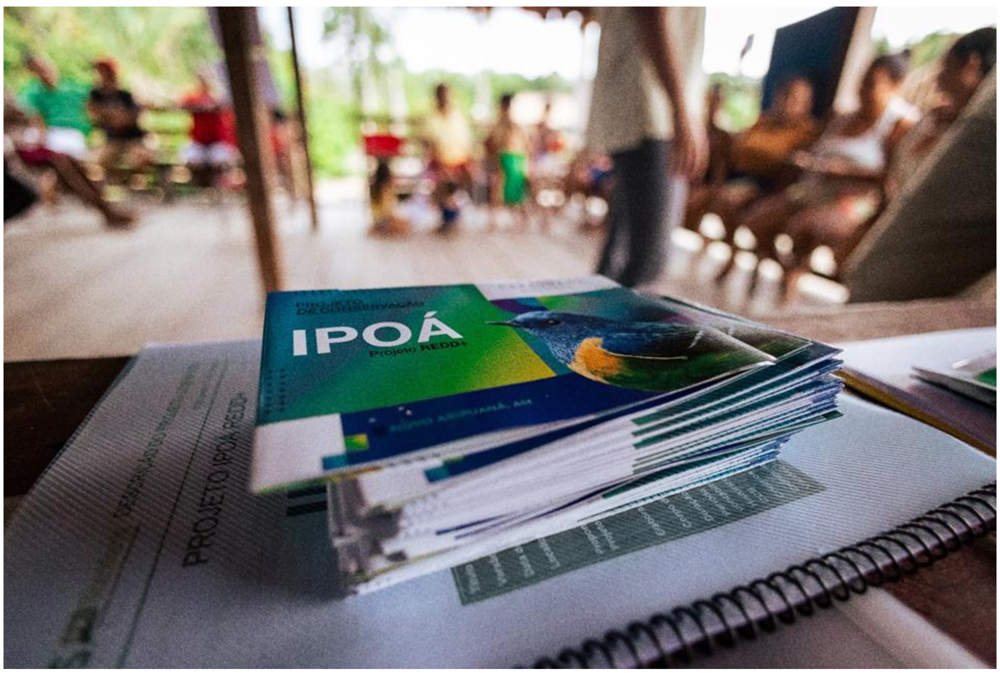

Figure 4 - Delivery of an informative folder and a copy of the IPOA Project Summary in Portuguese

Figure 5 - Social Analyst in charge of the project handing out information material to community members.

2.3.3 Informational Meetings with Stakeholders (G3.1)

Several meetings with the community were held during the present monitoring period year. In these meetings, appropriate means of communication were used such as flyers and informative banners, as mentioned in 2.3.2. The team of social and technical analysts applied these informative materials at the homes of community members, where the residents were invited to participate in the meetings and received an informative invitation.

Below is a description of the meetings with stakeholders.

Table 4 - Meetings and activities with the IPOA Project stakeholders

Meetings Carried Out	Activities/Actions	Informative Material
Between March 14 and 21, 2022	Initial diagnosis in the vicinity of the Project Area, carried out by Carbonext's social analyst. The Acari and Suncunduri rivers and Igarapés Arara and Capinarana were all visited. First contact was made with the communities and the detais of each family was recorded.	N/A

March 16, 2022	An informative meeting on the REDD+ project was held at the Boa Esperança community. Carbonext's social analyst gave the presentation using clear communication, to discuss the main points of the project and answer questions raised by the inhabitants. For communication, the WhatsApp contact of the social analyst was given to the community leader who has a cell phone. The project was also presented to the leaders of Vila Batista and Vale da Benção communities, where both signed the consent form of the project.	Banner (short description of the project, contact email of Carbonext – Figure 8 e 10)
April 23 and 25, 2022	Socio-economic questionnaires were applied to families in all communities, and public consultation was carried out with the residents present in the three communities involved: Vila Batista, Boa Esperança and Vale da Benção. During the visit, 44 people were interviewed pertaining to 6 families and three different communities. In addition, an informative meeting was conducted in the Vila Batista community with 24 participants from the three communities.	Distribution of invitation flyers (Figure 9) for all residents presents socio-economic questionnaires
February 2023	A zoning workshop was conducted, led by Carbonext GIS professionals accompanied by the social and technical team with a total of 23 participants (16 Boa Esperança, 5 Vila Batista and 2 Vale da Benção)[39]. Project updates for the first monitoring period, summarizing all the activities carried out and the next steps in the project. For community awareness and to inform about upcoming project actions.	Banner Folders Delivery of the PD summary in Portuguese.

[39] See evidence folder "11. Participatory Mapping Activity"

Figure 6 – Consulting the leaders of the Vale da Benção and Vila Batista communities

Figure 7 - Presentation of the project in the Boa Esperança community.

Figure 8 - Invitation flyer for community consultation meeting.

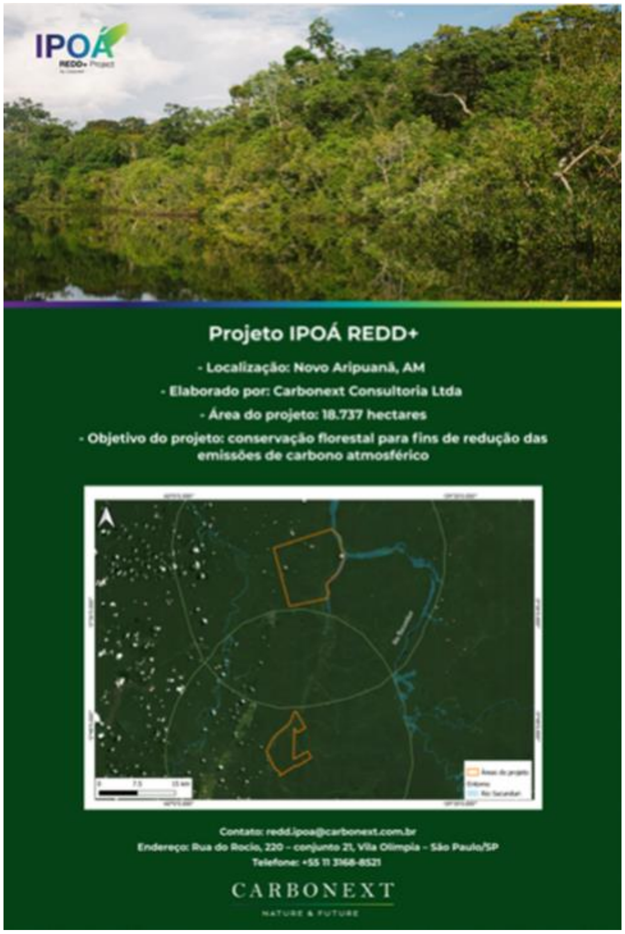

Figure 9 - Banner for the IPOÁ REDD+ Project.

2.3.4 Community Costs, Risks, and Benefits (G3.2)

The predicted costs risks and benefits of the project were assessed together with all three PZ communities during two field visits in 2022 (see section 2.3.3 above). On these occasions, techniques including active listening, visual aids and accessible language were used to achieve, in collaborative fashion together with the community, the following:

- present information regarding the demarcation of project boundaries, especially the Project Zone;
- Identify and answer the questions and concerns about the project;
- clarify its purpose;
- Identification of its potential costs, risks, and benefits.

Following these participatory sessions, the communities consented to participating in the project.

The project's vision is to maintain an open channel of communication, so that the mapped communities can inform the proponents of their own perceptions relating to project benefits, costs, and risks.

The content of the identified benefits, costs and risks are as follows:

- Community benefits achieved by the project's fourteen activity scopes during the MR1 period are reported in section 4.3.1;
- No community costs arising from the Ipoá REDD+ project were identified;
- Risks and procedures for their management are reported in section 4.1.2.

2.3.5 Information to Stakeholder on Verification Process (G3.3)

Regarding the CCB verification process conducted by an independent validation/verification body, there were two means of communication to stakeholders during the pre-validation and verification visits:

I. The Field Manager visited in October 22 the community stakeholders and provided information on the REDD project and the audit process.
II. An email was sent notifying institutional stakeholders about the public consultation.

I) On this occasion, the audit process was explained to the community, including the return audit visits for initial validation and verification, followed by annual verification visits. The communication methods were designed to be accessible and intelligible to the target audience, delivered in Portuguese, in person, in understandable language.

II) The institutional stakeholders (listed in section 2.1.9 of the P.D) were informed about the project by email, a summary document of the project was sent by email on the 17th of March of 2022.

2.3.6 Site Visit Information and Opportunities to Communicate with Auditor (G3.3)

The communities in the Project Zone were notified in advance of the Auditors' visit by the Project Field Agent. The invitation was personally delivered (Figure 9) by the project field agent by hand to the residents, inviting them to voluntarily participate in the Auditing visit.

Community leaders were also instructed to pass the message on to the residents. After the installation of the means of communication in Boa Esperança, now already in operation, community members will be notified in advance through a list of contacts on WhatsApp.

Visits to the project area will be informed at least one month in advance. So that the interested parties can plan and be available. The process of relationship building is underway with the community and will be reported on in subsequent reports.

Figure 10 - Invite delivered by the field agent in the communities.

2.3.7 Stakeholder Consultation (G3.4)

Strategy since the beginning of project development has been to incorporate the communities in the decision-making processes of the IPOÁ REDD+ Project. This occurred initially in March and April 2022 during meetings identify priority CCB activities, in which the communities participated fully and effectively in the process, as presented in the Project Description (section 2.3.7). During these meetings, the essential summary information about the project was transmitted, and residents were encouraged to comment, suggest, and participate in its development. Over the MR1 period, the residents' feedback was positive regarding the development of the project. Below is a list of the meetings held to inform the residents about the stages of the project:

It is notable that the public consultations for the present MR are specified, in the first table below regarding community stakeholders, and in the second table regarding institutional stakeholders.

Table 5. Project consultation meetings

Meetings	Objective
March 2022	Initial community public consultation and project development meeting, for presentation and consultation regarding the project, consent was received from all communities involved, through the signatures of the leaders of the three communities.
April 2022	Initial community public consultation and initial project development meeting, held in the Vila Batista community, maximizing the participation of residents of the PZ's three communities. 24 residents were in attendance (6 Boa Esperança, 12 Vale da Benção and 6 Vila Batista)[40], representing 25% of the total population of 96 people. The objective was to present project updates, encouraged residents to ask questions, suggest improvements, and participate in the decision-making process regarding the activities to be implemented in the region.
June 2022	Project development visit: A third-party agent accompanied the Carbonext field team to test the telephone signal frequency. On this trip, more information about deforestation detected by the CarbonEye remote monitoring system was also obtained.
September and October 2022	Project development and Forest Integrity field action visit: Project Field Manager dispatched to install signage at the project limits after an invasive deforestation event was detected. The sign installation took place in both months due to logistical problems, as there was no way to finish all the installations in a single trip. In October, the trip also had the objective of installing the communication antenna with the signal repeater identified in the test carried out previously, and to inform the community members of the planned validation/verification audit field visit involving the VVB and the Carbonext team. An antenna test was carried out with the active participation of the Boa Esperança community, who are its beneficiaries. They directed the equipment's installation and were instrumental in reaching the conclusion that the antenna signal test was unsuccessful. This was decisive in planning for the Starlink option, which was the successful solution implemented in February (below).
February 2023	MR1 first community public consultation with the members of the three communities, aiming to present the current stage of the project and updates on CCB actions. Residents were encouraged to ask questions and suggest

[40] See evidence folder "16. Public Consultation"

	improvements to the project; present the benefits that would be implemented; mediate a workshop of agreements regarding the use of the benefits. The meeting was attended (according to the meeting attendance list) by 22 residents.
July 2023	Meeting with the community members of the three communities with the aim of presenting the results of MR1, developing an association course, workshop for women, take support material for school and take personal protective equipment (PPE) to the community.
April 2024	MR1 second community public consultation and project development visit: various CCB actions involving the association and education are to be realized, along with the renewed community public consultation for MR1. The repeated public consultation was required in order to attend to a revised MR1 period and corresponding new field visit date audit.

Since the first contact with the communities, the contact details of Carbonext company and the Social Analyst in charge were passed to the communities. Throughout this monitoring period, contact was maintained with the communities through our communication channels. This was possible only when communities had access to telephone signal when traveling to larger cities with such infrastructure, because it is a remote region. After the installation of the antennas in February (02), this process was made easier, to a transformative degree.

Regarding institutional stakeholders, initially an email with a brief explanation of the IPOÁ project, location and summary was sent to the institutional stakeholders mapped (PD section 2.1.9) . Carbonext made contact details available in case any institution had any questions or comments, enabling them to have influence on the project as appropriate.

Meetings with the public sector actors were carried out as described in the table below. Carbonext keeps an open direct channel of communication for questions, comments and complaints as described in section 2.3.9 Stakeholder Consultation Channels.

Table 6. Institutional Stakeholder Consultation meetings

Meetings	Objective
17 March 2022	E-mail to stakeholders containing information about the project (Public consultation summary).
January 2023	Meeting between the technical, legal and social team of Carbonext and the municipal pedagogical coordinator to present the project and align the education activities

January 2023	Meeting between Carbonext's technical, legal and social team and municipal lawyer to present the project and discuss health and education activities.
January 2023	Meeting between political representatives and a lawyer from one of the project municipalities, and representatives of Carbonext’s technical, social and legal teams to explore the potential to collaborate on health and education activities.
March – April 2024	Second Institutional public consultation involving sending e-mail to stakeholders containing information about the project (Public consultation summary), and in-person visits to key stakeholders. The repeated public consultation was required in order to attend to a revised MR1 period and corresponding new field visit date audit.

Carbonext remains committed to seeking forms of collaboration with the public sector, throughout the Ipoá project’s lifetime, to deliver its planned benefits to their full potential.

2.3.8 Continued Consultation and Adaptive Management (G3.4)

The activities implemented by the Ipoá project are designed to be collective, from their definition and design phases, through implementation, to their continuous adaptive management. Initially, the decisions about activity design arose from the needs and wishes declared by the communities themselves during the initial consultations. Their implementation includes consultation on a continuous basis, at each monitoring period as of MR2, through means including a satisfaction survey, and includes special attention for inclusion of women and youths. Any community feedback received is relayed to project management, in order to make necessary changes in a timely manner.

The project works to maintain an effective and continuous communication channel with the communities involved and other stakeholders. In this first monitoring period, communication occurred through the project team during field visits, meetings and during implementation of activities, as well as through the indicated communication channels between the communities/other stakeholders and the Carbonext social team.

Ongoing consultation processes (detailed in the table below) are key to adaptive management, and creating a project that successfully brings benefits to local people, climate and biodiversity. They are key to building relationships, trust and alignment with communities and other stakeholders and permit project activities to be continuously adapted and modified when needed.

Table 7. Communication actions for ongoing consultation

Item nº	Communicat-ion Action	Objective	Parties	Frequency	Target audience	Evidence
i	**Informative Meeting about the Project**	To inform communities and other stakeholders about the project, and consult communities about project implementation	In person	Annual	Community residents and any further interested stakeholders	Attendance list List of suggestions[41]
ii	**Consultation Meeting**	To consult with the communities about project implementation	In person	Annual	Community stakeholders	Attendance list List of suggestions
iii	**Request for Information**	To request information from community leaders about communities, forest, etc.	Telephone	Monthly	Community stakeholders	Questionnaires [42]
iv	**Notifications**	To notify the community about technical visits, auditor visits, etc.	Telephone, and occasionally in person	Whenever necessary	Community stakeholders	Project promotional material delivered personally[43]
v	**Submission of Project Summary**	To inform identified stakeholders by e-mail about public consultation periods and verifications	E-mail	Annual	Institutional stakeholders	Evidence of Sending and receipt[44]
Vi	**Possibility of public-private collaboration**	Meeting with lawyers and politicians from local authority to analyze the possibility of a health-sector PPP	In person	Once	Public sector authorities	Meeting notes[45]
Vii	**Possibility of partnership with NGO**	Meeting with NGO that carried out health-related activities, but the result was negative, causing other	Online meeting	Whenever necessary	Third-sector partners	Meeting notes[46]

[41] See evidence file: : “05.Communication and Electrification”, document: “Diagnostico_Social_22”
[42] See evidence folder: “09. Other activities\Questionnaires”
[43] See evidence folder: “10. Community informed of audit”
[44] See evidence folder: “16. Public Consultation/institutional”
[45] See evidence folder: “01. Healthcare”, files: 23.01.31 and 23.10.20
[46] See evidence folder: “01. Healthcare”, files: 22.05.20 and 22.06.30

Item nº	Communicat-ion Action	Objective	Parties	Frequency	Target audience	Evidence
		possibilities to be explored				

Community feedback received through all the means described in *i – v* above is collected and processed internally by Carbonext The continuous adaptive management process, from community input to resulting action, is described as follows:

Incoming information from communities obtained via:

- Whatsapp or telephone with Carbonext social team or Project Field Team
- Direct data collected in the Field by Carbonext Social Team or Project Field Team
- CarbonEye

Internal processing:

- Carbonext Social, technical, GIS and Directorate.
- In the case of deforestation activity, a counsel meeting with the project owner's team is involved in decision-making.

Resulting action:

- Internal decision-making, planning, and adaptive rethinking on existing activities are conducted. resulting implementation of appropriate action at the project site.

Regarding items i – iii in the table above, the resulting actions during MR1 were clear. The consultation meetings enabled the identification of actions desired and needed by communities, leading to their subsequent implementation. For instance, internet and solar energy, being the first and third most-voted activities for implementation, respectively, (see community vote table, section 2.3.10) were successfully installed in MR1.

Regarding public and third sector actors shown in items vi and vii above, the January 2023 consultation with public sector representatives and lawyers from one of the Project's municipalities had the objective of exploring the possibilities for public-private partnerships regarding health and education. The result was negative, causing other possibilities to be explored.

In October 2022, meetings were held at the project site with a Civil Society Organization leading in the health field in the region, and telephone meetings with a municipal health coordinator also had the objective of exploring partnerships for health activities. These activity areas of health and education were (second and sixth most-voted activity by the communities in terms of most desired for implementation.

Consultations vi – vii were unsuccessful in the sense that they did not result in development and implementation of collaboration between the parties. This is primarily due to legal considerations (see sections 2.4.5 and 2.5.6) and best practices described at the end of section 2.3.7. However, they did indeed influence the project through adaptive management, because they led to refinement of understanding and identification of the avenues that these activity scopes will take in future MRs, through partnership with NGOs or private sector actors, for example.

2.3.9 Stakeholder Consultation Channels (G3.5)

During the community visits of the IPOÁ REDD+ project, consultations were conducted directly with communities of the Project Zone (PZ), and other surrounding resident stakeholders. They took place at one of the PZ's communities, being easily accessible to the community and the project was presented in full to the participants by social experts.

During the community public consultation, all three community leaders were present. All of the latter play an important role in the PZ, as they are the leaders generally recognized by community members, exerting predominant influence on collective decision-making. However, given the currently low level of community organization and absence of official community organizations such as associations[47], the leaders' role is unofficial, to date.

Regarding the institutional stakeholders, the project was presented by email, with the project summary sent to them with key information regarding objectives, introduction, location, other items and the contact address were sent in attachment, and they were invited to comment on the project.

Carbonext has communication channels, which were widely released to stakeholders during the public and institutional consultations (section 2.3.7), in addition to the relationship email - redd.ipoa@carbonext.com.br, the Whatsapp contact and a 0800 channel, where access is direct with our

[47] The creation of which is a planned project action

social team, for complaints, suggestions, denunciations, etc. For the community, it was also shared with the leaders, the direct WhatsApp contact of the project's social analyst to facilitate communication.

At the beginning of the project, communication with the communities was a barrier, as the Project Area has no telephone signal and travel to the nearest city was necessary for any contact. But after the installation of solar energy and internet access, access to communication was facilitated.

Other Stakeholders were consulted during meetings and conversations about the project, as presented in the table below[48].

Table 8. Summary of meetings with other stakeholders

Consultation Channel	Date	Party Involved	Evidence
Meeting to present the project to the relevant government environmental institutions	30/06/2022	Carbonext and Public Institutions (environmental secretariat)	Meeting minutes [49]
Talks to enable partnerships for the proposed actions of the Ipoá project	18/07/2022	Carbonext and Public Institutions (education and health secretariat)	Meeting minutes[50]
Informing about schools and community demands	14/09/2022	Carbonext and Public Institutions (municipal Councillor)	Meeting minutes[51]
Discussions to make the activity viable and present the project.	30/06/2022	Carbonext and a Civil Society Organization	Meeting minutes[52]
Seeking ways with the coordinators responsible to enable a more active participation in the community, with the training of an agent to take care of the community's health.	22/10/2022	Carbonext, and a local health clinic coordinator	Meeting minutes[53]

[48] See evidence folder “16. Public Consultation”
[49] See evidence: “16. Public Consultation\Stakeholder Consultation Channels”, file: 22.06.30
[50] See evidence: “16. Public Consultation\Stakeholder Consultation Channels”, file: 22.07.18
[51] See evidence: “16. Public Consultation\Stakeholder Consultation Channels”, file: 22.09.14
[52] See evidence: “16. Public Consultation\Stakeholder Consultation Channels”, file: 22.06.30
[53] See evidence: “16. Public Consultation\Stakeholder Consultation Channels”, file: 22.10.20

2.3.10 Stakeholder Participation in Decision-Making and Implementation (G3.6)

In order to involve the communities in activity design and decision making from the beginning of the project, between March 14 – 21, 2022 several meetings were conducted with the three river-dwelling (ribeirinho[54]) communities located within the Project Zone (PZ) – Boa Esperança, Vila Batista and Vale da Benção – . as reported in section 2.3.3.

These included participatory workshops and socio-economic questionnaires applied to the families, to gather data and information from all the communities located within the Project Zone. These events constituted the community-driven selection process of project activities, designed to meet their own necessities and desires. The results are presented in the table below, see also the photos in the following figure, while the process is fully discussed in section 2.1.11 of the PD – Project Activities and Theory of Change. This demonstrates the communities' participation and ownership of project decion-making processes during MR1.

Table 9. Community vote on project activities

Categories	Votes
Internet	10
Medical care (more frequent)	9
Solar energy	8
Brushcutter/ hoe equipment	6
Health care	6
Improved Transportation	5
School refurbishment	4
Water pump motor	4
School supplies	4
Medical center	3
Handicrafts	3
Toiletries for women	3
City Hall course to work in catering and cleaning	2
Road transportation from the Arara community	2
Ground transportation - car	2
School transport "catraia"	1
Irrigation in the summer	1
Internet course	1
Trained and responsible professionals	0

[54] Ribeirinho populations: people who live on the banks of the rivers in the Amazon region.

Playground for children	0
Health agent - medicines	0

Communication channels with the community were put in place as of MR1[55], creating an open channel for information, facilitating transparency regarding the actions and the participation of the community throughout the life of the project.

The figure below presents the governance structure and the parties involved in the decision-making and implementation processes.

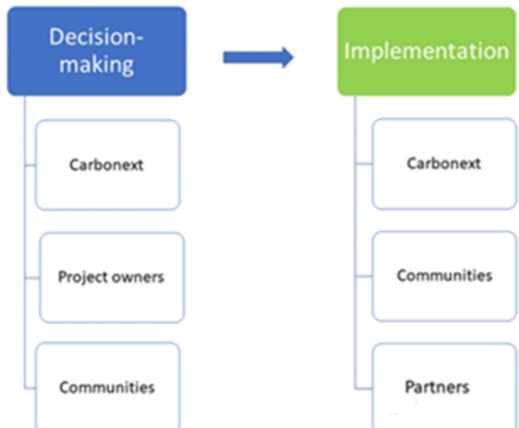

Figure 11 - Governance structure for decision-making and implementation

Given the facilitated communication, the project design and its activities follow the premise of respect for community decisions regarding the details, time and place of execution.

Cultural and gender information are important in the activity implementation, being measured for example by various indicators (see section 6), while participation of minority groups and the female population is encouraged.

[55] See evidence folder “16. Public Consultation”

Figure 12 - Community involvement in the design of the Ipoá REDD Project activities, April 2022, and complementary hygiene kits to be distributed to communities.

2.3.11 Anti-Discrimination Assurance (G3.7)

Carbonext has a Code of Ethics and Organizational Conduct in which prohibits any type of abuse or harassment, whether moral, sexual, or discriminatory. It does not allow:

- Harassment of a sexual or moral nature.
- Exposure of inappropriate material or any other inappropriate conduct.
- Verbal or written offenses.
- Belittling treatment or threats.
- Jokes, nicknames or any offensive reference to sex, race, age, colour and religion.

Owners and others involved the project design go through internal due diligence involving the appropriate administrative and judicial bodies, and through certificates and declarations, in order to verify any involvement with inappropriate conduct that could implicate any risk to the project.

2.3.12 Grievances (G3.8)

A communication channel was established for stakeholders to express, at any time, their concerns and to solve eventual conflicts and grievances that arise during project planning, implementation, and monitoring. The main communication channel is direct contact with the Social Analyst, the company channel[56] and the project's own email (redd.ipoa@carbonext.com.br), which is managed by the Carbonext team.

No grievances were registered, in the communications channel, during the monitoring period.

2.3.13 Worker Training (G3.9)

During the monitoring period, community members were trained in first aid and safety at work as well as training in association and cooperative formation.

The first aid training took place in July 2023 and was attended by 12 participants lasting 6 hours. A professional was hired with experience in training residents of isolated riverside communities with difficult access to immediate health care in the event of an accident at work. At the end of the training, participants were given a kit of Personal Protective Equipment (PPE) that will help community members work safer, and avoid accidents (Figure 12).

Figure 12 - First aid and work safety training

[56] See evidence folder "16. Public Consultation"

Also in July 2023, community members participated in training in associations and cooperative work. Likewise, an experienced professional was hired to speak to community members about the importance of social organization for communities and present what associations and cooperativism are. The training lasted two days, 6 hours each day, and included the participation of 17 community members (Figure 13).

Figure 13 - Training in associations and cooperativism

2.3.14 Community Employment Opportunities (G3.10)

As predicted in item 2.1.11 of the Project Description, training in association and cooperative work was provided to community members of the Ipoá REDD+ project during this monitoring period. In practical terms a community association permits a rural community to: defend the interests and rights of the local community; to achieve better productive efficiency, through professional training and technical assistance, incorporating technologies and better economic-financial management of agricultural activity; and to produce and transact goods within a recognized legal framework, which are likely to include non-timber forest products, for example.

In this training event, a specialized consultant, hired by the project developer, provided training in the following areas:

- What are associations and cooperatives and their differences;
- History and importance of associations and cooperativism;
- Fundamentals of community organization; and
- How to formalize an association.

As a product of his activity, the third-party contractor prepared a report showing the current context of the communities in relation to the level of organizational maturity for the creation of a residents' association. Furthermore, he indicated necessary actions to formalize the residents' association.[57]

In accordance with the PD's Theory of Change, the final objective (outcome and impact) of this activity is that community governance is strengthened, enabling and improving the living conditions of community members through access to public services and income generation through entrepreneurship and employability.

A gender capacity strengthening activity was also carried out, involving a consultant with extensive experience in carrying out activities with river-dwelling women in the Amazon was hired. One of the objectives of the activity was to identify possible activities that could generate income for women in the communities (Figure 14).

As with the activity on associations and cooperatives, the consultant also produced a final report on the activity. Some activities that could generate income for women were identified, including handicrafts, cutting and sewing, training as a community health agent, pedagogy, and working with natural remedies. The report also indicated possible activities that could be carried out with the aim of making it easier for women in the communities to obtain income, including:

- Training in crafts and entrepreneurship;
- Virtual health community agent training course;
- Workshop on digital marketing via cell phone;
- Seed processing course; and
- Training in making natural dyes and syrups for sale.

As indicated above, and in accordance with policies indicated in PD section 2.3.15, the project's work on employment opportunities was conducted without any discrimination based on gender, age, religion, marital status or ethnicity. With this, it is expected to strengthen the effective participation of community members in political, social and economic decisions, increasing local esteem and representation. In this way, the aim is that all participants in the training may be participants in community income generation, and the participation of women and senior citizens is encouraged.

2.3.15 Relevant Laws and Regulations Related to Worker's Rights (G3.11)

[57] See evidence folder "19. Employment"

The IPOÁ Project operates in full compliance with the labor laws of Brazil and strives to set an example of best practice regarding labor terms, conditions, and practices. Specifically, the project has met and exceed each of the following laws and regulations:

Decree-Law No. 5,452, May 1, 1943, is approval of the Consolidation of Labor Laws (CLT), which the then Brazilian president Getúlio Vargas and a group of lawyers and legislators drafted to ensure a series of titles and regulations in the relationship between employers and employees. Since then, the CLT has undergone a series of modifications and improvements.

Reduction to a condition analogous to slavery. Art. 149 - Reducing someone to a condition analogous to slavery, either by subjecting them to forced labor or exhausting working hours, or by subjecting them to degrading working conditions, or by restricting, by any means, their movement due to a debt contracted with the employer or agent:

Art. 60: Any work for minors under the age of fourteen is forbidden, except as apprentices." The Consolidation of Labor Laws (CLT) establishes the conditions for young people aged 14 to 17 to work in Brazil in articles 402, 403, 428, 429, 430, 431, 432 and 433.

Art. 5 Everyone is equal before the law, without distinction of any kind, and Brazilians and foreigners residing in the country are guaranteed the inviolability of the right to life, liberty, equality, security and property, under the following terms:

Art. 60: Any work for minors under the age of fourteen is forbidden, except as apprentices."

The Consolidation of Labor Laws (CLT) establishes the conditions for young people aged 14 to 17 to work in Brazil in articles 402, 403, 428, 429, 430, 431, 432 and 433.

Art. 7 These are the rights of urban and rural workers, in addition to others aimed at improving their social condition.

The main and biggest changes occurred with the Labor Reform under Law No. 13,467 of July 13, 2017, i to adapt the legislation to the new labor relations. The project proponent complies with all laws and regulations covering workers' rights and has no claims against them.

2.3.16 Occupational Safety Assessment (G3.12)

No activities in this scope were performed during this verification period .

During the MR1 period, there was an occurrence of invasion and illegal deforestation registered as police report on 06/07/2022. This type of event involves a substantial security risk for the project's field team and the owner's legal team who went to the invasion site and for the responsible command and control bodies, respectively.

To manage these risks (which are listed in the 2.1.11 section of the PD, Carbonext uses the following OP's and internal procedures:

OP - Deforestation Event
OP - Occupational Health and Safety
FIP - Forest Integrity Plan

2.4 Management Capacity

2.4.1 Required Technical Skills (G4.2)

In order to carry out REDD+ project planning, development, and monitoring, the key skills required for project implementation include:

I. Specialized REDD+ standards (VCS + CCB) and implementation knowledge;
II. Carbon accounting and monitoring;
III. Forestry/forest engineering;
IV. GIS mapping specialized in the Amazon region;
V. Social field skills and theory; and
VI. Biodiversity field skills and theory.

In the fields I to III above, Carbonext's technical team is responsible, led by the Technical Director and Managers; field IV is led by the GIS team; and regarding point V, Carbonext has a specialized social team with extensive education and prior experience in key areas. Furthermore, continuing on point V, the Project's Field Team, led by the Field Manager, is under the project owner's responsibility.

Regarding point V, Carbonext's a team of social experts, who have the necessary skills to involve the community and stakeholders, are important to ensure the community participation, activity design, and monitoring required within the context of CCB projects.

It is important to note that certain aspects of the work relating to point IV are expected to be carried out by specialized outsourced third-parties, and this was the case in this MR relating to the installation of satellite internet and solar panels.

On point VI, the fauna and flora inventory, a third-party team will be hired to collect data in the field and generate the technical report. Preference will be given to local teams that have knowledge of local conditions, but this activity will only take place in future MR periods and has not been implemented in MR1.

2.4.2 Management Team Experience (G4.2)

Name	Janaina Dallan
Position	Carbonext CEO
Education	Degree in Forestry Engineering from Luiz de Queiroz College of Agriculture (ESALQ) and a Master's in business of the environment
Resume summary	Janaína Dallan has been working with the carbon market since 2001 on projects around the world. As CEO of Carbonext, she is directly involved in coordinating the development and implementation of carbon credit projects. Throughout her career, she has managed UNFCCC CDM (Clean Development Mechanisms) projects through several international companies such as Golder Associates, Global Energy Partners, Orbeo / Societé Generale, One Carbon and Ecofys. Janaína's involvement Carbonext's projects spans from their very conception, prospecting, and initial groundwork, through technical development, and leading at the CEO level. She is currently a member of the United Nations Framework Convention on Climate Change (UNFCCC), in which she has been a member of the Registration and Issuance Team (RIT) since 2013, she is president of the Brazil Alliance for Nature-Based Solutions and is a member of the experts for the Taskforce for Scaling Voluntary Carbon Markets (TSVCM).
Contact Information	janaina@carbonext.com.br
LinkedIn	https://www.linkedin.com/in/janainadallan/

Name	Luiz Fernando de Moura
Position	Technical Director

Education	Forestry Engineer by Luiz de Queiroz College of Agriculture (ESALQ) the M.Sc. and Ph.D. in Wood technology by the Université Laval (Quebec, Canada).
Resumé summary	Luiz Fernando de Moura coordinates Carbonext's technical team. Luiz's direction has seen all of Carbonext's projects through validation and verification, being heavily involved in their technical development and leading at the Direction level. He participated in the preparation of "Energia Verde Carbonization Project - Mitigation of Methane Emissions in the Charcoal Production of Grupo Queiroz Galvão, Maranhão, Brazil", registered on March 21, 2011. Dr. de Moura also participation as project designer in Florestal Santa Maria REDD+ Project, Fortaleza Ituxi REDD+ project, UNITOR REDD+ project and Evergreen REDD+ project.
Contact Information	luiz.moura@carbonext.com.br
LinkedIn	https://www.linkedin.com/in/luiz-fernando-de-moura-6077089/

Name	Franciele Salvador
Position	Legal Manager e Compliance Officer
Education	Lawyer by the Cenecista College of Joinville
Resumé summary	Franciele Salvador manages Carbonext's legal group, providing strategic support for projects on the brazilian voluntary carbon market. Previous experience working in large companies, as well as specializations in Business Law, Compliance, Contract: Business Vision and Practice.
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LinkedIn	https://www.linkedin.com/in/franciele-salvador/

Name	David Swallow
Position	Technical REDD+ Coordinator

Education	Degree in Anthropology from the University of Edinburgh and an MSc in Ecology, Evolution and Conservation from Imperial College London.
Resumé summary	David Swallow has twelve years' experience conducting technical development of REDD+ projects on the Brazilian voluntary carbon market. During his three years at Carbonext, he has to-date managed two REDD+ projects through development, validation and verification with a technical team of 2 – 4 members. He specializes in REDD+ project's technical development applying VCS and CCB standards, and their social development impacts.
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LinkedIn	https://www.linkedin.com/in/david-swallow-6b960b14/

Name	Fernando Camargo da Silva
Position	REDD+ Analyst
Education	Degree in Forest Engineering by Federal University Of Paraná (UFPR) and MBA in Project Management by Positivo University. Currently Master's degree student in Nature Conservation (UFPR).
Resume summary	Ten years' experience in environmental studies, such as flora inventories, environmental licensing, biodiversity conservation projects, land use, research and development, ecological restoration and environmental disasters, and also working in different biomes of Brazil, such as Amazon Rainforest, Atlantic Rainforest, Pantanal and Pampa. Currently working with REDD+ project development.
Contact Information	fernando.silva@carbonext.com.br
LinkedIn	https://www.linkedin.com/in/fernando-camargo-626164113/

Name	Rui Almeida
Position	Social Project REDD+ Coordinator
Education	Graduate degree in Education, currently attending a post-graduate course in project management, and a master's course in Conflict Management.

Resume summary	Rui has extensive experience in Amazon-based projects with traditional communities, including quilombolas and indigenous peoples. Experience with the implementation of a financial fund in quilombola communities. Sustainable development, social dialogue, and land and conflict mediation. Training in Nonviolent Communication and advanced knowledge of social dialogue tools, digital technologies, communication methodologies and development of support material for communication projects.
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Name	Thiago Manoel Sozinho dos Santos
Position	REDD+ Social Analyst
Education	Degree in Forestry Engineering from Amazonia Federal Rural University and a Master's in forestry economy in Paraná Federal University
Resume summary	Thiago Sozinho has ten years' experience working on the follow themes: protected areas, environment, carbon market, community relations, research, geotechnologies, customer success, project management, BI and Data Science.
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Name	Mateus Trez
Position	GIS and Remote Sensing Coordinator
Education	Environmental Engineer from the Piracicaba Engineering School
Resume summary	Mateus Trez is responsible for coordinating team of analysts in the development of REDD+ projects and the prospection and feasibility analysis of new carbon projects in the Amazon. Experience in the development of REDD+ projects with participation in the elaboration of the VCS Santa Maria REDD+ Forest Project, UNITOR REDD+ project and Evergreen REDD+ project.
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LinkedIn	https://www.linkedin.com/in/mateustrez/

Name	Janaína Guidolini
Position	GIS Analyst
Education	Postdoctoral researcher in Environmental Science (Aeronautics Institute of Technology); PhD in Earth System Science (National Institute for Space Research); MSc in Agronomy-Soil Science (São Paulo State University); Graduated in Agronomy and Environmental Management (Federal Institute of Triângulo Mineiro).
Resume summary	Janaína Guidolini is an environmental scientist with over ten years of experience in the environmental area, especially in soil and water management and conservation, geoprocessing for environmental analysis and sustainability.
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LinkedIn	https://www.linkedin.com/in/janainaguidolini

Name	Rodrigo Ramirez Frederico
Position	GIS Analyst
Education	Bachelor in Biology from University of Sao Paulo (USP) and Master Student in environmental science at the same institution
Resume summary	Experience in ecosystem services modelling associated with GIS analysis, watershed management and payments for environmental services (PES). Also, I worked more than 4 years in environmental licensing, associated with regulatory bodies and environmental legislation. In addition, I also have experience in biodiversity conservation and environmental education projects.
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LinkedIn	https://www.linkedin.com/in/rodrigo-ramirez-frederico-572254137

Name	Valdenir Lemos Ribeiro
Position	Field technician
Education	Completed high school (2009)
Resume summary	Extensive experience as field service provider in the Amazon Forest environments and their communities; security guard experience for four years; Municipal secretary of environment in Novo Aripuanã, Amazonas (2010 – 2012); Municipal secretary of housing and land regularization in Novo Aripuanã, Amazonas (2017 – 2018).
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2.4.3 Project Management Partnerships/Team Development (G4.2)

Installation of solar panels – For the installation of solar panels and power generation, a specialized company, which operates throughout the northern region of the country, was contracted, as fully proven at audit.[58] Primarily, the electricity supplied by the solar panels is used to run the satellite internet, but it also manages to serve a small range of school equipment.

Satellite internet installation – high speed and stability satellite internet, designed to bring connectivity to remote places, meeting the needs of the Ipoá project communities. The purpose of this activity is to connect the community to the world wide web, giving them access to information, public services, friends and family members in an unprecedented and transformative way.

A specific company that supplies feminine hygiene products to promote wellness and health activities[59].

2.4.4 Financial Health of Implementing Organization(s) (G4.3)

Carbonext’s financial department is responsible for ensuring the company's financial health, so that it can run its operation in the most effective fashion possible, in addition to being able to execute projects without running financial risks.

[58] See evidence folder “05. Communication and Electrification”
[59] See evidence folder “09. Other activities”

To this end, the company has an annual budgeting process that is periodically reviewed to keep managers updated as to the main expenses for the period, including those related to the execution of projects. In this way, a monetary sum is set aside for each project to be executed, to avoid any cash flow risk.

Thus, the project proponents have the necessary funds to maintain the project activities until the start of the GHG revenues, as proven at audit and duly presented in the Non-Permanence Risk analyses.

2.4.5 Avoidance of Corruption and Other Unethical Behavior (G4.3)

In Brazil, the two main legal instruments that aim to prevent acts of corruption and unethical attitudes are (i) Law 12,846/2013 (Anti-Corruption Law); and (i) Law 8,429/1992 (Administrative Misconduct Law).

Law no. 12,846/2013, known as the Anti-Corruption Law, establishes rules regarding the administrative liability and civil liability of legal entities in the case of practice of acts against the public administration, national and/or foreign. Under Law 8,429/1992, administrative improbity is an immorality qualified by the dishonesty and acts of bad faith by the public agent.

Both practices, in addition to the legal risks, could represent enormous reputational risks to the project and to all that are involved. Due to this risks, Carbonext's Legal Due Diligence aims to verify and, if possible, mitigate any involvement of the landowner (partner) with such practices.

The clearance certificates examined by Carbonext's legal team (applicable for all the project proponents/partners and real estate assets) during the Legal Due Diligence process cover a large spectrum of jurisdictional and administrative matters, such as, for example, consultation before State Courts of Justice and Superior Courts, State and Federal Public Prosecutors, Environmental Agencies, Protest Registry Offices, Real Estate Registry Offices, etc., significantly reducing the chances of both debts and/or acts of corruption (regarding the landowner partner or the real estate assets) go unnoticed. Eventual discovery of relevant debts and/or acts of corruption involving the landowner (partner), is one of the many reasons for the termination of the intended partnership, as provided for in the contract.

Carbonext has an Anti-corruption Policy and a code of ethics and organizational conduct in place to ensure non-involvement with corruption[60]. So that employees, customers, partners, and other stakeholders can, with complete security and anonymity, report situations that are not in accordance with our values or with our Code of Ethics and Organizational Conduct and / or Compliance Policy, we created a channel for

[60] Anti-corruption policy and code of ethics and organizational conduct in force: https://carbonext.com.br/governanca-corporativa/

complaints to record, investigate and promote decisions involving the complaints received, to ensure compliance and continuous improvement of processes and internal controls. To register complaints, an e-mail can be sent to compliance@carbonext.com.br.

To the extent that Carbonext's anti-corrutption policy also includes partners, it along with the project contracts, are the legal instruments that ensure avoidance of corruption of all the project proponents, listed in section 2.1.3.

2.4.6 Commercially Sensitive Information (*Rules* 3.5.13 – 3.5.14)

Some information required by the VCS and CCB standards is considered confidential or commercially sensitive and cannot be made public by the project proponent. This information will be supplied to the audit team, but not available for the public. The documents are:

- Agreements and contracts between the parties involved.
- Financial evidence of the proponents' funds for the project.
- Property Rights Documents.

2.5 Legal Status and Property Rights

2.5.1 Recognition of Property Rights (G5.1)

All the lands involved in the project are private property owned by the project proponents, and their property rights are recognized, respected, and supported by all relevant legal institutions. All the properties have the relevant property documents attesting to their full land-rights, which were made available at audit.

The national public registry of rural properties is carried out with the Rural Environmental Registry ("Cadastro Ambiental Rural "- CAR). An experienced professional was hired by the proponent to carry out the CAR registration for both properties. The CAR of the initial Campos farm was cancelled by the previous owner, the numbering was updated and regularized before the public organizations on behalf of the representative (Silvio Balen). In the same way, the Maanaim property's CAR for was evaluated at the beginning of the documentation verification process. Despite the identification of unconsolidated overlapping areas – a common reality in the Amazon region – the CAR was successfully updated and ratified. It is important to note that the proprietor's ownership is fully backed up by the relevant documentation in the SIGEF system, which differentiates it from unconsolidated claims such as those identified in the Maanaim area.

The measures designed and implemented by the project to help to secure full community property rights are as follows: Carbonext applied the community questionnaire relevant to land tenure situation during the initial community diagnosis which covered the MR1 period. The same process will be repeated at each successive monitoring period, ensuring Carbonext's commitment to guaranteeing the rights of all those with legitimate and legally approved land claims, and the corresponding project benefits. This is further illustrated by the specific case discussed in section 2.5.5.

2.5.2 Free, Prior and Informed Consent (G5.2)

The areas involved in the Ipoá Project are privately owned, and the project is the result of the owner's express interest in carrying out REDD+ activities.

The communities "Vila Batista", "Boa Esperança" and "Vale da Benção" which are in the Project Zone, were consulted at the beginning of the project, which is repeated at each MR to meet the CCB requirements for Free Prior and Informed consent (FPIC). Questionnaires[61] were applied in the communities three times, in March, April, and October.

In March all the communities were represented at the meetings, and 42% of the total population attended the consultations regarding the beginning of the project. Such consultations will be repeated at each monitoring period as of MR2.

The questionnaires[62] applied by Carbonext for the development of the current CCB project - as well as the consent statements of the people involved in the consultation process - were made available at audit.

Table 10. Community attendance at initial consultations

Community	Nº attendants	Nº respondents
Vila Batista	35	13
Nova Esperança	40	5
Vale da Benção	30	26
Total	**105**	**44**

The consent forms signed by communities in March 2022 declare and consent to the following: the use of their image; that the requirements regarding free, prior, and informed consent were respected; and that a communication channel for complaints and incidents involving the project had been presented.

[61] See evidence folder 09. Other Activities
[62] See evidence folder "09. Other Activities"

Property rights have not been affected by the Ipoá REDD+ Project up to the present time, and there is no removal or displacement of housing or way of working of the community because of the project activities. To this extent, no restitution or compensation is required for the community members.

2.5.3 Property Right Protection (G5.3)

As described in section 2.5.1 of this document, all lands in the project are private property owned by the project proponents, and their property rights are recognized, respected, and supported by all relevant legal institutions. All the properties have the relevant property documents, which were made available at audit, attesting to their full land rights. The upholding of the proprietors' property rights is intrinsic to the REDD+ project, which monitors the integrity of its forest cover remotely, and has numerous Operational Procedures to deal with the various types of threat to their land right.

On the community side, during the March 2022 field visit, agreements with the continuation of the REDD+ (as presented at audit) were freely signed by the leaders the three communities who live in and around the property, and during this exercise no land-tenure issues were reported by communities. This process was the result of previous workshops and the commitment of both the community and proponent parties to identify and define the activities that will be developed in the area, and the community representatives' agreement with these objectives as well as the overarching one of forest conservation.

This consultation process ensures that the communities have not been forced to relocate their homes or their activities, which are central to their culture and livelihoods. The monitoring of community property rights, as well as those of the project owners, will be conducted on an ongoing basis throughout the project lifetime.

2.5.4 Identification of Illegal Activity (G5.4)

In the without-project scenario, the area is confronted with the threat of illegal logging followed by installation of cattle ranching, the predominant economic activity in the region. To address this threat, the project has numerous processes and protocols in place during the present MR, described below.

Carbonext's continuous remote monitoring system, CarbonEye, runs with an expected frequency of every 7 days. This allows the project proponents to identify any illegal transgressions and take the necessary measures. As such, CarbonEye is a key tool to combat the illegal dynamic of deforestation.

Following this, the implementation of activities on the ground is governed by a number of protocols which are fully described in MR section 4.3.1. Those that deal directly with illegal activity, are numbers 6 and 8:

1. OP for community conflict resolution
2. OP for stakeholder communication

3. OP for irregular or illegal community activity
4. OP for community health and safety at work
5. OP for definition of alternative land-uses
6. FIP – Forest Integrity Plan
7. Responsibility Agreement and Workshop
8. OP for deforestation events

Regarding the deforestation event detected in May 2022, the protocols' relevance is as follows:

6. The FIP is the responsibility of the project field team under the proprietor's control, and has as its strategy the implementation of preventive, combat and mitigation measures for events that may cause GHG emissions. Such measures include those implemented in the field including sign installation and field measurements conducted in September 2022 following the deforestation event. The FIP provides a clear and objective definition of the stages of the monitoring process, generation of records, quantifications and reports of corrective actions employed with the aim of mitigating emissions, as well as compliance with the VCS and CCB Standards. In future MR periods, the project will train at between 2 to 4 community monitors in up to five and to patrol the area according to the Forest Integrity Plan (FIP).

8. The Protocol for Deforestation Events provides an overview of the process and responsibilities of various parties, including the proprietor's legal team and the field team, and relevant formulae to report results.

Further information on avoidance of illegal activity is also found in PD Section *2.4.1. Project Governance Structures;* and MR sections*: 2.4.5. Avoidance of Corruption and Other Unethical Behavior*; 3.1.3 *Monitoring Plan;* and 4.3.1. *Community Monitoring Plan*.

Operational Procedures are also in place to address broader illegal activities that may occur such as: theft, vandalism, embezzlement of resources and/or equipment, etc. Due to being internal procedures and materials of Carbonext's intellectual property, they were made available at validation to the audit team.

2.5.5 Ongoing Disputes (G5.5)

There are no legally formalized ongoing disputes in the Project Area, regarding land rights, resource access rights, or other issues.

Carbonext's internal legal procedure for the MR1 period included, among others, due diligence of records from the district where the properties are located. This specific survey covers any ongoing dispute (if any) that are discussed in federal and state courts, regardless of the commencement date of the dispute. No ongoing or unresolved disputes that compromise ownership of the property in the last twenty years were identified during this survey, conducted for MR1[63].

In relation to public entities, whether state, municipal or federal, of direct or indirect administration, the certificates analyzed indicate the legitimacy of the project's land-titles, ruling out the possibility of illegal appropriation of public lands (*grilagem*). The latter would make the development of the project impossible because public lands cannot be legally acquired via squatting, except with the effective participation of the public entity that owns the area, which would require some legislative measure (law, decree, ordinance, etc.) expressly legalizing the area.

Under Brazilian legislation, it can be stated that the certificates from environmental, administrative, jurisdictional, and certifying bodies indicate that possession. property rights, and use of the natural resources therein, duly belong to the proponent, being fully legal and extensively documented and proven. These premises constitute the proprietor's legal basis to develop the present REDD+ project.

Regarding community land disputes, as reported in section 2.5.6 of the PD, that during the pre-validation visit, a community member claimed to have documentation of part of the Maanaim property however there is no evidence to support this. This situation did not change during MR1. Carbonext commits to ongoing monitoring of this situation and to facilitate any legitimate claims to land rights if they do arise. Carbonext's general policy regarding such disputes and resulting changes to carbon credit rights is expounded in section 2.5.1.

The same approach as that given to land disputes will be applied to resource disputes: monitoring by project field personnel at least at each monitoring event.

2.5.6 National and Local Laws (G5.6)

There is no specific governing legislation in Brazil for carbon credit projects or even for the carbon credits generated. Therefore, projects must be aligned with the laws and principles in diverse areas of the legal sphere.

There is a hierarchy between norms in Brazilian legislation, starting with the Federal Constitution. All branches of law applicable to carbon projects and the applicable rules have their origin, foundation, and validity in the Federal Constitution.

[63] See evidence folder "17. Legislation analysis"

In Environmental Law, Article 225 of the Constitution is the basis for all norms, principles, objectives, and policies. The National Land Policy begins with article 184. Furthermore, rights such as possession, property, and free enterprise are also based on the Federal Constitution. Considering the sources in the brief explanation above, there are several directly and indirectly applicable rules, regarding carbon credit projects, each one regulating a specific aspect.

Law 12,651/2012 (Forest Code) - The Forest Code establishes general rules regarding the protection of vegetation, Permanent Preservation areas and Legal Reserve areas, forest exploitation, the supply of raw materials, control of the origin of forest products and the control and prevention of forest fires. In addition, it stipulates economic and financial instruments to achieve its objectives. It creates the CAR system, which was later regulated by MMA Normative Instruction No. 2 of May 5, 2014 .

Regarding the Brazilian Forest Code, the following definitions stand out as being relevant:

“Article 41 - The Federal Executive Power is authorized to institute, without prejudice to the compliance with the environmental legislation, a support and incentive program for environmental conservation, as well as for the adoption of technologies and good practices that reconcile agricultural, livestock, and forest productivity, with a reduction in environmental impacts, as a way of promoting ecologically sustainable development, always observing the criteria of progressiveness, comprising the following categories and lines of action.

The significant law updates for the monitoring period are:

- ART. 29, § 4 The owners and possessors of rural properties with an area of more than 4 (four) fiscal modules who register them in the CAR by December 31, 2023, as well as the owners and possessors of rural properties with an area of up to 4 (four) fiscal modules or that meet the provisions of art. 3 of Law No. 11.326, of July 24, 2006, who register them in the CAR by December 31, 2025. (Edited by Law No. 14.595, of 2023);

- ART. 59, § 2 The registration of rural property in the CAR is a mandatory condition for joining the PRA, which will be requested by the owner or possessor of the rural property within one (1) year of notification by the competent body, which will previously validate the registration and identify environmental liabilities, subject to the provisions of § 4 of art. 29 of this Law. (Edited by Law 14.595 of 2023);

- ART. 59, § 4 During the period between the publication of this Law and the expiry of the deadline for the interested party to join the PRA, and while the term of commitment is being complied with, the owner or possessor may not be fined for infractions committed before July 22, 2008, relating to the irregular suppression of vegetation in Permanent Preservation Areas, Legal Reserves and Restricted Use Areas. (Edited by Law No. 14.595, of 2023);

- ART. 59, § 9 The competent environmental bodies must guarantee financial institutions access to data from the CAR and the PRA that allows them to verify the environmental regularity of the owner or possessor of rural property. (Included by Law No. 14.595, of 2023);

§ 4º - The maintenance activities of the Permanent Preservation Areas, Legal Reserves and Restricted Use Areas are eligible for any payments or incentives for environmental services, constituting additionality for the purposes of national and international markets for certified reductions in greenhouse gas emissions".

Law 9,433/1997 (The National Policy on Water Resources) - The National Policy on Water Resources is based on the premise that water is a limited natural resource, with economic value and pertains to the public domain. Similarly, the management of water resources must always provide for the multiple uses of water, and the management of water resources must be decentralized and count on the participation of the Public sector, users and communities .

Law 12,305/2010 (National Solid Waste Policy) - The National Solid Waste Policy brings together the set of principles, objectives, instruments, guidelines, goals and actions adopted by the Federal Government, alone or in cooperation with states, the Federal District, municipalities or individuals, with a view to the integrated management and environmentally appropriate management of solid waste. This is a milestone in Brazilian environmental legislation, as it is the first federal standard created with a focus on the problem of solid waste. Thus, it deals with relevant issues related to social, environmental, and economic interests in practically all activities. It includes as instruments environmental, sanitary, and agricultural monitoring and inspection, technical and financial cooperation between the public and private sectors for the development of research on new products, methods, processes and technologies of management, recycling, reuse, waste treatment and final disposal environmentally sound tailings; scientific and technological research; environmental education, among others .

Law 12,187/09 (National Policy on Climate Change - PNMC) - The PNMC's objectives are to make economic and social development compatible with the protection of the climate system, the reduction of anthropogenic greenhouse gas emissions in relation to their different sources , the strengthening of human removals by sinks of greenhouse gases across the national territory, the implementation of measures to promote adaptation to climate change by the 3 (three) spheres of the Federation, with the participation and collaboration of economic and social agents stakeholders or beneficiaries, in particular those especially vulnerable to its adverse effects; the preservation, conservation and recovery of environmental resources, with particular attention to the great natural biomes considered National Heritage; the consolidation and expansion of legally protected areas and the encouragement of reforestation and recomposition of vegetation cover in degraded areas; and encouraging the development of the Brazilian Emissions Reduction Market – MBRE .

Decree no. 9,578/2018 - Provides for the National Fund on Climate Change (FNMC), dealt with in Law no. 12,114, of December 9, 2009, and the National Policy on Climate Change, dealt with in Law n. 12,187, of December 29, 2009. Among the topics covered by the decree is the application of FNMC resources to projects to reduce carbon emissions from deforestation and forest degradation, with priority for natural areas threatened with destruction and relevant to conservation strategies biodiversity (art. 7, V) .

The significant law update for the monitoring period is:

- Art. 5 The National Climate Change Fund - FNMC, of an accounting nature, established by Law No. 12.114 of 2009 and regulated by this Decree, linked to the Ministry of the Environment and Climate Change, aims to ensure resources to support projects or studies and finance undertakings aimed at mitigating climate change and adapting to climate change and its effects;

Resolution no. 001/1986 under CONAMA - Deals with Environmental Licensing. It is an administrative procedure through which the Public Administration establishes conditions and limits certain activities .

Law 9,985/2000 (SNUC Law) - This Law establishes the National System of Nature Conservation Units - SNUC, establishes criteria and norms for the creation, implementation and management of "Conservation Units" (that is, conservation areas) and presents a series of important concepts for the proper understanding of Conservation Units .

Law 6,938/1981 (National Environmental Policy) - The National Environmental Policy has the objective of preserving, improving and recovering the environmental quality conducive to life, aiming to ensure conditions for socio-economic development, the interests of national security and the protection of human dignity .

Law 10,406/2002 (Civil Code) - Deals with various rights and obligations, including possession, property and legal business .

Law 6,015/1973 (Public Records Law) - The law deals with public records and, especially, in its chapter V it refers to the registration of rural properties, through which the ownership of rural property is demonstrated .

Land Squatting Laws: Law Nº10.406, Art 1.238 of the Civil Code, Art 191 of the Federal Constitution among other laws which govern Land Squatting are fully discussed under section 2.5.1 .

Consolidation of labour laws (CLT): Regulates provisions relating to labour legislation and institutes the Permanent Program for Consolidation, Simplification and bureaucratization of Integral Labour Standards and the National Labour Award, and amends Decree No. 9,580 of November 22, 2018

Regulatory norms (NR) such as: NR-5 - Internal Commission for Accident Prevention NR-12 - Safety At Work In Machinery And Equipment; NR-24 - Sanitary And Comfort Conditions In The Workplace; NR-26- Safety Signs.

State Laws:

Law 1,532/82: Disciplines the State Policy for the Prevention and Control of Pollution, Improvement and Recovery of the Environment and Protection of Natural Resources .

Law 2,416/96: Provides for the requirements for granting a license for the exploration, processing and industrialization of forest products and by-products for timber purposes .

Law 3,135/07: Establishes Amazonas State Policy on Climate Change, Environmental Conservation and Sustainable Development .

Law 3,785/12: Provides for environmental licensing in the State of Amazonas, revokes Law n. 3,219, of December 28, 2007 .

Law 4,406/16: Establishes the State Policy for Regularization, provides for the Rural Environmental Registry - CAR, the Rural Environmental Registry System - Sicar-AM, the Environmental Regularization Program - PRA, in the State of Amazonas .

Decree no. 10,028/87: State System for Licensing Activities with Potential Impact on the Environment .

Decree no. 15,780/94: Amends Art. 57 of State Decree No. 10,028, of February 4, 1987, which regulated Law No. 1,532, of July 6, 1982, which provides for the State Licensing System for Activities with Potential of Impact on the Environment .

Decree no. 15,842/94: Amends Art. 44 of State Decree No. 10,028, of February 4, 1987, which regulated Law No. 1,532 of July 6, 1982, which provides for the State Licensing System for Activities with Potential of Impact on the Environment .

Decree no. 44.716/2021 – Assumes the compromise of the states of Amazonas with the neutrality of carbon, reducing emissions by fifty percent until 2050

As such, the Project IPOÁ REDD+ is in compliance with all identified applicable laws, and no cases of non-compliance with laws, statutes and other regulations were identified.

Decree 11,550/2023 Provides for the Inter-ministerial Committee on Climate Change.

Specific laws dealing with the carbon market in Brazil are still under discussion in the national congress.

3 CLIMATE

3.1 Monitoring GHG Emission Reductions and Removals

3.1.1 Data and Parameters Available at Validation

Data / Parameter	CF
Data unit	tCt/td.m-1
Description	Carbon fraction of dry matter in t Ct-1 d.m
Source of data	Values from the literature (e.g. IPCC 2006 INV GLs AFOLU Chapter 4 Table 4.3[64]) shall be used if available, otherwise default value of 0.47 t C t-1 d.m. can be used
Value applied	0.47
Justification of choice of data or description of measurement methods and procedures applied	The default value was used for conservativeness purposes.
Purpose of data	Indicate one of the following: • Determination of baseline scenario (AFOLU projects only) • Calculation of baseline emissions • Calculation of project emissions • Calculation of leakage Calculation of baseline emissions Calculation of project emissions Calculation of leakage

[64] https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/4_Volume4/V4_04_Ch4_Forest_Land.pdf

Comments	Provide any additional comments If new and more accurate carbon fraction data become available, these can be used to estimate the net anthropogenic GHG emission reduction of the subsequent fixed baseline period.

Data / Parameter	44/12
Data unit	Indicate the unit of measure Dimensionless
Description	Provide a brief description of the data/parameter Carbon mass to CO2e mass conversion factor
Source of data	Indicate the source(s) of data From scientific literature: 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 AFOLU[65]
Value applied	Provide the value applied 44/12
Justification of choice of data or description of measurement methods and procedures applied	Justify the choice of data source, providing references where applicable. Where values are based on measurement, include a description of the measurement methods and procedures applied (e.g., what standards or protocols have been followed), indicate the responsible person/entity that undertook the measurement, the date of the measurement and the measurement results. More detailed information may be provided in an appendix. Conversion from C to CO2e based on molecular weights
Purpose of data	Indicate one of the following: • Determination of baseline scenario (AFOLU projects only) • Calculation of baseline emissions • Calculation of project emissions • Calculation of leakage Calculation of baseline emissions

[65] https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/4_Volume4/V4_04_Ch4_Forest_Land.pdf

	Calculation of project emissions Calculation of leakage
Comments	Provide any additional comments IPCC standard value

Data / Parameter	SLFty
Data unit	Indicate the unit of measure Dimensionless
Description	Provide a brief description of the data/parameter Fraction of wood products that will be emitted to the atmosphere within 5 years of production by class of wood product ty
Source of data	Indicate the source(s) of data As per CP-W: Sawnwood, page 13[66]
Value applied	Provide the value applied 0.20
Justification of choice of data or description of measurement methods and procedures applied	Justify the choice of data source, providing references where applicable. Where values are based on measurement, include a description of the measurement methods and procedures applied (e.g., what standards or protocols have been followed), indicate the responsible person/entity that undertook the measurement, the date of the measurement and the measurement results. More detailed information may be provided in an appendix. Default conservative value prescribed by CP-W.
Purpose of data	Indicate one of the following: • Determination of baseline scenario (AFOLU projects only) • Calculation of baseline emissions • Calculation of project emissions

[66] https://verra.org/wp-content/uploads/2018/03/VMD0005-CP-W-v1.1.pdf

	• Calculation of leakage Calculation of baseline emissions
Comments	Provide any additional comments Parameter values to be updated if new empirically based peer-reviewed findings become available

Data / Parameter	OFty
Data unit	Indicate the unit of measure Dimensionless
Description	Provide a brief description of the data/parameter Fraction of wood products that will be emitted to the atmosphere between 5 and 100 years of timber harvest by class of wood product ty
Source of data	Indicate the source(s) of data The OFty is the complementary number of SLFty: the sum of both parameters must equal 1 (i.e., 100%).
Value applied	Provide the value applied 0.80
Justification of choice of data or description of measurement methods and procedures applied	Justify the choice of data source, providing references where applicable. Where values are based on measurement, include a description of the measurement methods and procedures applied (e.g., what standards or protocols have been followed), indicate the responsible person/entity that undertook the measurement, the date of the measurement and the measurement results. More detailed information may be provided in an appendix. As per CP-W.
Purpose of data	Indicate one of the following: • Determination of baseline scenario (AFOLU projects only) • Calculation of baseline emissions

	• Calculation of project emissions • Calculation of leakage Calculation of baseline emissions
Comments	Provide any additional comments Parameter values to be updated if new empirically based peer-reviewed findings become available.

Data / Parameter	PCOMi
Data unit	Indicate the unit of measure Dimensionless
Description	Provide a brief description of the data/parameter Commercial volume as a percent of total aboveground volume in stratum i.
Source of data	Indicate the source(s) of data For calculation of this parameter, the following parameters and sources were used: - Volume of merchantable wood in exploitation (35.08 m3/ha) from FSM_REDD_MR2_04_03_2021[67] - Dmm value (0.59 t d.m.m-3) from Brown et al 1989. - Above ground carbon stock from MCTI 2015[68] - CF value (0.47 t Ct-1 d.m) from IPCC 2006 V4_04_Ch4_Forest_Land[69]
Value applied	Provide the value applied

[67]https://registry.verra.org/mymodule/ProjectDoc/Project_ViewFile.asp?FileID=52137&IDKEY=9lksjoiuwqowrnoiuomnckjashoufifmln902309ksdflku098l71896923
[68] http://redd.mma.gov.br/images/FREL/RR_LULUCF_Mudana-de-Uso-e-Floresta.pdf
[69] https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/4_Volume4/V4_04_Ch4_Forest_Land.pdf

	Submontane Dense Tropical Rainforest: 0,07814 Lowland Dense Tropical Rainforest: 0,07791 Alluvial Dense Tropical Raiforest: 0,06863 Pioneer Formation with fluvial influence without palm trees: 0,1161
Justification of choice of data or description of measurement methods and procedures applied	Justify the choice of data source, providing references where applicable. Where values are based on measurement, include a description of the measurement methods and procedures applied (e.g., what standards or protocols have been followed), indicate the responsible person/entity that undertook the measurement, the date of the measurement and the measurement results. More detailed information may be provided in an appendix. Volume of merchantable wood in exploitation (m3/ha) has been based on “Florestal Santa Maria Project” (35.08 m3/ha), whose Project Area is located about 120 km South of this Project Area and has the same predominat forest class represented within it (According to Figure 22 of FSM VCS-PD[70]): Submontane dense tropical forest (“Ds”), in the IBGE classification, which are the same as those predominating in this project. Age classes are assumed to be similar, as both sites had not been deforested or managed until the date of their forest inventories (i.e., both were primary forests).
Purpose of data	Indicate one of the following: • Determination of baseline scenario (AFOLU projects only) • Calculation of baseline emissions • Calculation of project emissions • Calculation of leakage Calculation of baseline emissions
Comments	Provide any additional comments Updated at the time of baseline revision (at least every 10 years). Application of the commercial percentage of total volume introduces the simplifying assumption (and conservative, as it is

[70]https://registry.verra.org/mymodule/ProjectDoc/Project_ViewFile.asp?FileID=52137&IDKEY=9lksjoiuwqowrnoiuomnckjashoufifmln902309ksdflku098l71896923 (last visited in 10/09/2021)

	only used in the ex-ante baseline calculations) that all commercial stocks are extracted (i.e., perfect efficiency).

Data / Parameter	Cab_tree
Data unit	tCO2-e ha-1
Description	Mean aboveground biomass carbon stock in stratum i
Source of data	National GHG Inventory (Ministério da Ciência, Tecnologia e Inovação, 2015)[71], Table 14. Values multiplied by 44/12 conversion parameter. Regarding data per stratum, the carbon stock numbers were as follows: - Dense Tropical Submontane Rainforest (code "Ds"): 140.69 tC/ha. - Dense Tropical Lowland Rainforest (code "Db"): 141.10 tC/ha. - Dense Tropical Alluvial Rainforest (code "Da"): 160.18 tC/ha. - Pioneer Formation with fluvial influence without palm trees (code "Pa"): 91.70 tC/ha.
Value applied	- Submontane Dense Tropical Rainforest: 515.86. - Lowland Dense Tropical Rainforest: 517.37. - Alluvial Dense Tropical Rainforest: 587.33. - Pioneer Formation with fluvial influence without palm trees: 336.23.

[71] http://redd.mma.gov.br/images/FREL/RR_LULUCF_Mudana-de-Uso-e-Floresta.pdf

Justification of choice of data or description of measurement methods and procedures applied	Official data from Brazilian government.
Purpose of data	• Determination of baseline scenario (AFOLU projects only) • Calculation of baseline emissions • Calculation of project emissions • Calculation of leakage Calculation of baseline emissions
Comments	N/A

Data / Parameter	Cbb_tree
Data unit	tCO2-e ha-1
Description	Mean belowground biomass carbon stock in stratum i
Source of data	Indicate the source(s) of data National GHG Inventory (Ministério da Ciência, Tecnologia e Inovação, 2015)[72], Table 14. Values multiplied by 44/12 conversion parameter. Regarding data per stratum, the carbon stock numbers were as follows: - Dense Tropical Submontane Rainforest (code "Ds"): 39.7 tC/ha. - Dense Tropical Lowland Rainforest (code "Db"): 39.83 tC/ha. - Dense Tropical Alluvial Rainforest (code "Da"): 45.22 tC/ha. - Pioneer Formation with fluvial influence without palm trees (code "Pa"): 17.01 tC/ha.

[72] http://redd.mma.gov.br/images/FREL/RR_LULUCF_Mudana-de-Uso-e-Floresta.pdf

Value applied	- Submontane Dense Tropical Rainforest: 145.60. - Lowland Dense Tropical Rainforest: 146.04. - Alluvial Dense Tropical Raiforest: 165.81. - Pioneer Formation with fluvial influence without palm trees: 62.37.
Justification of choice of data or description of measurement methods and procedures applied	Official data from Brazilian government.
Purpose of data	Calculation of baseline emissions
Comments	N/A

Data / Parameter	Pasture carbon pool
Data unit	tCO2e
Description	Pasture carbon pool in the baseline scenario
Source of data	National GHG Inventory (Ministério da Ciência, Tecnologia e Inovação, 2015)[73], Table 5. Value (7.57 tC/ha) multiplied by 44/12 conversion parameter.
Value applied	27.76
Justification of choice of data or description of measurement	Official data from Brazilian government.

73 http://redd.mma.gov.br/images/FREL/RR_LULUCF_Mudana-de-Uso-e-Floresta.pdf)

methods and procedures applied	
Purpose of data	Calculation of baseline emissions
Comments	Calculation based on country-specific values.

Data / Parameter	BCEF
Data unit	Dimensionless
Description	Biomass conversion and expansion factor for conversion of commercial wood volume per unit area to total aboveground tree biomass per unit área.
Source of data	According to CP-AB[74]- page 14, being the average of the three proposed factors.
Value applied	1.32
Justification of choice of data or description of measurement methods and procedures applied	BCEF was applied for conversion of merchantable volume to total aboveground tree biomass
Purpose of data	Calculation of baseline emissions Calculation of project emissions Calculation of leakage
Comments	N/A.

[74] https://verra.org/wp-content/uploads/2017/11/VMD0001v1.1.pdf

Data / Parameter	COMF
Data unit	Dimensionless
Description	Combustion factor for stratum i (vegetation type)
Source of data	E-BPB refers to Table 2.6 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 4 Agriculture, Forestry and Other Land Use, Chapter 2, “Primary tropical moist forest”[75]
Value applied	0.50
Justification of choice of data or description of measurement methods and procedures applied	Value was applied according to E-BPB: Table 2.6 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 4 Agriculture, Forestry and Other Land Use, Chapter 2, “Primary tropical moist forest”
Purpose of data	Calculation of baseline emissions Calculation of project emissions
Comments	N/A.

Data / Parameter	Ggi
Data unit	g kg-1 dry matter burnt
Description	Emission factor for stratum i for gas g

[75] https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/4_Volume4/V4_02_Ch2_Generic.pdf

Source of data	Defaults can be found in Volume 4, Chapter 2, of the IPCC 2006 Inventory Guidelines in table 2.5 (see Annex 2: emission factors for various types of burning for CH4 and N2O).[76]
Value applied	GCH4 = 6.8 GN20 = 0.2
Justification of choice of data or description of measurement methods and procedures applied	Default values form IPCC 2006.
Purpose of data	Calculation of baseline emissions Calculation of project emissions Calculation of leakage emissions
Comments	N/A.

Data / Parameter	Deforestation
Data unit	Ha
Description	Maps of forest cover areas converted into non-forest areas
Source of data	Measured through data from Mapbiomas project
Value applied	Yearly variable: deforestation values are presented for the Reference Region, Leakage Belt and Project Area (projections) in this VCS/CCB-PD

[76] https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/4_Volume4/V4_02_Ch2_Generic.pdf

Justification of choice of data or description of measurement methods and procedures applied	The Mapbiomas project contributes to understand the land use dynamics in Brazil. The data generated by this program is used in this project. Mapbiomas data are applicable for use in this project, according to the criteria listed below (Methodology VM0007): i) Mapbiomas data covers the entire project area, leakage belt and reference region. ii) Mapbiomas data cover the entire reference period (beginning, middle and end) of the fixed baseline period. iii) Mapbiomas monitors conversion of forest land to non-forest land. iv) Monitoring occurred during the entire fixed baseline period. In case of unavailability of Mapbiomas data for the monitoring period, other sources will be consulted such as PRODES, or a classification of imagery (Landsat8) will be carried out to measure deforested area: Land use and land cover mapping is assessed using images with spatial resolution superior to 30 meters. Image acquisition is performed during the period of low incidence of clouds and rainfall in the region, as closer as possible to the end of the monitoring period. Images undergo geometric correction by means of geo-referencing, using topographic maps as reference or USG-NASA orthorectified images. For analysis of areas with cloud cover, visual interpretation of radar image would be performed. Evaluation of classification accuracy is performed by analyzing the overall accuracy and kappa index obtained from a confusion matrix. The minimum accuracy of the classification mapping should be over than 90%, considered very high.
Purpose of data	Calculation of baseline emissions
Comments	N/A.

Data / Parameter	Dmm
Data unit	t d.m.m-3
Description	Mean wood density of commercially harvested species
Source of data	Brown, S., A. J. R. Gillespie, and A. E. Lugo, 1989. Biomass estimation methods for tropical forests with applications to forest inventory data. Forest Science, 35:881- 902. See pg. 890, Table 4, Moist
Value applied	0.59
Justification of choice of data or description of measurement methods and procedures applied	Country-specific data obtained from the same biome.
Purpose of data	Calculation of baseline emissions Calculation of project emissions Calculation of leakage emissions
Comments	N/A.

Data / Parameter	LDF
Data unit	t C m-3
Description	Factor for calculating the biomass of dead wood created during logging operations per cubic meter extracted (i.e. Logging Damage Factor)

Source of data	VMD0011 LK-ME[77]
Value applied	0.53
Justification of choice of data or description of measurement methods and procedures applied	It is the default value for broadleaf and mixed forests, according to module VMD0011 LK-ME. Default value for broadleaf and mixed forests of 0.53 t C m-3 from 774 logging gaps measured by Winrock International in Bolivia, Belize, the Republic of Congo, Brazil, and Indonesia may be used for tropical broadleaf forests
Purpose of data	Calculation of leakage emissions Calculation of project emissions
Comments	N/A.

Data / Parameter	LIF
Data unit	t C m-3
Description	Factor for calculating the emissions arising from the creation of logging infrastructure (roads, skid trails and decks) during logging operations per cubic meter extracted
Source of data	LK-ME, pg 17[78]
Value applied	0.29

[77] https://verra.org/wp-content/uploads/2018/03/VMD0011-LK-ME-v1.1.pdf
[78] https://verra.org/wp-content/uploads/2018/03/VMD0011-LK-ME-v1.1.pdf

Justification of choice of data or description of measurement methods and procedures applied	Conservative default value of 0.29 t CO2-e m-3 calculated from 1,839 hectares of logging concessions analysed by Winrock International in the Republic of Congo and Brazil, may be used for tropical broadleaf forests.
Purpose of data	Calculation of leakage emissions Calculation of project emissions
Comments	N/A.

Data / Parameter	LFme
Data unit	Dimensionless
Description	Leakage factor for market effects calculations
Source of data	VMD0011 (LK-ME)
Value applied	0.2
Justification of choice of data or description of measurement methods and procedures applied	The mean merchantable biomass as a proportion of total aboveground tree biomass for each forest type (PML) is more than 15% greater than the merchantable biomass as a proportion of total aboveground tree biomass within the project boundaries (PMP).
Purpose of data	Calculation of leakage
Comments	N/A.

Data / Parameter	C_{LB}
Data unit	tCO2e ha-1
Description	Area-weighted average aboveground tree carbon stock for forests available for unplanned deforestation inside the leakage belt
Source of data	Project Area data
Value applied	470.19
Justification of choice of data or description of measurement methods and procedures applied	Based on similarity analysis, data from the Project Area was applied to Leakage Belt area. It was taken a weighted average of aboveground biomass inside the Project Area.
Purpose of data	Calculation of leakage emissions
Comments	N/A.

Data / Parameter	C_{OLB}
Data unit	tCO2e ha-1
Description	Area-weighted average aboveground tree carbon stock for forests available for unplanned deforestation outside the leakage belt
Source of data	Saatchi, R.A. Houghton, R.C. dos Santos Alvalá, J.V. Soares, and Yifan Yu. Distribution of Aboveground Live Biomass in the Amazon Basin. 2007. As per LK-ASU, the number is derived from peer-reviewed literature that is appropriate, as it considers the same biome

Value applied	578.1
Justification of choice of data or description of measurement methods and procedures applied	The value of 578,1 tCO2/ha is the total mean biomass for floodplain forest in the Amazon basin of 157.66 tC/ha (according to Table 7 the study mentioned before), multiplied by 44/12, the conversion factor from tC to tCO2.
Purpose of data	Calculation of leakage emissions
Comments	N/A.

Data / Parameter	DLF
Data unit	%
Description	Displacement Leakage Factor
Source of data	Indicate the source(s) of data Local assessment
Value applied	Provide the value applied 10
Justification of choice of data or description of measurement methods and procedures applied	If deforestation agents do not participate in leakage prevention activities and project activities, the Displacement Factor shall be 100%. Where leakage prevention activities are implemented, the factor shall be equal to the proportion of the baseline agents estimated to be given the opportunity to participate in leakage prevention activities and project activities. The project design team estimates that all communities in Project Zone will participate on deforestation prevention activities, as Sustaintability and Conservation Courses, that will disseminate sustentable practices in the use of forest and agriculture resources, preventing leakage

	emissions. Thus, the "Displacement Leakage Factor" (DLF) was conservatively defined as 10%.
Purpose of data	Calculation of leakage
Comments	This value is an ex-ante estimate. Accurate and actual values will be monitored and reported in verification periods

Data / Parameter	**GWP**
Data unit	Dimensionless
Description	GWP of CH_4 and N_2O
Source of data	Sixth Report IPCC (AR6) IPCC, 2022: Climate Change 2022: Impacts, Adaptation, and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [H.-O. Pörtner, D.C. Roberts, M. Tignor, E.S. Poloczanska, K. Mintenbeck, A. Alegría, M. Craig, S. Langsdorf, S. Löschke, V. Möller, A. Okem, B. Rama (eds.)]. Cambridge University Press. Cambridge University Press, Cambridge, UK and New York, NY, USA, 3056 pp., doi:10.1017/9781009325844.
Value applied	27 for CH_4 273 for N_2O
Justification of choice of data or description of measurement methods and procedures applied	The IPCC default value was used for conservativeness purposes
Purpose of Data	Calculation of baseline emissions Calculation of project emissions Calculation of leakage emissions
Comments	N/A

3.1.2 Data and Parameters Monitored

Data / Parameter	Aburn,i,t
Data unit	ha
Description	Area burnt in stratum i at time t
Source of data	Remote sensing data
Description of measurement methods and procedures to be applied	The measurement methods are via remote sensing as described in section 3.1.3. Aburn is considered to be the entire area deforested in a given stratum, on the basis that burning is the common practice in the region, and that all deforested area undergoes burning at some point after deforestation[79]. Further context on this assumption is provided in PD section 3.3.2.
Frequency of monitoring/recording	Areas burnt will be monitored yearly, examination will occur prior to any verification event.
Value monitored	PA: 9.4 LB: 17.0
Monitoring equipment	Remote sensing and GIS tools, applying methodology as described in section 3.1.2.
QA/QC procedures to be applied	Best practices in remote sensing Land use / land change map for monitoring period; Overlay land use / land change map with fire alert's location data from INPE-BDQUEIMADAS

[79] https://cnr.ncsu.edu/news/2019/09/amazon-rainforest-fires-everything-you-need-to-know/ (last visited in 20/09/2022)

	(http://www.inpe.br/queimadas/abasFogo.php) into the period; Quantify pixels of deforested areas over fire alerts. Monitor burned forest areas
Purpose of data	All of the following: • Calculation of baseline emissions. • Calculation of project emissions. • Calculation of leakage.
Calculation method	As burning of biomass is common practice in the region, it is considered that Aburn is equivalent to 100% of deforested areas in a given stratum.
Comments	N/A

Data / Parameter	ADistPA
Data unit	Ha
Description	Area impacted by natural disturbance in the project stratum i converted to natural disturbance stratum q in year t; ha
Source of data	Remote Sensing imagery combined with ground verification or GPS coordinates
Description of measurement methods and procedures to be applied	Specify the measurement methods and procedures, any standards or protocols to be followed, and the person/entity responsible for the measurement. Include any relevant information regarding the accuracy of the measurements (e.g., accuracy associated with meter equipment or laboratory tests). Minimum monitoring unit shall be equal to a minimum of 11 Landsat pixels or one hectare.

Frequency of monitoring/recording	Must be monitored at least every 5 years or if verification occurs on a frequency of less than every 5 years examination must occur prior to any verification event
Value monitored	0
Monitoring equipment	Remote sensing and GIS tools, applying methodology as described in section 3.1.2.
QA/QC procedures to be applied	The Mapbiomas project contributes to understand the land use dynamics in Brazil. The data generated by this program is used in this project. Mapbiomas data are applicable for use in this project, according to the criteria listed below (Methodology VM0007): i) Mapbiomas data covers the entire project area, leakage belt and reference region. ii) Mapbiomas data cover the entire reference period (beginning, middle and end) of the fixed baseline period. iii) Mapbiomas monitors conversion of forest land to non-forest land. iv) Monitoring occurred during the entire fixed baseline period. In case of unavailability of Mapbiomas data for monitoring period, other souces will be consulted such as PRODES or a classification of imagery (Landsat8) will be carried out to measure deforested area: Images undergo geometric correction by means of geo-referencing, using topographic maps as reference or USG-NASA orthorectified images. For analysis of areas with cloud cover, visual interpretation of radar image would be performed. Evaluation of classification accuracy is performed by analyzing the overall accuracy and kappa index obtained from a confusion matrix. The minimum accuracy of the classification mapping is 90%, considered very high. Land-use and land cover mapping is assessed using images

	with spatial resolution superior to 30 meters. Image acquisition is performed during the period of low incidence of clouds and rainfall in the region, as closer as possible to the end of the monitoring period.
Purpose of data	Calculation of project emissions.
Calculation method	Ex-ante, estimations of emissions from natural disturbances shall be based on historic incidence of such event in the Project region.
Comments	N/A

Data / Parameter	ABSLPAt
Data unit	Ha
Description	Annual area of baseline deforestation in the Project Area at year t
Source of data	Remote sensing data, and GIS tools
Description of measurement methods and procedures to be applied	Remote sensing data, GIS, and calculation methods presented in PD section 3.2.1.
Frequency of monitoring/recording	Annual
Value monitored	**AUD – Annual baseline unplanned deforestation for the Project Area, in the sum of strata** 2022 — 0.26 2023 — 0.00***** **APD – Annual baseline planned deforestation for the Project Area, in the sum of strata**

	2022 — 501.70 2023 — 0.00* * baseline values adjusted duly adjusted to the MR1 period
Monitoring equipment	Remote sensing and GIS tools
QA/QC procedures to be applied	The QA/ QC procedures relating to remote sensing and GIS are the same as other parameters that utilize these procedures, see the ADistPA parameter above.
Purpose of data	Calculation of baseline emissions.
Calculation method	Remote sensing data, GIS, and calculation methods presented in PD section 3.2.1.
Comments	N/A

Data / Parameter	Leakage outside
Data unit	tCO2/year
Description	Net CO2 emissions due to unplanned deforestation dis outside the leakage belt up to year t*.
Source of data	Remote sensing data and GIS. Secondary data.
Description of measurement methods and procedures to be applied	Remote sensing data, GIS, and calculation methods presented in PD section 3.2.3.
Frequency of monitoring/recording	Annual

Value monitored	2022 — 0.00 2023 — 0.00
Monitoring equipment	Remote sensing and GIS tools, applying methodology as described in section 3.1.2.
QA/QC procedures to be applied	N/A
Purpose of data	Calculation of leakage
Calculation method	As per LK-ASU, Step 4.
Comments	N/A

Data / Parameter	MANFOR
Data unit	ha
Description	Total area of forests under active management nationally
Source of data	Official country-specific data from IBAMA.[80]
Description of measurement methods and procedures to be applied	As per LK-ASU, a demonstration is required that areas will be protected against deforestation. Such a demonstration must include the existence of forest guards in sufficient numbers to prevent illegal colonization and an active management plan detailing harvest plans and return intervals, and/or evidence that the concession owner has previously evicted illegal colonists/squatters from the forest areas.

80 http://www.ibama.gov.br/ultimas/2142-manejo-sustentavel-autorizado-pelo-ibama-em-2019-totalizou-39-mil-hectares#:~:text=Manejo%20sustent%C3%A1vel%20autorizado%20pelo%20Ibama%20em%202019%20totalizou%2039%20mil%20hectares,-Publicado%3A%20Sexta%2C%2021&text=Bras%C3%ADlia%20(21%2F02%2F2020,metros%20c%C3%BAbicos%20de%20madeira%20nativa

	Ex-post estimates of MANFOR must be recalculated at least every 5 years.
Frequency of monitoring/recording	Every five years
Value monitored	1,400,000
Monitoring equipment	N/A
QA/QC procedures to be applied	As per Section 9.3 of REDD+ MF or other VCS methodology that uses this module.
Purpose of data	Calculation of leakage.
Calculation method	N/A
Comments	N/A

Data / Parameter	PROTFOR
Data unit	ha
Description	Total area of fully protected forests nationally
Source of data	Official country-specific data from ISA[81]
Description of measurement methods and procedures to be applied	As per the LK-ASU module, a demonstration is required that areas will be protected against deforestation. Such a demonstration is made by government mechanisms and national polices. Ex-ante it can be assumed that PROTFOR must remain constant.

[81] https://uc.socioambiental.org/pt-br/paineldedados

Frequency of monitoring/recording	Every five years
Value monitored	63,755,583.00
Monitoring equipment	N/A
QA/QC procedures to be applied	As per Section 9.3 of REDD+ MF or other VCS methodology that uses this module.
Purpose of data	Calculation of leakage
Calculation method	N/A
Comments	N/A

Data / Parameter	TOTFOR
Data unit	ha
Description	Total available national forest area
Source of data	The total available national forest is estimated by National Forest Information System SNIF[82]
Description of measurement methods and procedures to be applied	As per the LK-ASU module, forest areas suitable for conversion to livestock. Ex ante it can be conservatively assumed that TOTFOR must remain constant for the baseline period.
Frequency of monitoring/recording	Every five years

[82] https://snif.florestal.gov.br/pt-br/os-biomas-e-suas-florestas

Value monitored	497,962,509.00
Monitoring equipment	N/A
QA/QC procedures to be applied	As per Section 9.3 of REDD+ MF or other VCS methodology that uses this module.
Purpose of data	Calculation of leakage.
Calculation method	N/A
Comments	N/A

Data / Parameter	ΔCP,LB
Data unit	tCO2e
Description	Net greenhouse gas emissions within the leakage belt in the project case
Source of data	Measurements of deforestation in the LB as described in section 3.2.3 and defined by Module M-REDD
Description of measurement methods and procedures to be applied	As per Module M-REDD equations 1 and 6, and descriptions in 3.2.3
Frequency of monitoring/recording	At each monitoring event.
Value monitored	2022: 8,536.07 2023: 2,237.00
Monitoring equipment	*N/A*

QA/QC procedures to be applied	*As per Section 9.3 of REDD+ MF or other VCS methodology that uses this module.*
Purpose of data	*Calculation of leakage*
Calculation method	*N/A*
Comments	*N/A*

Data / Parameter	*PROPIMM*
Data unit	Proportion
Description	Estimated proportion of baseline deforestation caused by immigrating population
Source of data	Official country-specific (government) data. The proportion of baseline deforestation caused by immigrating population (PROPIMM) was estimated for a period from 2014 to 2022. For calculating PROPIMM, local data[83] were used for births, deaths, and population. It is then assumed that the total annual population growth in a given municipality is attributed to: i) births and ii) immigration. Thus, by subtracting the number of annual births from the total annual population growth, it is possible to infer the number of immigrants.
Description of measurement methods and procedures to be applied	Estimated as proportion of the area deforested in the past 5 years by population that migrated into the leakage belt and project area in the past 5 years (all areas within 2 km of the boundaries of the project area and the leakage belt must be considered here).

[83] http://tabnet.datasus.gov.br/cgi/deftohtm.exe?sinasc/cnv/nvam.def
https://www.ibge.gov.br/estatisticas/sociais/populacao/9103-estimativas-de-populacao.html

Frequency of monitoring/recording	At each monitoring event
Value monitored	0.002023806
Monitoring equipment	*N/A*
QA/QC procedures to be applied	As per Section 9.3 of REDD+ MF or other VCS methodology that uses this module.
Purpose of data	Calculation of leakage emissions
Calculation method	N/A
Comments	N/A

Data / Parameter	*ADefLB,i,t*
Data unit	ha
Description	Area of recorded deforestation in the leakage belt in the project case in stratum i in year t
Source of data	As per Module M-REDD. Satellite imagery.
Description of measurement methods and procedures to be applied	As per Module M-REDD. Satellite imagery analysis.
Frequency of monitoring/recording	At each verification event
Value monitored	2022: 13.47 2023: 3.53

Monitoring equipment	N/A
QA/QC procedures to be applied	As per Section 9.3 of REDD+ MF or other VCS methodology that uses this module.
Purpose of data	Calculation of leakage emissions
Calculation method	N/A
Comments	N/A

Data / Parameter	*ADefPA,I,u,t*
Data unit	ha
Description	Area of recorded deforestation in the Project Area in the project case in stratum i converted to land use u in year t
Source of data	As per Module M-REDD. Satellite imagery.
Description of measurement methods and procedures to be applied	As per Module M-REDD. Satellite imagery analysis.
Frequency of monitoring/recording	At each monitoring event
Value monitored	2022: 7.45 2023: 1.95
Monitoring equipment	Remote sensing and GIS tools, applying methodology as described in section 3.1.2.

QA/QC procedures to be applied	As per Section 9.3 of REDD+ MF or other VCS methodology that uses this module.
Purpose of data	Calculation of project emissions
Calculation method	N/A
Comments	N/A

Data / Parameter	*RFt*
Data unit	*%*
Description	*Provide a brief description of the data/parameter* *Risk factor used to calculate VCS buffer credits*
Source of data	All sources as reported to the VCS Non-Permanence Risk Report (v3.1), including: Remote sensing data and GIS; Supervisor report; Literature data, etc.
Description of measurement methods and procedures to be applied	*All measurement methods are as described in the VCS Non-Permanence Risk Report.*
Frequency of monitoring/recording	At each verification event
Value monitored	*10*
Monitoring equipment	*Remote sensing, GIS and other equipment as described in the VCS-approved AFOLU Non-Permanence Risk Tool*
QA/QC procedures to be applied	*Literature data from reputed sources will be used and critically checked. When possible, the average of two or more sources will be used.*
Purpose of data	*Calculation of baseline emissions*

Calculation method	*All the risk factors described in the VCS Risk Report were assessed.*
Comments	*N/A*

Data / Parameter	Project Forest Cover Monitoring Map
Data unit	Ha
Description	Map showing the location of forest land within the project area at the beginning of each monitoring period. If within the Project Area some forest land is cleared, the benchmark map must show the deforested areas at each monitoring event
Source of data	Remote sensing in combination with GPS data collected during ground-truthing
Description of measurement methods and procedures to be applied	The minimum map accuracy must be 90% for the classification of forest/non-forest in the remote sensing imagery. If the classification accuracy is less than 90% then the map is not acceptable for further analysis. More remote sensing data and ground truthing data will be needed to produce a product that reaches the 90% minimum mapping accuracy
Frequency of monitoring/recording	Must be monitored at least every 5 years or if verification occurs on a frequency of less than every 5 years examination must occur prior to any verification event
Value monitored	17,329.73
Monitoring equipment	Remote sensing and GIS tools, applying methodology as described in section 3.1.2.
QA/QC procedures to be applied	The QA/QC procedures relating to remote sensing and GIS are the same as other parameters that utilize these procedures, see the ADistPA parameter above.

Purpose of data	Calculation of Project Emissions
Calculation method	Remote sensing and GIS
Comments	N/A

Data / Parameter	Leakage Belt Forest Cover Monitoring Map
Data unit	ha
Description	Map showing the location of forest land within the leakage belt area at the beginning of each monitoring period. Only applicable where leakage is to be monitored in a leakage belt
Source of data	Remote sensing in combination with GPS data collected during ground truthing
Description of measurement methods and procedures to be applied	The minimum map accuracy must be 90% for the classification of forest/non-forest in the remote sensing imagery. If the classification accuracy is less than 90% then the map is not acceptable for further analysis. More remote sensing data and ground truthing data will be needed to produce a product that reaches the 90% minimum mapping accuracy.
Frequency of monitoring/recording	Must be monitored at least every 5 years or if verification occurs on a frequency of less than every 5 years examination must occur prior to any verification event
Value monitored	21.895,26
Monitoring equipment	Remote sensing and GIS tools, applying methodology as described in section 3.1.2.

QA/QC procedures to be applied	The QA/QC procedures relating to remote sensing and GIS are the same as other parameters that utilize these procedures, see the ADistPA parameter above.
Purpose of data	Calculation of leakage emissions
Calculation method	Remote sensing and GIS
Comments	N/A

Data / Parameter	*ADefLB,i,u,t*
Data unit	*ha*
Description	*Area of recorded deforestation in the leakage belt in stratum i converted to land use u in year t*
Source of data	*As per Module M-REDD v2.2. Remote sensing imagery.*
Description of measurement methods and procedures to be applied	*As per Module M-REDD. Satellite imagery analysis.*
Frequency of monitoring/recording	*At each verification event.*
Value monitored	2022: 13.47 2023: 3.53
Monitoring equipment	*N/A*

QA/QC procedures to be applied	*As per Section 9.3 of REDD+ MF or other VCS methodology that uses this module.*
Purpose of data	*Calculation of leakage emissions*
Calculation method	*N/A*
Comments	*N/A*

3.1.3 Monitoring Plan

The Monitoring Plan below was developed according to Methodology VM0007 "Methodology for Avoided Unplanned Deforestation", v1.1, and continues what is described in section 3.3.3. of the PD. The methodology encompasses four main monitoring tasks:

i) Monitoring of project implementation
ii) Monitoring of actual carbon stock changes and greenhouse gas emissions
iii) Monitoring of leakage carbon stock changes and greenhouse gas emissions
iv) Estimation of ex post net carbon stock changes and greenhouse gas emissions

This Monitoring Plan describes how these tasks were implemented. The following procedures were applicable to each task,:

a) Technical description of monitoring tasks;
b) Data to be collected;
c) Overview of data collection procedures;
d) Quality control and assurance procedures;
e) Results
f) Inspection and accountability policies for monitoring activities;
g) Data archiving;
h) Organization and responsibilities of the parties involved;
i) Procedures for dealing with non-conformities with the validated monitoring plan.

Monitoring project implementation

a) Technical description of monitoring tasks

All project development and execution activities set out in this document are being monitored in compliance with the standards established in accordance with the document "VM0007 REDD+ Methodology Framework (REDD+ MF)".

All essential data adopted in the preparation and execution of the project is being documented and archived throughout the 30 years of the project, remaining available for audits and other actors involved in the process.

b) Data to be collected

The geographic data from primary or secondary sources that make up the project database are:

Legenda : Geographic data used in project implementation.

Theme	Source	Format	Type
Project Area	Carbonext	Vector	Primary
Leakage Belt	Carbonext	Vector	Primary
Reference Region Location (RRL)	Carbonext	Vector	Primary
Reference Region Deforestation (RRD)	Carbonext	Vector	Primary

Other data can be generated throughout the project, so the list will be updated during verifications.

c) Overview of data collection procedures

The procedures used by the Carbonext technical team to generate the primary data are described in the table below. Secondary data is acquired from its sources and processed to meet the demands of the project.

Legenda: General description of the primary geographic data produced in the implementation of the project.

Theme	Description
Project Area	Produced using GIS tools to analyze the area covered by forest for at least 10 years before the project start date
Leakage Belt	Produced using GIS tools and analysis of the area covered by forest along with similarity criteria with the project area
Reference Region Location (RRL)	Delimited through analysis of the area covered by forest and similarity criteria with the project area
Reference Region Deforestation (RRD)	Delimited through analysis of the area covered by forest and similarity criteria with the project area

d) Quality control and assurance procedures

All procedures and records generated strictly follow the operational protocols established by the company, ensuring the consistency and integrity of the data produced throughout the development of the project. This practice establishes a standard of quality and process management and contributes to transparency and credibility.

e) Inspection and accountability policies for monitoring activities

Supervision and accountability policies for monitoring activities will include:

a) Use of Carbonext's internal expertise to manage the review of monitoring plan documents and data, ensuring that it complies with applicable standards;
b) Regular review meetings between project proponents to evaluate the monitoring plan;
c) To ensure transparency and accountability, the monitoring plan and its results will be available to VVB and VCS verifiers with each verification.

f) Data archiving

All maps and records generated during project implementation and monitoring are being stored and made available to all involved parties and VCS verifiers for audit purposes. Backup copies of the files will be available at the project proponent's facilities, as well as at Carbonext's facilities, in a cloud environment, following the best information security standards. All documents and records are kept in a secure and retrievable manner for at least two years after the end of the project credit period.

g) Organization and responsibilities of the parties involved

Carbonext, as one of the project proponents, is responsible for providing the project's data governance guidelines, defining the procedures for collection, processing, storage, availability and information security. The other parties involved in the project participate in these definitions and jointly decide the strategies to be adopted. In this way, we guarantee that the process is constructed and consolidated jointly between the parties.

h) Procedures for dealing with non-conformities with the validated monitoring plan

Project proponents will establish the necessary procedures to address nonconformities. All necessary procedures and guidelines will be developed with the aim of meeting the main levels of control.

One of the main objectives will be to minimize the risk of error. To this end, we always seek to establish the best procedures, using technologies and training, making it possible to obtain reliable data to support monitoring results and thus reduce the possibility of non-conformities. This includes training staff on the various responsibilities to be performed in the IPOÁ REDD+ project; Field review, in order to monitor the procedures determined in the methodological guidelines and; Careful review of all information relevant to the project, including monitoring reports, before its release to interested parties (VVB, governments, institutions...), in order to verify and confirm that all calculations, information, analyzes and conclusions are measurable and reliable. These actions will be led by Carbonext.

If non-conformities arise during the processes, the data will be reviewed and the non-conformities must be dealt with. All deviations will be reported in section 3.1.3 of the Monitoring Report.

Monitoring real changes in carbon stock and GHG emissions

The change categories subject to MRV-A (monitoring, reporting, verification, and accounting) are “Forest area converted to non-forest”, “Forest area in the process of reducing carbon stock” and “Forest area in process of increasing carbon stock”).

Monitoring land use and cover change

a) Technical description of monitoring tasks

Monitoring of actual changes in carbon stocks and GHG emissions in the project area was carried out using satellite images and remote sensing techniques, with the aim of identifying and quantifying deforestation that occurred during the monitoring period in the reference region, leakage belt and project area. The analysis of land use changes was based on primary information since secondary data such as data from Prodes or Mapbiomas is not available. Based on secondary data from the database of the BD/Queimadas (INPE) system, we evaluated the fire occurrences within the reference region during the monitoring period

b) Data to be collected

To measure deforestation during the MR period, images from the Sentinel-2 satellite were used. As there were no images from the end of monitoring with low cloud cover, it was necessary to look at previous dates. To cover possible deforestation after the dates of the optical images, until the end of the monitoring period, we revisited, through visual inspection, the entire area with radar images from the Sentinel-1 satellite. In this context, deforestation that occurred in the period from 17/03/2022 (March 17, 2022) to 31/12/2023 (December 31, 2023) was mapped through semi-automatic classification of supervised multispectral optical images and images from the radar from visual inspection.

For the semi-automatic classification, the first step was to obtain an image from the Sentinel 2 – MultiSpectral Instrument (MSI) dated 04/09/2023, which covers the entire reference region with the lowest possible cloud cover index for the period. We perform false color composition, combining the R8 G4 B3 bands to highlight the vegetation through band 8, near infrared. With the help of high-resolution imagery (WMS imaging services such as ESRI and Planet basemaps), LU/LC samples were collected for forest, non-forest, and water areas throughout the reference region. In addition to the multispectral optical image, an NDVI (Normalized Difference Vegetation Index) image was created using the band algebra (B8-B4)/(B8+B4), where band 4 is the red spectral band and the 8 is the infrared spectral band, feeding the classifier with one more variable and increasing the separation of the forest class from any other non-forest class.

The conclusion of the deforestation measurements for the monitoring period was carried out through the visual interpretation of the SAR radar (Synthetic Aperture Radar), operating in the C band (between 8 and 4 GHz or 3.8 - 7.5 cm), double VV polarization and VH, spatial resolution 20 meters, from the Sentinel 1 satellite (European Space Agency) with image from 30/12/2023. This strategy was necessary to map deforestation in a period of predominance of clouds that prevents the use of optical images to classify land use. Microwave remote sensors like Sentinel 1 (C-band) are widely adopted for this type of mapping because they can penetrate clouds and help identify deforestation.

c) Overview of data collection procedures

The classification of the selected optical image used the Random Forest algorithm, in the Google Earth Engine (GEE) environment. The samples for training the Random Forest classifier were divided into two blocks, 30% for training and evaluating the classifier; and 70% for use in the classification process. The classifier was run with 100 decision trees.

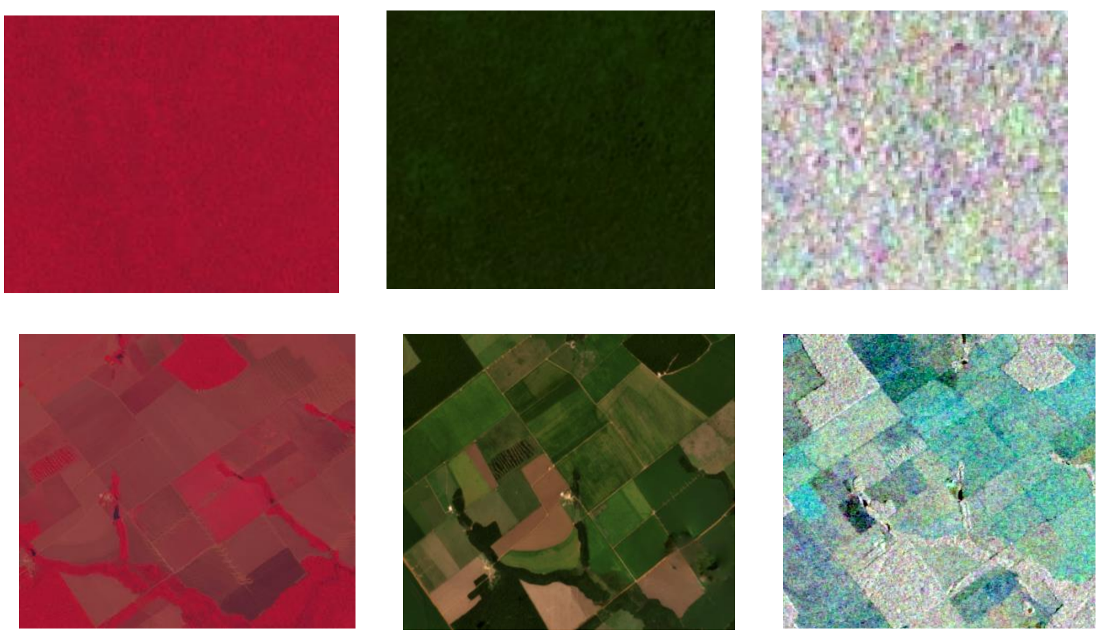
The initial classification result was processed in a GIS environment (ArcGIS), in order to reduce waste and classifier errors. Pixel filters such as Majority (ArcGIS) were applied to reduce the "salt and pepper" effect of classification by identifying isolated pixels and recategorizing them according to the majority value of their eight neighbors. Furthermore, a visual inspection of the classification product superimposed on the sentinel-2 image was carried out with the aim of correcting the main errors of the classifier via visual interpretation. Due to the high cloud cover in the region, we used SAR images from Sentinel-1 in the process of quantifying deforestation. These images were used in the GIS environment (ArcGIS) for vectorization by visual interpretation of the increase in deforested areas omitted from visual images due to high cloud cover until the end of the monitoring period.

Mapping deforestation semi-automatically and through visual interpretation was carried out through the following steps:

Target identification: non-forest areas;

Training: areas of deforestation observed in optical satellite images, of equal or higher resolution, are superimposed to identify patterns of hue, texture/roughness, shape and location of deforested areas;

Photointerpretation: identification and vectorization in the radar image of areas of deforestation omitted by clouds in the optical image, using as a mask all deforestation already mapped.



Forest

Non-forest

Sentinel-2 imagery – (R8 G4 B3) false color composition – 10 m pixel

Planet imagery (web map service) true color view – 3m pixel

Sentinel 1 SAR Imagery (VH) Polarization – 20m pixel

Figure 13. Land use sampling frame for photointerpretation and deforestation measurement

d) Quality control and quality assurance procedures

Validation of the land use data used for measurement was carried out using the confusion matrix, in order to calculate the general success rate per period and class. Three specific classes were used: forest, deforestation and water.

The analysis of the accuracy of measuring land use changes was carried out using the confusion matrix, global accuracy and Kappa index. Sampling was stratified, proportionally by area, based on a total of 310 points randomly distributed in the reference region, leakage belt and project area, using the Create Accuracy Assessment Points tool in ArcGIS.

The accuracy evaluation point map assigns value to the land use map generated in the classification process in a column. This column has been omitted and the interpreter populates the ground truth field based on the interpretation of the reference data used, such as the raw optical image itself, maps from ESRI and Planet web services with higher spatial resolution than the input image. Thus, the table of accuracy evaluation points is the input to the Compute Confusion Matrix tool in ArcGIS.

Legenda: Confusion Matrix and Kappa Accuracy Assessment – Sentinel 2 imagery

	REFERENCE					
CLASSIFIED		**Water**	**Forest**	**Deforestation**	**Total**	**User Accuracy**
	Water	10	0	0	10	100%
	Forest	1	278	2	281	98,93%
	Deforestation	0	1	18	19	94,74%
	Total	11	279	20	310	Kappa
	Prod. Accuracy	90,91%	99,64%	90%		92,80%

The overall classification accuracy for all land use and land cover classes was 92.80%in the reference region, higher than the 90% suggested by the VM0007 methodology. The specific accuracy for the Forest and Deforestation classes were 98.93 and 94.74% respectively.

The accuracy analysis of the classification of the radar image was performed in the same way as the validation of the product classified from the optical image.

Legenda: Confusion Matrix and Kappa Accuracy Assessment - SAR (radar) imagery

REFERENCE

CLASSIFIED		Water	Forest	Deforestation	Total	User Accuracy
	Water	10	0	0	10	100%
	Forest	0	93	1	94	98,94%
	Deforestation	0	1	9	10	90%
	Total	10	94	10	114	Kappa
	Prod. Accuracy	100%	98,94%	90%		94,24%

The overall accuracy of the classification of all land-use and land cover classes was 94.24% in the Reference Region, higher than the 90% suggested by the VM00007 methodology. The specific accuracy for the Forest and Deforestation classes were 98.94 and 90%, respectively.

e) Results

The results obtained through the procedures were overlaid and cropped to the limits of the project area. In this way, we obtained the following deforestation results in the monitoring period.

Legenda: Deforestation mapped during the monitoring period in the Project Area.

IPOÁ Monitoring Report #1
Deforested Area- PA (ha)
11.61

11.61 ha of deforestation were detected in the Project Area over the monitoring period. The deforestation all occurred in the Maanaim property portion of the PA, as shown in the Figure below.

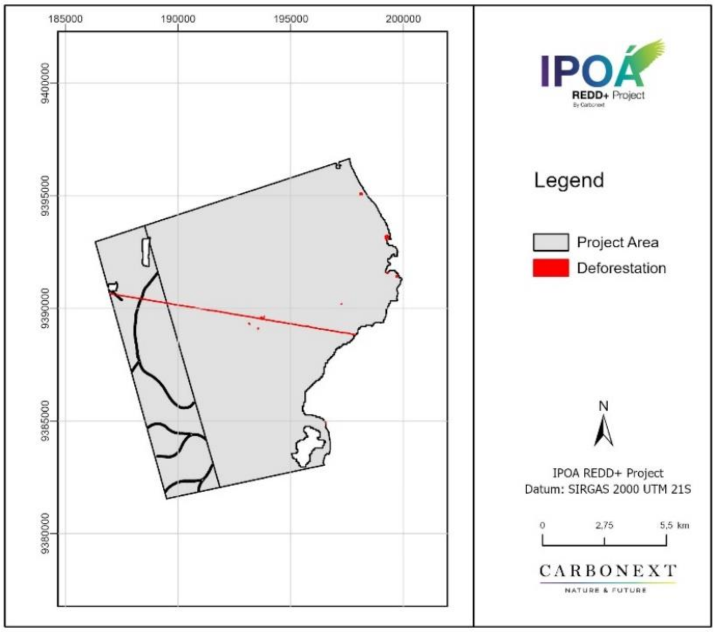


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Figure 14. Deforestation in PA during the monitoring period.

f) Inspection and accountability policies for monitoring activities

Oversight and accountability policies for monitoring activities include:

- Use of Carbonext's internal expertise to manage the review of monitoring plan documents and data, ensuring that it complies with applicable standards;
- Periodic review meetings between project proponents to evaluate the monitoring plan;

To ensure transparency and accountability, the monitoring plan and its results are available to VVB and VCS verifiers at each verification.

g) Data archiving

All maps and records generated during project implementation and monitoring are retained and made available to VCS verifiers upon verification. Backup copies of the files must be available at the project proponent facilities, as well as at Carbonext facilities. All documents and records are kept in a secure and retrievable form for at least two years after the end of the project credit period.

h) Organization and responsibilities of the parties involved in all of the above.

All satellite image assessments are carried out by the Carbonext SIG specialist, also responsible for issuing reports and archiving data (digital version of original documents), and providing assistance during verification audits.

i) Procedures for handling non-conformities with the validated monitoring plan

Project proponents will establish the necessary procedures to deal with non-conformities. All necessary procedures and guidelines are designed with the aim of meeting the main levels of control.

One of the main objectives will be to minimize the risk of error. To this end, a quality management system for project information will be developed, seeking reliable data to support monitoring results and, thus, reduce the possibility of non-conformities. This includes training staff on the various responsibilities to be performed under the IPOÁ REDD+ Project; Field review, in order to follow the procedures determined in the methodological guidelines and; Careful review of all information relevant to the project, including monitoring reports, before its dissemination to interested parties (VVB, governments, institutions...), in order to verify and confirm that all calculations, information, analyzes and conclusions are measurable and reliable. These actions will be led by Carbonext.

If non-conformities arise during the processes, the data will be reviewed and the non-conformities must be addressed. All deviations will be reported in section 3.1.3 of the Monitoring Report.

Monitoring land use and cover change within the leakage belt

a) Technical description of monitoring tasks.

Monitoring of actual changes in carbon stock and GHG emissions within the project area was carried out using satellite images and remote sensing techniques, with the aim of identifying and quantifying deforestation that occurred during the monitoring period in the region. reference, leakage belt and project area. The analysis of changes in land use was based on primary information due to the unavailability of secondary data, such as data from Prodes or MapBiomas.

Based on secondary data from the database of the BD/Queimadas (INPE) system, we evaluated the fire occurrences within the reference region during the monitoring period

b) Data to be collected

To measure deforestation during the MR period, images from the Sentinel-2 satellite were used. As there were no images from the end of monitoring with low cloud cover, it was necessary to look at previous dates. To cover possible deforestation after the dates of the optical images, until the end of the monitoring period, we revisited, through visual inspection, the entire area with radar images from the Sentinel-1 satellite. In this context, deforestation that occurred in the period from 17/03/2022 (March 17, 2022) to 31/12/2023 (December 31, 2023) was mapped through semi-automatic classification of supervised multispectral optical images and images from the radar from visual inspection.

For the semi-automatic classification, the first step was to obtain an image from the Sentinel 2 – MultiSpectral Instrument (MSI) dated 04/09/2023, which covers the entire reference region with the lowest possible cloud cover index for the period. We perform false color composition, combining the R8 G4 B3 bands to highlight the vegetation through band 8, near infrared. With the help of high-resolution imagery (WMS imaging services such as ESRI and Planet basemaps), LU/LC samples were collected for forest, non-forest, and water areas throughout the reference region. In addition to the multispectral optical image, an NDVI (Normalized Difference Vegetation Index) image was created using the band algebra (B8-B4)/(B8+B4), where band 4 is the red spectral band and the 8 is the infrared spectral band, feeding the classifier with one more variable and increasing the separation of the forest class from any other non-forest class.

The conclusion of the deforestation measurements for the monitoring period was carried out through the visual interpretation of the SAR radar (Synthetic Aperture Radar), operating in the C band (between 8 and 4 GHz or 3.8 - 7.5 cm), double VV polarization and VH, spatial resolution 20 meters, from the Sentinel 1 satellite (European Space Agency) with image from 30/12/2023. This strategy was necessary to map deforestation in a period of predominance of clouds that prevents the use of optical images to classify land use. Microwave remote sensors like Sentinel 1 (C-band) are widely adopted for this type of mapping because they can penetrate clouds and help identify deforestation.

c) Overview of data collection procedures.

The classification of the selected optical image used the Random Forest algorithm, in the Google Earth Engine (GEE) environment. The samples for training the Random Forest classifier were divided into two blocks, 30% for training and evaluating the classifier; and 70% for use in the classification process. The classifier was run with 100 decision trees.

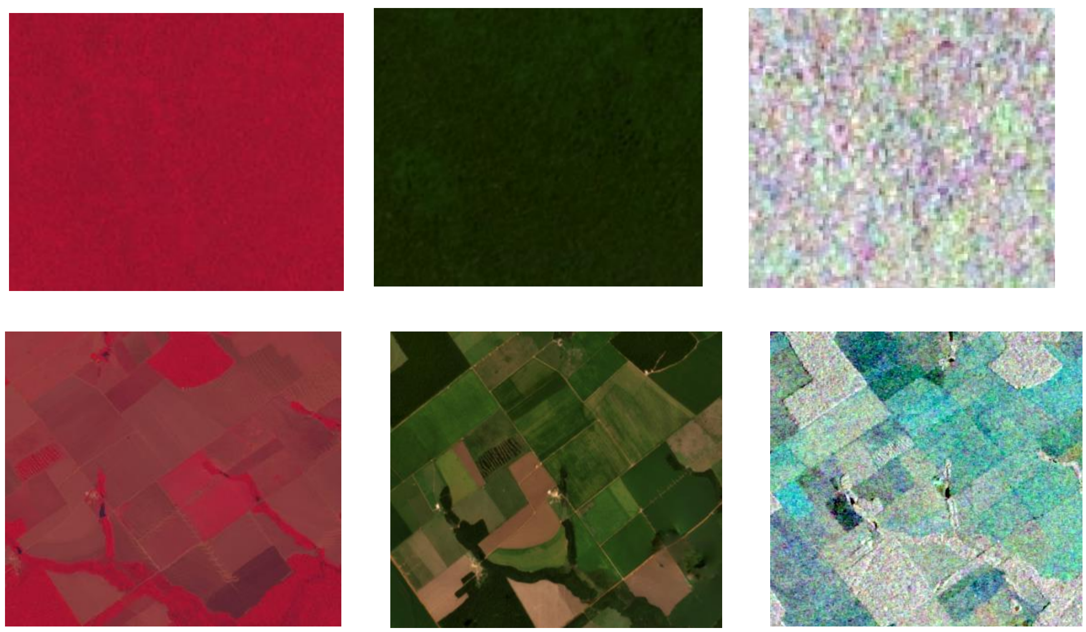
The initial classification result was processed in a GIS environment (ArcGIS), in order to reduce waste and classifier errors. Pixel filters such as Majority (ArcGIS) were applied to reduce the "salt and pepper" effect of classification by identifying isolated pixels and recategorizing them according to the majority value of their eight neighbors. Furthermore, a visual inspection of the classification product superimposed on the sentinel-2 image was carried out with the aim of correcting the main errors of the classifier via visual interpretation. Due to the high cloud cover in the region, we used SAR images from Sentinel-1 in the process of quantifying deforestation. These images were used in the GIS environment (ArcGIS) for vectorization by visual interpretation of the increase in deforested areas omitted from visual images due to high cloud cover until the end of the monitoring period.

Mapping deforestation semi-automatically and through visual interpretation was carried out through the following steps:

Target identification: non-forest areas;

Training: areas of deforestation observed in optical satellite images, of equal or better resolution, are superimposed to identify patterns of hue, texture/roughness, shape and location of deforested areas;

Photointerpretation: identification and vectorization in the radar image of areas of deforestation omitted by clouds in the optical image, using as a mask all deforestation already mapped;



Forest

Non-forest

Sentinel-2 imagery – (R8 G4 B3) false color composition – 10 m pixel	Planet imagery (web map service) true color view – 3m pixel	Sentinel 1 SAR Imagery (VH) Polarization – 20m pixel

Figure 15. Land use sampling frame for photointerpretation and deforestation measurement.

d) Quality control and assurance procedures.

Validation of the land use data used for measurement was carried out using the confusion matrix, in order to calculate the general success rate per period and class. Three specific classes were used: forest, deforestation and water.

The analysis of the accuracy of measuring land use changes was carried out using the confusion matrix, global accuracy and Kappa index. Sampling was stratified, proportionally by area, based on a total of 309 points randomly distributed in the reference region, leakage belt and project area, using the Create

Accuracy Assessment Points tool in ArcGIS. The accuracy evaluation point map assigns value to the land use map generated in the classification process in a column. This column has been omitted and the interpreter fill the ground truth field based on the interpretation of the reference data used, such as the raw optical image itself, maps from ESRI and Planet web services with higher spatial resolution than the input image. Thus, the accuracy evaluation point table is the input to the Compute Confusion Matrix tool in ArcGIS (

Table 11).

Table 11 - Confusion Matrix and Kappa Accuracy Assessment - Sentinel imagery.

	REFERENCE					
CLASSIFIED		Water	Forest	Deforestation	Total	User Accuracy
	Water	10	0	0	10	100%
	Forest	1	278	2	281	98,93%
	Deforestation	0	1	18	19	94,74%
	Total	11	279	20	310	Kappa
	Prod. Accuracy	90,91%	99,64%	90%		92,80%

The overall classification accuracy for all land use and land cover classes was 92.80% in the reference region, higher than the 90% suggested by the VM0007 methodology. The specific accuracy for the Forest and Deforestation classes were 98.93 and 94.74% respectively.

The accuracy analysis of the classification of the radar image was performed in the same way as the validation of the product classified from the optical image.

The overall accuracy of the classification of all land-use and land cover classes was 94.24% in the Reference Region, higher than the 90% suggested by the VM00007 methodology. The specific accuracy for the Forest and Deforestation classes were 98.94 and 90%, respectively (Table 12).

Table 12 - Confusion Matrix and Kappa Accuracy Assessment - SAR (radar) imagery.

	REFERENCE					
CLASSIFIED		Water	Forest	Deforestation	Total	User Accuracy
	Water	10	0	0	10	100%
	Forest	0	93	1	94	98,94%

	Deforestation	0	1	9	10	90%
	Total	10	94	10	114	Kappa
	Prod. Accuracy	100%	98,94%	90%		94,24%

e) Results

The results obtained through the procedures were superimposed and cropped to the limits of the leakage belt. In this way, we obtained the following deforestation results during the monitoring period.

Table 13 - Deforestation mapped in the monitoring period in the Leakage Belt.

IPOÁ Monitoring Report#1
Deforested Area- LB (ha)
17.0

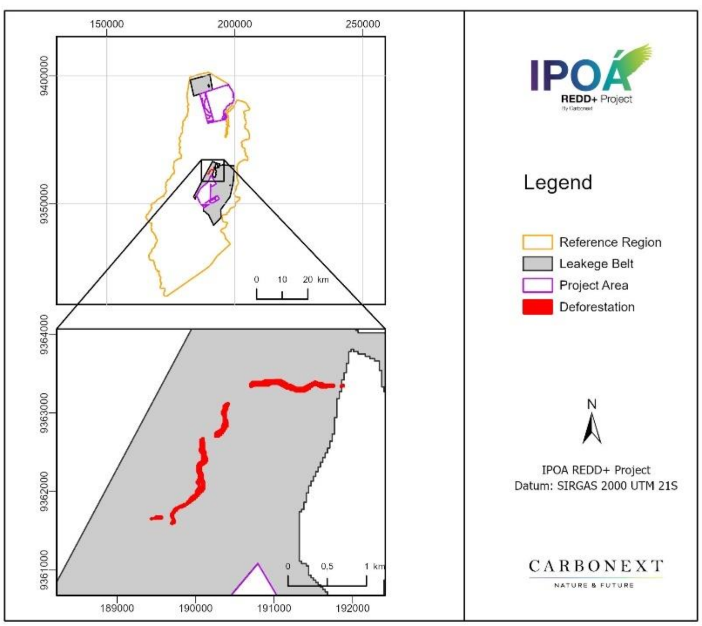


Figure 16. Deforestation in LB during the monitoring period.

f) Inspection and accountability policies for monitoring activities

Oversight and accountability policies for monitoring activities include:

- Use of Carbonext's internal expertise to manage the review of monitoring plan documents and data, ensuring that it complies with applicable standards;
- Periodic review meetings between project proponents to evaluate the monitoring plan;

To ensure transparency and accountability, the monitoring plan and its results are available to VVB and VCS verifiers at each verification.

g) Data archiving.

All maps and records generated during project implementation and monitoring are retained and made available to VCS verifiers upon verification. Backup copies of the files must be available at the project proponent facilities, as well as at Carbonext facilities. All documents and records are kept in a secure and retrievable form for at least two years after the end of the project credit period.

h) Organization and responsibilities of the parties involved in all of the above items.

All satellite image assessments are carried out by the Carbonext SIG specialist, also responsible for issuing reports and archiving data (digital version of original documents) and providing assistance during verification audits.

i) Procedures for handling non-conformities with the validated monitoring plan

Project proponents will establish the necessary procedures to deal with non-conformities. All necessary procedures and guidelines are designed with the aim of meeting the main levels of control.

One of the main objectives will be to minimize the risk of error. To this end, a quality management system for project information will be developed, seeking reliable data to support monitoring results and, thus, reduce the possibility of non-conformities. This includes training staff on the various responsibilities to be performed under the IPOÁ REDD+ project; Field review, in order to follow the procedures determined in the methodological guidelines and; Careful review of all information relevant to the project, including monitoring reports, before its dissemination to interested parties (VVB, governments, institutions...), in order to verify and confirm that all calculations, information, analyzes and conclusions are measurable and reliable. These actions will be led by Carbonext.

If non-conformities arise during the processes, the data will be reviewed and the non-conformities must be addressed. All deviations will be reported in section 3.1.3 of the Monitoring Report.

Monitoring changes in carbon stocks and non-CO2 emissions from wildfires

a) Technical description of monitoring tasks

Monitoring carbon stocks is mandatory in the following cases:

Within the project area:

- Areas subject to significant unplanned reduction in carbon stock due to uncontrolled wildfires and other catastrophic events. In these areas, carbon stock losses must be estimated as quickly as possible after the catastrophic event.

Within the leakage area:

- Areas subject to planned and significant carbon stock reduction in the project scenario according to the ex-ante assessment. In these areas, carbon stocks must be estimated at least once after the planned event that caused the carbon stock to decrease.

b) Data to be collected

The mapping of fire outbreaks was carried using the BD/Queimadas (INPE) system database, from which we exported data on fire outbreaks for the entire Reference Region covering the whole monitoring period.

c) Overview of data collection procedures

Heat Points: point geospatial features that indicate areas of hotspots on forest land use for the entire reference region during the monitoring report period. The impact of fire on the PA and LB will be measured by the overlap of the fire scar layer with the boundaries of the PA and LB.

d) Quality control and assurance procedures

As mentioned, the BDQueimadas/INPE database was used to assess forest heat points in the Reference Region over the monitoring period, as displayed in the Figure below. No heat points were detected in the Project Area for the present monitoring period, whereas in the RR 2254 heat points were detected, which follows the pattern observed over the historical period. Furthermore, no heat points were detected close to the Project Area. For this reason, when observing the spatial distribution of heat points during the MR1 period, the assessment of fire risk is maintained as low.

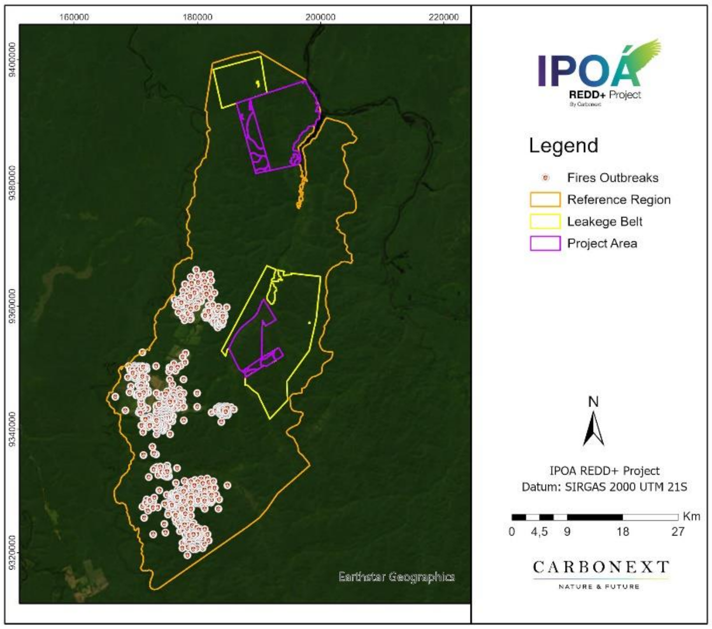


Figure 17 - Fire outbreaks in RR during the monitoring period

e) Supervision and accountability policies for monitoring activities

Supervision and accountability policies for monitoring activities include:

- Use of Carbonext's internal expertise to manage the review of monitoring plan documents and data, ensuring that it complies with applicable standards;
- Regular review meetings between project proponents to evaluate the monitoring plan;
- To ensure transparency and accountability, the monitoring plan and its results will be available to VVB and VCS verifiers at each verification.

f) Data archiving

All maps and records generated during project implementation and monitoring are retained and made available to VCS verifiers upon verification for inspection to demonstrate that the AUD project activity has actually been implemented. All documents and records will be maintained in a secure and retrievable form for at least two years after the end of the project credit period.

g) Organization and responsibilities of the parties involved in all of the above

All fire impact assessments are carried out by Carbonext's team of GIS experts, also responsible for reporting and data archiving (digital version of original documents) and providing assistance during verification audits.

h) Procedures for dealing with non-conformities with the validated monitoring plan

Project proponents will establish the necessary procedures to address non-conformities. All necessary procedures and guidelines will be developed with the aim of meeting the main levels of control.

One of the main objectives will be to minimize the risk of error. To this end, a quality management system for project information will be developed, seeking reliable data to support monitoring results and thus reduce the possibility of non-conformities. This includes training staff on the various responsibilities to be performed in the IPOÁ REDD+ Project; Field review, in order to monitor the procedures determined in the methodological guidelines and; Careful review of all information relevant to the project, including monitoring reports, before its release to interested parties (VVB, governments, institutions...), in order to verify and confirm that all calculations, information, analyzes and Conclusions are measurable and reliable. These actions will be led by Carbonext.

If non-conformities arise during the processes, the data will be analyzed and the non-conformities must be dealt with.

Ex post calculation of the net reduction of anthropogenic GHG emissions

a) Technical description of monitoring tasks

The ex-post calculation of net reductions in anthropogenic GHG emissions is similar to the ex-ante calculation with the only difference being that the estimated ex post changes in carbon stock and GHG emissions must be used in the case of the project and leakage scenario.

b) Data to be collected

Table 14 - Data collected to monitor ex-post net reductions for the IPOÁREDD+ project.

Parameter	Unit	Source	Frequency
Reduction in net GHG emissions attributable to project activities for the year	tCO_2e	Calculated	Annually
Number of Verified Carbon Units (VCUs) to be made available for sale in the year	tCO_2e	Calculated	Annually

c) Overview of data collection procedures

The data collection procedures are the same as those described in the previous steps. This step involves compiling data from previous procedures to calculate the ex-post reduction of net anthropogenic GHG emissions.

d) Quality control and assurance procedures

A map showing the cumulative deforested areas within the project area and leakage belt must be updated and presented to VCS verifiers at each verification event. The cumulative area cannot generate additional VCUs in future periods.

e) Supervision and accountability policies for monitoring activities

Supervision and accountability policies for monitoring activities will include:

- Use of Carbonext's internal expertise to manage the review of monitoring plan documents and data, ensuring that it complies with applicable standards;
- Regular review meetings between project proponents to evaluate the monitoring plan;
- To ensure transparency and accountability, the monitoring plan and its results will be available to VVB and VCS verifiers at each verification.

f) Data archiving

All maps and records generated during project implementation and monitoring will be retained and made available to VCS verifiers upon verification. Backup copies of the files must be available at the project proponent's facilities as well as at Carbonext's facilities. All documents and records will be maintained in a secure and retrievable form for at least two years after the end of the project credit period.

g) Organization and responsibilities of the parties involved in all of the above

All database assessments for calculations of net emissions reductions will be carried out by Carbonext experts, also responsible for reporting and data archiving (digital version of original documents) and for providing assistance during verification audits.

h) Non-conformities

Project proponents will establish the necessary procedures to address non-conformities. All necessary procedures and guidelines will be developed with the aim of meeting the main levels of control.

One of the main objectives will be to minimize the risk of error. To this end, a quality management system for project information will be developed, seeking reliable data to support monitoring results and thus reduce the possibility of non-conformities. This includes training staff on the various responsibilities to be performed in the IPOÁ REDD+ Project; Field review, in order to monitor the procedures determined in the methodological guidelines and; Careful review of all information relevant to the project, including monitoring reports, before its release to interested parties (VVB, governments, institutions...), in order to verify and confirm that all calculations, information, analyzes and Conclusions are measurable and reliable. These actions will be led by Carbonext.

If non-conformities arise during the processes, the data will be analyzed and the non-conformities must be dealt with. All deviations will be reported in section 3.1.3 of the Monitoring Report.

In complement to the results of the GIS-based Monitoring Plan, the results of the activities during MR1 under the Climate pillar are described below.

Table 15. Activities within the climate pillar carried out during MR1

Pillar	Scope	Activity	Date	Indicador/ Output	Details	Data Collection Method	Risk
CLIMATE	**Forest Conservation**	Remote Monitoring	Jul 2022	2 deforestation events detected remotely	The CarbonEye system detected two events in July and august 2022 (described below).[84]	GIS methods described in section below	Invasive deforestation events are managed by OP 3, described in section 4.3.1
		Deforestation prevention and mitigation actions	Aug 2022	2 deforestation events visited in the field. 5 meetings carried out to	The following deforestation events were visited in the field: 1) Deforestation creating a road in Maanaim Farm in July 2022 - Signs were installed to warn about the prohibition of	As described in section below	Irregular community activity risks are managed by OP 8, described in section 4.3.1

[84] See evidence folder: “08. Deforestation”; files: “22.05.12”; and “22.08.16”

				combat deforestation.[85] 2 protocol applied regarding deforestation events deforestation[86]	deforestation.[87] 2) Community agricultural area, Maanaim property, August, 2022.		

3.1.4 Dissemination of Monitoring Plan and Results (CL4.2)

The results of the first monitoring period will be disseminated on the internet via the present monitoring report as of February 2024, when the latter will be uploaded to the Verra website for public comments [88]. After successful verification, MR1 will be included on the Ipoá project's own Verra Project page (ID 3324)[89]. This is the primary means of dissemination to the market in general, while the results of MR1 will be sent to the project's acknowledged institutional stakeholders during MR2 consultation.

Regarding community stakeholders, during the period of the first year of monitoring March/2022 to March/2023 there was no disclosure of results, as the actions were still in the planning and implementation phases.

From the implementation of the actions, the results will be presented in the community in field visits, with banners and informative materials of easy understanding.

For the contact with the community, the communication channels are open for any questions or feedback on the results or the process of the monitoring report since the technical visit carried out

[85] See folder: 08. Deforestation\22.05.12 - Estrada Maanaim; files:" 22.06.30 - Reunião proprietários_Invasão"; and "22.06.30 – Explanatory note"
[86] See evidence folder: "12. Operational Procedures\Applied"
[87] See folder "08.Deforestation"
[88] https://verra.org/consultations/projects-open-for-public-comment/
[89] https://registry.verra.org/app/projectDetail/VCS/3324

in March 2022. At this time, the direct contact of the Carbonext Analyst for contact with the community has been made available, and the company's telephone number for contact.

After this first moment, in October 2022 a direct contact number was made available for complaints, claims and suggestions.

After this first moment, in October 2022 a direct contact number was made available for complaints, claims and suggestions.

3.2 Quantification of GHG Emission Reductions and Removals

3.2.1 Baseline Emissions

As described in the Project Design (PD), the net baseline carbon stock changes and greenhouse gas emissions of the project were calculated through two different modules of the VM0007 Methodology. In order to estimate the baseline emissions relating to APD, the VMD0006 module (BL-PL) "Estimation of Baseline Carbon Stock Changes and Greenhouse Gas Emissions from Planned Deforestation and Planned Degradation" was applied. Regarding the estimation of baseline emissions from unplanned deforestation, AUD scenario, the VMD0007 module (BL-UP) was applied: "Estimation of baseline carbon stock changes and greenhouse gas emissions from unplanned deforestation and unplanned wetland degradation".

The present section 3 is accompanied by a calculation spreadsheet[90], attached as a separate file, for the purposes of illustration of the calculation methodology.

Estimation of the annual areas of unplanned baseline deforestation in the PA

The accumulated AUD baseline deforestation projected to occur within the Project Area is 652.48 hectares over the 30-year project lifetime, while over the present MR period, it is 0.26 hectares. The estimated average annual rate of deforestation in the Project Area over 30 years is therefore 21.75 hectares.

For the calculation, the module VMD0007 for estimation of baseline carbon stock changes and greenhouse gas emissions from unplanned deforestation and unplanned wetland degradation (BL-UP) was applied.

Table 16 - Projection of annual baseline unplanned deforestation for the Project Area (Location Analysis), per stratum (1)

Avoided Unplanned Deforestation Annual baseline deforestation for the Project Area Dense Tropical Submontane Rainforest		
Year	**Total yearly area (ha/year)**	**Total cumulative area (ha)**

[90] Excel file name: IPOA_SPREADSHEET_IPOA_MR_1.x"

2022	0.00	0.00
2023	0.00	0.00
2024	0.00	0.00
2025	0.00	0.00
2026	0.00	0.00
2027	66.17	66.17
2028	48.47	114.64
2029	0.00	114.64
2030	0.00	114.64
2031	0.00	114.64
2032	31.75	146.39
2033	29.02	175.41
2034	61.09	236.50
2035	2.88	239.38
2036	0.00	239.38
2037	34.38	273.76
2038	0.00	273.76
2039	30.15	303.91
2040	65.88	369.79
2041	2.74	372.52
2042	0.00	372.52
2043	0.00	372.52
2044	0.00	372.52
2045	0.76	373.29
2046	0.00	373.29
2047	0.36	373.65
2048	0.00	373.65
2049	0.00	373.65
2050	21.44	395.08
2051	0.00	395.08
2052	0.00	395.08

Table 17- Projection of annual baseline unplanned deforestation for the Project Area (Location Analysis), per stratum (2)

Avoided Unplanned Deforestation Annual baseline deforestation for the Project Area Dense Tropical Lowland Rainforest		
Year	**Total yearly area (ha/year)**	**Total cumulative area (ha)**
2022	0.00	0.00
2023	0.00	0.00
2024	0.00	0.00
2025	0.00	0.00
2026	0.00	0.00
2027	0.00	0.00
2028	0.00	0.00
2029	0.00	0.00
2030	0.00	0.00
2031	0.00	0.00
2032	0.00	0.00
2033	0.00	0.00
2034	0.00	0.00
2035	0.00	0.00
2036	0.00	0.00

2037	0.00	0.00
2038	0.00	0.00
2039	0.00	0.00
2040	0.00	0.00
2041	0.00	0.00
2042	0.00	0.00
2043	0.00	0.00
2044	0.00	0.00
2045	0.00	0.00
2046	0.00	0.00
2047	0.00	0.00
2048	0.00	0.00
2049	0.00	0.00
2050	0.00	0.00
2051	0.00	0.00
2052	0.00	0.00

Table 18 - Projection of annual baseline unplanned deforestation for the Project Area (Location Analysis), per stratum (3)

Avoided Unplanned Deforestation Annual baseline deforestation for the Project Area Dense Tropical Alluvial Rainforest		
Year	**Total yearly area (ha/year)**	**Total cumulative area (ha)**
2022	0.00	0.00
2023	0.00	0.00
2024	0.00	0.00
2025	0.00	0.00
2026	0.00	0.00
2027	0.00	0.00
2028	0.00	0.00
2029	0.00	0.00
2030	0.00	0.00
2031	0.00	0.00
2032	0.00	0.00
2033	0.00	0.00
2034	0.00	0.00
2035	0.00	0.00
2036	0.00	0.00
2037	0.00	0.00
2038	0.00	0.00
2039	0.00	0.00
2040	0.00	0.00
2041	0.00	0.00
2042	0.00	0.00
2043	0.00	0.00
2044	0.00	0.00
2045	0.00	0.00
2046	0.00	0.00
2047	0.00	0.00

2048	0.00	0.00
2049	0.00	0.00
2050	0.00	0.00
2051	0.00	0.00
2052	0.00	0.00

Table 19 - Projection of annual baseline unplanned deforestation for the Project Area (Location Analysis), per stratum (4)

Avoided Unplanned Deforestation Annual baseline deforestation for the Project Area Pioneer Formation with fluvial influence without palm trees		
Year	**Total yearly area (ha/year)**	**Total cumulative area (ha)**
2022	0.26	0.26
2023	0.00	0.26
2024	0.00	0.26
2025	0.00	0.26
2026	0.00	0.26
2027	26.35	26.61
2028	27.01	53.62
2029	0.00	53.62
2030	0.00	53.62
2031	0.00	53.62
2032	1.10	54.72
2033	1.49	56.21
2034	38.78	94.99
2035	0.00	94.99
2036	0.00	94.99
2037	0.00	94.99
2038	9.20	104.18
2039	15.51	119.69
2040	50.18	169.87
2041	32.19	202.06
2042	3.49	205.55
2043	6.51	212.05
2044	0.00	212.05
2045	26.77	238.83
2046	0.00	238.83
2047	2.05	240.87
2048	0.00	240.87
2049	0.00	240.87
2050	14.20	255.08
2051	0.00	255.08
2052	2.32	257.40

Table 20 - Projection of annual baseline unplanned deforestation for the Project Area (Location Analysis), sum of strata

Avoided Unplanned Deforestation Annual baseline deforestation for the Project Area Sum of Strata		
Year	**Total yearly area (ha/year)**	**Total cumulative area (ha)**
2022	0.26	0.26
2023	0.00	0.26

2024	0.00	0.26
2025	0.00	0.26
2026	0.00	0.26
2027	92.52	92.78
2028	75.48	168.26
2029	0.00	168.26
2030	0.00	168.26
2031	0.00	168.26
2032	32.85	201.11
2033	30.51	231.62
2034	99.86	331.49
2035	2.88	334.37
2036	0.00	334.37
2037	34.38	368.75
2038	9.20	377.94
2039	45.66	423.60
2040	116.06	539.66
2041	34.93	574.59
2042	3.49	578.07
2043	6.51	584.58
2044	0.00	584.58
2045	27.54	612.11
2046	0.00	612.11
2047	2.41	614.52
2048	0.00	614.52
2049	0.00	614.52
2050	35.64	650.16
2051	0.00	650.16
2052	2.32	652.48

Avoided Planned Deforestation (APD)

The baseline net GHG emissions for planned deforestation (APD) were determined as:

$$\Delta C_{BSL,planned} = \sum_{t=1}^{t*} \sum_{i=1}^{M} (\Delta C_{BSL,i,t} + GHG_{BSL-E,i,t})$$

Where:

ΔCBSL, planned Net greenhouse gas emissions in the baseline from planned deforestation up to year t*; t CO2-e

ΔCBSL, i,t Net carbon stock changes in all pools in the baseline stratum i in year t; t CO2-e

GHGBSL-E, i,t Greenhouse gas emissions as a result of deforestation activities within the project boundary in the baseline stratum i in year t; t CO2-e yr-1

i 1, 2, 3, … M strata

t 1, 2, 3, … t* years elapsed since the projected start of the project activity.

The agent of deforestation in APD scenario is a specific individual: The landowner intended to deforest the area for development of agricultural activity, in the absence of the project activity.

As Brazilian law requires government approval for deforestation to occur, the intention to deforest within the Project Area is demonstrated by evidence that a request for approval has been filed with the relevant government department for permission to deforest and convert to an alternative land use.

The proportion of the total area planned to be deforested, in accordance with the law, does not overlap areas protected by the Brazilian forest code, such as Legal Reserves and Permanent Preservation Areas.

The scheduled plan for deforestation, per property, is as presented in PD section 3.1.4, under the Avoided Planned Deforestation (APD) heading. The deforestation values for each stratum, presented in the tables below, were obtained through analysis of forest types affected per year in the deforestation schedule using IBGE (2021) data, as shown in Figure 8 in the PD's physical parameters section. The resulting baseline deforestation areas per stratum from planned deforestation are presented below.

Table 21 - Annual baseline deforestation for the Project Area from planned deforestation, per stratum (1)

Avoided Planned Deforestation Annual baseline deforestation for the Project Area Dense Tropical Submontane Rainforest		
Year	**Total yearly area (ha/year)**	**Total cumulative area (ha)**
2022	132.63	132.63
2023	0.00	132.63
2024	940.51	1,073.14
2025	941.29	2,014.43
2026	225.94	2,240.37
2027	0.00	2,240.37
2028	0.00	2,240.37
2029	0.00	2,240.37
2030	0.00	2,240.37
2031	0.00	2,240.37
2032	0.00	2,240.37
2033	0.00	2,240.37
2034	0.00	2,240.37
2035	0.00	2,240.37
2036	0.00	2,240.37
2037	0.00	2,240.37
2038	0.00	2,240.37
2039	0.00	2,240.37

2040	0.00	2,240.37
2041	0.00	2,240.37
2042	0.00	2,240.37
2043	0.00	2,240.37
2044	0.00	2,240.37
2045	0.00	2,240.37
2046	0.00	2,240.37
2047	0.00	2,240.37
2048	0.00	2,240.37
2049	0.00	2,240.37
2050	0.00	2,240.37
2051	0.00	2,240.37
2052	0.00	2,240.37

Table 22 - Annual baseline deforestation for the Project Area from planned deforestation, per stratum (2)

Avoided Planned Deforestation Annual baseline deforestation for the Project Area Dense Tropical Lowland Rainforest		
Year	**Total yearly area (ha/year)**	**Total cumulative area (ha)**
2022	369.07	369.07
2023	0.00	369.07
2024	0.00	369.07
2025	0.00	369.07
2026	0.00	369.07
2027	0.00	369.07
2028	0.00	369.07
2029	0.00	369.07
2030	0.00	369.07
2031	0.00	369.07
2032	0.00	369.07
2033	0.00	369.07
2034	0.00	369.07
2035	0.00	369.07
2036	0.00	369.07
2037	0.00	369.07
2038	0.00	369.07
2039	0.00	369.07
2040	0.00	369.07
2041	0.00	369.07
2042	0.00	369.07
2043	0.00	369.07
2044	0.00	369.07
2045	0.00	369.07
2046	0.00	369.07
2047	0.00	369.07
2048	0.00	369.07
2049	0.00	369.07
2050	0.00	369.07
2051	0.00	369.07
2052	0.00	369.07

Table 23 - Annual baseline deforestation for the Project Area from planned deforestation, per stratum (3)

Avoided Planned Deforestation Annual baseline deforestation for the Project Area Dense Tropical Alluvial Rainforest		
Year	**Total yearly area (ha/year)**	**Total cumulative area (ha)**
2022	0.00	0.00
2023	0.00	0.00
2024	0.00	0.00
2025	0.00	0.00
2026	0.00	0.00
2027	0.00	0.00
2028	0.00	0.00
2029	0.00	0.00
2030	0.00	0.00
2031	0.00	0.00
2032	0.00	0.00
2033	0.00	0.00
2034	0.00	0.00
2035	0.00	0.00
2036	0.00	0.00
2037	0.00	0.00
2038	0.00	0.00
2039	0.00	0.00
2040	0.00	0.00
2041	0.00	0.00
2042	0.00	0.00
2043	0.00	0.00
2044	0.00	0.00
2045	0.00	0.00
2046	0.00	0.00
2047	0.00	0.00
2048	0.00	0.00
2049	0.00	0.00
2050	0.00	0.00
2051	0.00	0.00
2052	0.00	0.00

Table 24 - Annual baseline deforestation for the Project Area from planned deforestation, per stratum (4)

Avoided Planned Deforestation Annual baseline deforestation for the Project Area Pioneer Formation with fluvial influence without palm trees		
Year	**Total yearly area (ha/year)**	**Total cumulative area (ha)**
2022	0.00	0.00
2023	0.00	0.00
2024	0.00	0.00
2025	0.00	0.00
2026	8.54	8.54
2027	0.00	8.54
2028	0.00	8.54

2029	0.00	8.54
2030	0.00	8.54
2031	0.00	8.54
2032	0.00	8.54
2033	0.00	8.54
2034	0.00	8.54
2035	0.00	8.54
2036	0.00	8.54
2037	0.00	8.54
2038	0.00	8.54
2039	0.00	8.54
2040	0.00	8.54
2041	0.00	8.54
2042	0.00	8.54
2043	0.00	8.54
2044	0.00	8.54
2045	0.00	8.54
2046	0.00	8.54
2047	0.00	8.54
2048	0.00	8.54
2049	0.00	8.54
2050	0.00	8.54
2051	0.00	8.54
2052	0.00	8.54

Table 25 - Total annual baseline deforestation for the Project Area from planned deforestation, sum of strata

Avoided Planned Deforestation Annual baseline deforestation for the Project Area Sum of Strata		
Year	**Total yearly area (ha/year)**	**Total cumulative area (ha)**
2022	501.70	501.70
2023	0.00	501.70
2024	940.51	1,442.20
2025	941.29	2,383.49
2026	234.49	2,617.98
2027	0.00	2,617.98
2028	0.00	2,617.98
2029	0.00	2,617.98
2030	0.00	2,617.98
2031	0.00	2,617.98
2032	0.00	2,617.98
2033	0.00	2,617.98
2034	0.00	2,617.98
2035	0.00	2,617.98
2036	0.00	2,617.98
2037	0.00	2,617.98
2038	0.00	2,617.98
2039	0.00	2,617.98
2040	0.00	2,617.98
2041	0.00	2,617.98
2042	0.00	2,617.98
2043	0.00	2,617.98

2044	0.00	2,617.98
2045	0.00	2,617.98
2046	0.00	2,617.98
2047	0.00	2,617.98
2048	0.00	2,617.98
2049	0.00	2,617.98
2050	0.00	2,617.98
2051	0.00	2,617.98
2052	0.00	2,617.98

Emissions from biomass burning in the baseline.

Based on the IPCC 2006 Inventory Guidelines and E-BPB Module (VMD0013), estimating greenhouse gas emissions from biomass burning was determined as follows. The same methodology was applied for both AUD and APD components of the project:

$$E_{BiomassBurn,i,t} = \sum_{g=1}^{G}\left(\left(\left(A_{burn,i,t} * B_{i,t} * COMF_i * G_{g,i}\right) * 10^{-3}\right) * GWP_g\right)$$

Where:

$E_{BiomassBurn,\ i,t}$ Greenhouse gas emissions due to biomass burning as part of deforestation activities in stratum *i* in year *t*; tCO_2-e of each GHG (CO_2, CH_4, N_2O)

$A_{burn,i,t}$ Area burnt for stratum *i* at time *t*; ha

$B_{i,t}$ Average aboveground biomass stock before burning stratum *i*, time *t*; tonnes d.m. ha-1

$COMF_i$ Combustion factor for stratum *i*; dimensionless (default value derived from Table 2.6 of IPCC, 2006)[91]

[91] E-BPB module refers to Table 2.6 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Agriculture, Forestry and Other Land Use, Chapter 2, "Primary tropical moist forest" https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/4_Volume4/V4_02_Ch2_Generic.pdf (last visited in 20/06/2022).

$G_{g,i}$ Emission factor for stratum i for gas g; kg t-1 dry matter burnt (default values derived from Table 2.5 of IPCC, 2006)[92]

GWP_g Global warming potential for gas g; t CO_2/t gas g (default values from IPCC, 2022: CH_4 = 27; N_2O = 273)[93]

g 1, 2, 3 ... *G* greenhouse gases

i 1, 2, 3 ... *M* strata

t 1, 2, 3 ... *t* years elapsed since the start of the REDD+ project activity

The table below shows the parameters used to calculate biomass burning for the baseline scenario.

Table 26. Parameters used to calculate biomass burning for the baseline scenario.

Parameter	Value	Unit
COMF	0.50	Dimensionless
G_{CH4}	6.80	g kg^{-1} dry matter burnt
GWP_{CH4}	27.00	Dimensionless
G_{N2O}	0.20	g kg^{-1} dry matter burnt
GWP_{N2O}	273.00	Dimensionless
B Ds	299.34	t d.m. ha^{-1}
B Db	300.21	t d.m. ha^{-1}
B Da	340.81	t d.m. ha^{-1}
B Pa	195.11	t d.m. ha^{-1}

Below, the summary of emissions of CH_4 and N_2O from biomass burning are presented separately, and then summing them to show the total biomass burning emissions. These are presented by component: AUD first followed APD. The burning emissions were quantified for each of the four strata in section 3.2.1 of the PD, but are presented here only in the sum of strata, for the sake of brevity.

Table 27. Summary of biomass burning emissions (CH_4) in the sum of strata within the Project Area in the baseline case – AUD component

Avoided Unplanned Deforestation Biomass Burn Emissions - CH_4 Sum of Strata				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	0.26	0.26	4.85	4.85

[92] E-BPB module refers to Table 2.5; page 2.47 "Tropical forest" of Volume 4, Chapter 2, of the IPCC 2006 Inventory Guidelines: For CH_4: 6.8 - 2 = 4.8 g kg-1 dry matter burnt (conservative); For N_2O: 0.20 g kg-1 dry matter burnt (unique value proposed): https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/4_Volume4/V4_02_Ch2_Generic.pdf (last visited in 20/06/2022).
[93] IPCC, 2022: Annex II: Definitions, Units and Conventions [Al Khourdajie et al. (eds)]. In IPCC, 2022: Climate Change 2022: Mitigation of Climate Change. Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [P.R. Shukla et al. (eds.)]. Cambridge University Press, Cambridge, UK and New York, NY, USA. https://www.ipcc.ch/report/ar6/wg3/downloads/report/IPCC_AR6_WGIII_Annex-II.pdf

2023	0.00	0.26	0.00	4.85
2024	0.00	0.26	0.00	4.85
2025	0.00	0.26	0.00	4.85
2026	0.00	0.26	0.00	4.85
2027	92.52	92.78	2,375.06	2,379.92
2028	75.48	168.26	1,882.98	4,262.90
2029	0.00	168.26	0.00	4,262.90
2030	0.00	168.26	0.00	4,262.90
2031	0.00	168.26	0.00	4,262.90
2032	32.85	201.11	925.20	5,188.10
2033	30.51	231.62	854.68	6,042.79
2034	99.86	331.49	2,461.07	8,503.86
2035	2.88	334.37	82.07	8,585.93
2036	0.00	334.37	0.00	8,585.93
2037	34.38	368.75	979.73	9,565.67
2038	9.20	377.94	170.85	9,736.51
2039	45.66	423.60	1,147.24	10,883.76
2040	116.06	539.66	2,809.34	13,693.10
2041	34.93	574.59	675.91	14,369.01
2042	3.49	578.07	64.73	14,433.74
2043	6.51	584.58	120.84	14,554.58
2044	0.00	584.58	0.00	14,554.58
2045	27.54	612.11	519.06	15,073.64
2046	0.00	612.11	0.00	15,073.64
2047	2.41	614.52	48.28	15,121.91
2048	0.00	614.52	0.00	15,121.91
2049	0.00	614.52	0.00	15,121.91
2050	35.64	650.16	874.70	15,996.61
2051	0.00	650.16	0.00	15,996.61
2052	2.32	652.48	43.16	16,039.78

Table 28. Summary of biomass burning emissions (N_2O) in the sum of strata within the Project Area in the baseline case – AUD component

Avoided Unplanned Deforestation Biomass Burn Emissions - N_2O Sum of Strata				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	0.26	0.26	1.35	1.35
2023	0.00	0.26	0.00	1.35
2024	0.00	0.26	0.00	1.35
2025	0.00	0.26	0.00	1.35
2026	0.00	0.26	0.00	1.35
2027	92.52	92.78	661.13	662.48
2028	75.48	168.26	524.15	1,186.63
2029	0.00	168.26	0.00	1,186.63

2030	0.00	168.26	0.00	1,186.63
2031	0.00	168.26	0.00	1,186.63
2032	32.85	201.11	257.54	1,444.17
2033	30.51	231.62	237.91	1,682.08
2034	99.86	331.49	685.07	2,367.15
2035	2.88	334.37	22.85	2,389.99
2036	0.00	334.37	0.00	2,389.99
2037	34.38	368.75	272.72	2,662.71
2038	9.20	377.94	47.56	2,710.27
2039	45.66	423.60	319.35	3,029.62
2040	116.06	539.66	782.01	3,811.63
2041	34.93	574.59	188.15	3,999.78
2042	3.49	578.07	18.02	4,017.79
2043	6.51	584.58	33.64	4,051.43
2044	0.00	584.58	0.00	4,051.43
2045	27.54	612.11	144.49	4,195.92
2046	0.00	612.11	0.00	4,195.92
2047	2.41	614.52	13.44	4,209.36
2048	0.00	614.52	0.00	4,209.36
2049	0.00	614.52	0.00	4,209.36
2050	35.64	650.16	243.48	4,452.84
2051	0.00	650.16	0.00	4,452.84
2052	2.32	652.48	12.01	4,464.85

Table 29. Summary of total biomass burning emissions (CH_4 and N_2O) in the sum of strata within the Project Area in the baseline case – AUD component.

Avoided Unplanned Deforestation Biomass Burn Emissions Sum of Strata				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	0.26	0.26	6.20	6.20
2023	0.00	0.26	0.00	6.20
2024	0.00	0.26	0.00	6.20
2025	0.00	0.26	0.00	6.20
2026	0.00	0.26	0.00	6.20
2027	92.52	92.78	3,036.19	3,042.40
2028	75.48	168.26	2,407.13	5,449.53
2029	0.00	168.26	0.00	5,449.53
2030	0.00	168.26	0.00	5,449.53
2031	0.00	168.26	0.00	5,449.53
2032	32.85	201.11	1,182.74	6,632.27
2033	30.51	231.62	1,092.60	7,724.86
2034	99.86	331.49	3,146.14	10,871.00

2035	2.88	334.37	104.92	10,975.92
2036	0.00	334.37	0.00	10,975.92
2037	34.38	368.75	1,252.45	12,228.38
2038	9.20	377.94	218.41	12,446.78
2039	45.66	423.60	1,466.59	13,913.37
2040	116.06	539.66	3,591.36	17,504.73
2041	34.93	574.59	864.05	18,368.78
2042	3.49	578.07	82.75	18,451.53
2043	6.51	584.58	154.48	18,606.01
2044	0.00	584.58	0.00	18,606.01
2045	27.54	612.11	663.54	19,269.55
2046	0.00	612.11	0.00	19,269.55
2047	2.41	614.52	61.71	19,331.27
2048	0.00	614.52	0.00	19,331.27
2049	0.00	614.52	0.00	19,331.27
2050	35.64	650.16	1,118.19	20,449.45
2051	0.00	650.16	0.00	20,449.45
2052	2.32	652.48	55.18	20,504.63

Below the biomass burning emissions in the baseline continue with those corresponding to the APD component.

Table 30. Summary of biomass burning emissions (CH_4) in the sum of strata within the Project Area in the baseline case – APD component

Avoided Planned Deforestation Biomass Burn Emissions - CH_4 Sum of Strata				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	501.70	501.70	14,327.57	14,327.57
2023	0.00	501.70	0.00	14,327.57
2024	940.51	1,442.20	26,801.80	41,129.38
2025	941.29	2,383.49	26,824.19	67,953.56
2026	234.49	2,617.98	6,597.44	74,551.00
2027	0.00	2,617.98	0.00	74,551.00
2028	0.00	2,617.98	0.00	74,551.00
2029	0.00	2,617.98	0.00	74,551.00
2030	0.00	2,617.98	0.00	74,551.00
2031	0.00	2,617.98	0.00	74,551.00
2032	0.00	2,617.98	0.00	74,551.00
2033	0.00	2,617.98	0.00	74,551.00
2034	0.00	2,617.98	0.00	74,551.00
2035	0.00	2,617.98	0.00	74,551.00
2036	0.00	2,617.98	0.00	74,551.00
2037	0.00	2,617.98	0.00	74,551.00
2038	0.00	2,617.98	0.00	74,551.00

2039	0.00	2,617.98	0.00	74,551.00
2040	0.00	2,617.98	0.00	74,551.00
2041	0.00	2,617.98	0.00	74,551.00
2042	0.00	2,617.98	0.00	74,551.00
2043	0.00	2,617.98	0.00	74,551.00
2044	0.00	2,617.98	0.00	74,551.00
2045	0.00	2,617.98	0.00	74,551.00
2046	0.00	2,617.98	0.00	74,551.00
2047	0.00	2,617.98	0.00	74,551.00
2048	0.00	2,617.98	0.00	74,551.00
2049	0.00	2,617.98	0.00	74,551.00
2050	0.00	2,617.98	0.00	74,551.00
2051	0.00	2,617.98	0.00	74,551.00
2052	0.00	2,617.98	0.00	74,551.00

Table 31. Summary of biomass burning emissions (N_2O) in the sum of strata within the Project Area in the baseline case – APD component

Avoided Planned Deforestation Biomass Burn Emissions Sum of Strata				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	501.70	501.70	18,315.81	18,315.81
2023	929.97	1,431.66	33,886.67	52,202.48
2024	940.51	2,372.17	34,262.39	86,464.87
2025	941.29	3,313.46	34,291.00	120,755.88
2026	234.49	3,547.95	8,433.91	129,189.79
2027	0.00	3,547.95	0.00	129,189.79
2028	0.00	3,547.95	0.00	129,189.79
2029	0.00	3,547.95	0.00	129,189.79
2030	0.00	3,547.95	0.00	129,189.79
2031	0.00	3,547.95	0.00	129,189.79
2032	0.00	3,547.95	0.00	129,189.79
2033	0.00	3,547.95	0.00	129,189.79
2034	0.00	3,547.95	0.00	129,189.79
2035	0.00	3,547.95	0.00	129,189.79
2036	0.00	3,547.95	0.00	129,189.79
2037	0.00	3,547.95	0.00	129,189.79
2038	0.00	3,547.95	0.00	129,189.79
2039	0.00	3,547.95	0.00	129,189.79
2040	0.00	3,547.95	0.00	129,189.79
2041	0.00	3,547.95	0.00	129,189.79
2042	0.00	3,547.95	0.00	129,189.79
2043	0.00	3,547.95	0.00	129,189.79
2044	0.00	3,547.95	0.00	129,189.79
2045	0.00	3,547.95	0.00	129,189.79
2046	0.00	3,547.95	0.00	129,189.79

2047	0.00	3,547.95	0.00	129,189.79
2048	0.00	3,547.95	0.00	129,189.79
2049	0.00	3,547.95	0.00	129,189.79
2050	0.00	3,547.95	0.00	129,189.79
2051	0.00	3,547.95	0.00	129,189.79
2052	0.00	3,547.95	0.00	129,189.79

Table 32. Summary of total biomass burning emissions (CH_4 and N_2O) in the sum of strata within the Project Area in the baseline case – APD component.

Avoided Planned Deforestation Biomass Burn Emissions Sum of Strata				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	501.70	501.70	18,315.81	18,315.81
2023	0.00	501.70	0.00	18,315.81
2024	940.51	1,442.20	34,262.39	52,578.21
2025	941.29	2,383.49	34,291.00	86,869.21
2026	234.49	2,617.98	8,433.91	95,303.12
2027	0.00	2,617.98	0.00	95,303.12
2028	0.00	2,617.98	0.00	95,303.12
2029	0.00	2,617.98	0.00	95,303.12
2030	0.00	2,617.98	0.00	95,303.12
2031	0.00	2,617.98	0.00	95,303.12
2032	0.00	2,617.98	0.00	95,303.12
2033	0.00	2,617.98	0.00	95,303.12
2034	0.00	2,617.98	0.00	95,303.12
2035	0.00	2,617.98	0.00	95,303.12
2036	0.00	2,617.98	0.00	95,303.12
2037	0.00	2,617.98	0.00	95,303.12
2038	0.00	2,617.98	0.00	95,303.12
2039	0.00	2,617.98	0.00	95,303.12
2040	0.00	2,617.98	0.00	95,303.12
2041	0.00	2,617.98	0.00	95,303.12
2042	0.00	2,617.98	0.00	95,303.12
2043	0.00	2,617.98	0.00	95,303.12
2044	0.00	2,617.98	0.00	95,303.12
2045	0.00	2,617.98	0.00	95,303.12
2046	0.00	2,617.98	0.00	95,303.12
2047	0.00	2,617.98	0.00	95,303.12
2048	0.00	2,617.98	0.00	95,303.12
2049	0.00	2,617.98	0.00	95,303.12
2050	0.00	2,617.98	0.00	95,303.12
2051	0.00	2,617.98	0.00	95,303.12
2052	0.00	2,617.98	0.00	95,303.12

Wood products carbon pool in the baseline

As described in the PD, to estimate the biomass carbon of the commercial volume extracted in the process of deforestation, the following equation was applied, which is applicable both for IPOÁ's APD and AUD components, according to "Option 2: Commercial inventory estimation", as recommended in the CP-W:

$$C_{XB,i} = C_{AB_{tree},i} * \frac{1}{BCEF} * Pcom_i$$

Where:

$C_{XB,i}$	Mean stock of extracted biomass carbon from stratum *i*; t CO_2-e ha^{-1}
$C_{ABtree,i}$	Mean aboveground biomass carbon stock in stratum *i*; tCO_2-e ha^{-1}
$BCEF$	Biomass conversion and expansion factor (BCEF) for conversion of merchantable volume to total aboveground tree biomasss; dimesnionless
$Pcom_i$	Commercial volume as a percent of total aboveground volume in stratum i; dimensionless
i	1,2,3, … *M* strata

For calculation of P_{COMi} parameter, the following parameters and sources were used: volume of merchantable wood in exploitation (35.08 m^3/ha) from FSM_REDD_MR2_04_03_2021[94]; Dmm value (0.59 t d.m.m^{-3}) from Brown et al 1989; above ground carbon stock from MCTI 2015[95]; CF value (0.47 t Ct-1 d.m) from IPCC 2006 V4_04_Ch4_Forest_Land[96]. For the BCEF parameter, the average of the three proposed factors in CP-AB[97] was applied.

To calculate the carbon entering the wood products pool at the time of deforestation, the following equation was applied.

[94] https://registry.verra.org/mymodule/ProjectDoc/Project_ViewFile.asp?FileID=52137&IDKEY=9lksjoiuwqowrnoiuomnckjashoufifmln902309ksdflku098l71896923
[95] http://redd.mma.gov.br/images/FREL/RR_LULUCF_Mudana-de-Uso-e-Floresta.pdf
[96] https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/4_Volume4/V4_04_Ch4_Forest_Land.pdf
[97] https://verra.org/wp-content/uploads/2017/11/VMD0001v1.1.pdf

$$C_{WP,i} = \sum_{ty=s,w,oir,p,o} C_{XB,ty,i} * (1 - WW_{ty})$$

Where:

$C_{WP,i}$	Carbon stock entering the wood products pool from stratum i; t CO_2-e ha^{-1}
$C_{XB,ty,i}$	Mean stock of extracted biomass carbon by class of wood product ty from stratum i; t CO_2-e ha^{-1}
WW_{ty}	Wood waste. The fraction immediately emitted through mill inefficiency by class of wood product ty; dimensionless (0.24 for developing countries; Winjum et al. 1998 cited by CP-W)
Ty	'Wood product class – defined here as sawnwood (s)
i	1,2, 3,… *M* strata

To calculate the amount of wood products entering the pool at the time of deforestation (CWP,i, calculated in C-WP) that is expected to be emitted over a 100-year timeframe, the following equation was applied.

$$C_{WP100,i} = C_{WP,i} - C_{WP,i} * (1 - SLFp) * (1 - Ofp)$$

Where:

$C_{WP100,i}$	Carbon stock entering the wood products pool at the time of deforestation that is expected to be emitted over 100-years from stratum i; t CO_2-e ha^{-1}
$C_{WP,i}$	Carbon stock entering wood products pool at time of deforestation from stratum i; t CO_2-e ha^{-1}

SLF_{ty} Fraction of wood products that will be emitted to the atmosphere within 5 years of timber harvest by class of wood product ty; dimensionless (0.2 for sawnwood; Winjum et al. 1998 cited by CP-W)

OF_{ty} Fraction of wood products that will be emitted to the atmosphere between 5 and 100 years of timber harvest by class of wood product ty; dimensionless (0.80 for sawnwood in tropical forests; Winjum et al. 1998 cited by CP-W)

Ty 'Wood product class – defined here as sawnwood (s)

i 1,2, 3,… *M* strata

The parameters used in calculation of wood products carbon pool in the baseline, are displayed in the table below.

Table 33. Parameters used to calculate of wood products carbon pool in the baseline scenario

Parameter	Value	Unit
Cab Ds	515.86	t C m^{-3}
Cab Db	517.37	t C m^{-3}
Cab Da	587.33	t C m^{-3}
Cab Pa	336.23	t C m^{-3}
PCOM Ds	0.07	Dimensionless
PCOM Db	0.07	Dimensionless
PCOM Da	0.06	Dimensionless
PCOM Pa	0.11	Dimensionless
BCEF	1.32	Dimensionless
WWp	0.24	Dimensionless
SLFp	0.20	Dimensionless
Ofp	0.80	Dimensionless

The results of the estimates regarding the wood products carbon pool in the baseline are demonstrated in the table below, by component: AUD first followed APD. The wood products carbon pool for each of the four strata in section 3.2.1 of the PD, but are presented here only in the sum of strata, for the sake of brevity.

Table 34. Summary of wood products in the sum of strata within the Project Area in the baseline case – AUD component

Avoided Unplanned Deforestation

Wood products carbon pool in baseline Sum of strata					
Year	ha/year	ha (cumulative)	CWP tCO_2/year	CWP tCO_2 (cumulative)	CWP_{100} tCO_2
2022	0.26	0.26	5.37	5.37	4.51
2023	0.00	0.26	0.00	5.37	0.00
2024	0.00	0.26	0.00	5.37	0.00
2025	0.00	0.26	0.00	5.37	0.00
2026	0.00	0.26	0.00	5.37	0.00
2027	92.52	92.78	1,899.99	1,905.36	1,595.99
2028	75.48	168.26	1,550.08	3,455.44	1,302.07
2029	0.00	168.26	0.00	3,455.44	0.00
2030	0.00	168.26	0.00	3,455.44	0.00
2031	0.00	168.26	0.00	3,455.44	0.00
2032	32.85	201.11	674.61	4,130.05	566.68
2033	30.51	231.62	626.56	4,756.62	526.31
2034	99.86	331.49	2,050.85	6,807.46	1,722.71
2035	2.88	334.37	59.14	6,866.60	49.68
2036	0.00	334.37	0.00	6,866.60	0.00
2037	34.38	368.75	706.04	7,572.64	593.07
2038	9.20	377.94	188.90	7,761.54	158.67
2039	45.66	423.60	937.65	8,699.18	787.62
2040	116.06	539.66	2,383.35	11,082.53	2,002.01
2041	34.93	574.59	717.29	11,799.82	602.52
2042	3.49	578.07	71.57	11,871.39	60.12
2043	6.51	584.58	133.61	12,004.99	112.23
2044	0.00	584.58	0.00	12,004.99	0.00
2045	27.54	612.11	565.51	12,570.50	475.03
2046	0.00	612.11	0.00	12,570.50	0.00
2047	2.41	614.52	49.43	12,619.93	41.52
2048	0.00	614.52	0.00	12,619.93	0.00
2049	0.00	614.52	0.00	12,619.93	0.00
2050	35.64	650.16	731.91	13,351.84	614.81
2051	0.00	650.16	0.00	13,351.84	0.00
2052	2.32	652.48	47.72	13,399.56	40.09

Table 35. Summary of wood products in the sum of strata within the Project Area in the baseline case – APD component

Avoided Planned Deforestation Wood products carbon pool in baseline Sum of strata					
Year	ha/year	ha (cumulative)	CWP tCO_2/year	CWP tCO_2 (cumulative)	CWP_{100} tCO_2
2022	501.70	501.70	10,302.93	10,302.93	8,654.46
2023	0.00	501.70	0.00	10,302.93	0.00
2024	940.51	1,442.20	19,314.45	29,617.38	16,224.13
2025	941.29	2,383.49	19,330.58	48,947.95	16,237.68
2026	234.49	2,617.98	4,815.46	53,763.41	4,044.98
2027	0.00	2,617.98	0.00	53,763.41	0.00
2028	0.00	2,617.98	0.00	53,763.41	0.00
2029	0.00	2,617.98	0.00	53,763.41	0.00
2030	0.00	2,617.98	0.00	53,763.41	0.00
2031	0.00	2,617.98	0.00	53,763.41	0.00
2032	0.00	2,617.98	0.00	53,763.41	0.00

2033	0.00	2,617.98	0.00	53,763.41	0.00
2034	0.00	2,617.98	0.00	53,763.41	0.00
2035	0.00	2,617.98	0.00	53,763.41	0.00
2036	0.00	2,617.98	0.00	53,763.41	0.00
2037	0.00	2,617.98	0.00	53,763.41	0.00
2038	0.00	2,617.98	0.00	53,763.41	0.00
2039	0.00	2,617.98	0.00	53,763.41	0.00
2040	0.00	2,617.98	0.00	53,763.41	0.00
2041	0.00	2,617.98	0.00	53,763.41	0.00
2042	0.00	2,617.98	0.00	53,763.41	0.00
2043	0.00	2,617.98	0.00	53,763.41	0.00
2044	0.00	2,617.98	0.00	53,763.41	0.00
2045	0.00	2,617.98	0.00	53,763.41	0.00
2046	0.00	2,617.98	0.00	53,763.41	0.00
2047	0.00	2,617.98	0.00	53,763.41	0.00
2048	0.00	2,617.98	0.00	53,763.41	0.00
2049	0.00	2,617.98	0.00	53,763.41	0.00
2050	0.00	2,617.98	0.00	53,763.41	0.00
2051	0.00	2,617.98	0.00	53,763.41	0.00
2052	0.00	2,617.98	0.00	53,763.41	0.00

Pasture carbon pools in the baseline

Regarding calculation of the carbon pool remaining in pasturelands after deforestation, a conservative value from the Third edition of the National GHG Inventory (Ministério da Ciência, Tecnologia e Inovação, 2015)[98] of 27.8 tCO_2/ha ($CABnon{-}treepost$,i + $CBBnon{-}tree,post$,i) was applied[99] for both AUD and APD components. The proportion of baseline deforestation converted to pasture was defined as 100%. The table below summarizes the results obtained for pasture carbon pools in the baseline scenario, over the 30 years of the project lifetime, per component: AUD and APD. Here, the results are in the sum of strata only, with full details per stratum are available in the PD, section 3.2.1.

Table 36. Summary of pasture carbon pool in the sum of strata within the Project Area in the baseline case – AUD component

Avoided Unplanned Deforestation Annual baseline pasture carbonpool for the Project Area Sum of Strata				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	0.26	0.26	7.25	7.25
2023	0.00	0.26	0.00	7.25

[98] https://repositorio.mcti.gov.br/handle/mctic/5279
[99] Source: http://redd.mma.gov.br/images/FREL/RR_LULUCF_Mudana-de-Uso-e-Floresta.pdf (Table 5, is the value of 7.57 tC/ha, for the Amazon biome, multiplied by the 44/12 conversion factor)

2024	0.00	0.26	0.00	7.25
2025	0.00	0.26	0.00	7.25
2026	0.00	0.26	0.00	7.25
2027	92.52	92.78	2,568.02	2,575.27
2028	75.48	168.26	2,095.09	4,670.36
2029	0.00	168.26	0.00	4,670.36
2030	0.00	168.26	0.00	4,670.36
2031	0.00	168.26	0.00	4,670.36
2032	32.85	201.11	911.81	5,582.16
2033	30.51	231.62	846.86	6,429.02
2034	99.86	331.49	2,771.91	9,200.93
2035	2.88	334.37	79.94	9,280.87
2036	0.00	334.37	0.00	9,280.87
2037	34.38	368.75	954.27	10,235.15
2038	9.20	377.94	255.31	10,490.46
2039	45.66	423.60	1,267.32	11,757.78
2040	116.06	539.66	3,221.32	14,979.10
2041	34.93	574.59	969.48	15,948.58
2042	3.49	578.07	96.73	16,045.32
2043	6.51	584.58	180.58	16,225.90
2044	0.00	584.58	0.00	16,225.90
2045	27.54	612.11	764.34	16,990.24
2046	0.00	612.11	0.00	16,990.24
2047	2.41	614.52	66.80	17,057.04
2048	0.00	614.52	0.00	17,057.04
2049	0.00	614.52	0.00	17,057.04
2050	35.64	650.16	989.25	18,046.29
2051	0.00	650.16	0.00	18,046.29
2052	2.32	652.48	64.50	18,110.79

Table 37. Summary of pasture carbon pool in the sum of strata within the Project Area in the baseline case – APD component

Avoided Planned Deforestation Annual baseline pasture carbonpool for the Project Area Sum of Strata				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2 (cumulative)**
2022	501.70	501.70	13,925.40	13,925.40
2023	0.00	501.70	0.00	13,925.40
2024	940.51	1,442.20	26,105.32	40,030.72
2025	941.29	2,383.49	26,127.12	66,157.84
2026	234.49	2,617.98	6,508.55	72,666.39
2027	0.00	2,617.98	0.00	72,666.39
2028	0.00	2,617.98	0.00	72,666.39
2029	0.00	2,617.98	0.00	72,666.39
2030	0.00	2,617.98	0.00	72,666.39
2031	0.00	2,617.98	0.00	72,666.39
2032	0.00	2,617.98	0.00	72,666.39
2033	0.00	2,617.98	0.00	72,666.39
2034	0.00	2,617.98	0.00	72,666.39
2035	0.00	2,617.98	0.00	72,666.39

2036	0.00	2,617.98	0.00	72,666.39
2037	0.00	2,617.98	0.00	72,666.39
2038	0.00	2,617.98	0.00	72,666.39
2039	0.00	2,617.98	0.00	72,666.39
2040	0.00	2,617.98	0.00	72,666.39
2041	0.00	2,617.98	0.00	72,666.39
2042	0.00	2,617.98	0.00	72,666.39
2043	0.00	2,617.98	0.00	72,666.39
2044	0.00	2,617.98	0.00	72,666.39
2045	0.00	2,617.98	0.00	72,666.39
2046	0.00	2,617.98	0.00	72,666.39
2047	0.00	2,617.98	0.00	72,666.39
2048	0.00	2,617.98	0.00	72,666.39
2049	0.00	2,617.98	0.00	72,666.39
2050	0.00	2,617.98	0.00	72,666.39
2051	0.00	2,617.98	0.00	72,666.39
2052	0.00	2,617.98	0.00	72,666.39

Uncertainty

The X-UNC Estimation of Uncertainty for REDD+ Project Activities module (VMD017) was duly applied to assess the following sources of uncertainty applicable to this project: i) Uncertainty in Projection of Baseline Rate of Deforestation or Degradation; ii): Uncertainty of Emissions and Removals in Project Area in Baseline Scenario.

The result was 2.41% of uncertainty. Given that uncertainty was estimated to be under 15% for all forest types, no deduction resulted for uncertainty, as the allowable uncertainty in this methodology REDD+ MF is +/- 15% at a 95% confidence level. This is fully presented in section 3.2.1 of the PD.

***Ex-ante* project emissions**

Finally, *ex-ante* project emissions were duly estimated in the PD, applying the VM0007 Methodology Framework guidelines. They were set at a conservative level of 10%, full details being provided in 3.2.1 of the PD.

The table below summarizes *ex-ante* in the baseline scenario for the sum of strata only, full details per stratum being available in the aforementioned section of the PD.

Table 38. Summary of annual ex ante project emissions estimates in the sum of strata within the Project Area

Avoided Unplanned Deforestation Annual ex ante emission of deforestation in PA Sum of Strata				
Year	**ha/Year**	**ha (cumulative)**	**tCO_2/Year**	**tCO_2**

2022	0.03	0.03	8.97	8.97
2023	0.00	0.03	0.19	9.16
2024	0.00	0.03	0.19	9.34
2025	0.00	0.03	0.19	9.53
2026	0.00	0.03	0.19	9.71
2027	9.25	9.28	4,420.34	4,430.06
2028	7.55	16.83	3,623.50	8,053.56
2029	0.00	16.83	214.88	8,268.43
2030	0.00	16.83	214.88	8,483.31
2031	0.00	16.83	214.88	8,698.19
2032	3.29	20.11	1,939.28	10,637.47
2033	3.05	23.16	1,857.45	12,494.92
2034	9.99	33.15	4,887.12	17,382.04
2035	0.29	33.44	585.03	17,967.07
2036	0.00	33.44	436.46	18,403.53
2037	3.44	36.87	2,150.25	20,553.78
2038	0.92	37.79	605.09	21,158.87
2039	4.57	42.36	2,430.09	23,588.96
2040	11.61	53.97	5,576.09	29,165.06
2041	3.49	57.46	1,741.17	30,906.22
2042	0.35	57.81	590.34	31,496.57
2043	0.65	58.46	653.35	32,149.92
2044	0.00	58.46	321.47	32,471.38
2045	2.75	61.21	1,277.07	33,748.46
2046	0.00	61.21	337.46	34,085.92
2047	0.24	61.45	368.82	34,454.74
2048	0.00	61.45	269.18	34,723.92
2049	0.00	61.45	215.61	34,939.53
2050	3.56	65.02	1,714.95	36,654.48
2051	0.00	65.02	107.48	36,761.96
2052	0.23	65.25	182.25	36,944.21

No emissions from logging are predicted in the project and therefore were not factored into baseline calculations, as presented in the PD, section 3.2.1.

Regarding other types of greenhouse gasses in the project scenario, in accordance with the methodology's VMD0013 module, "5 Procedures" section: "Where used in the baseline, accounting must occur under both the baseline and project scenarios in both the Project Area and in the Leakage Belt". Therefore, emissions of CH_4 and N_2O were accounted for *ex-ante* in the project scenario, in order to match the baseline scenario, as detailed in the tables below. In the present project, such GHG emissions from biomass burning are considered to originate from the following category, provided by the methodology: "2. Periodical burning of grassland or agricultural land after deforestation".

The applicable equation is as follows:

$$E_{biomassburn,i,t} = \sum_{g=1}^{G} \left(\left(\left(A_{burn,i,t} \times B_{i,t} \times COMF_i \times G_{g,i} \right) \times 10^{-3} \right) \times GWP_g \right) \qquad (1)$$

Where:

$E_{biomassburn,i,t}$ Greenhouse gas emissions due to biomass burning in stratum *i* in year *t* of each GHG (CO_2, CH_4, N_2O) (t CO_{2e})

$A_{burn,i,t}$ Area burnt for stratum *i* in year *t* (ha)

$B_{i,t}$ Average aboveground biomass stock before burning stratum *i,* year (t d.m. ha^{-1})

$COMF_i$ Combustion factor for stratum *i* (unitless)

$G_{g,i}$ Emission factor for stratum *i* for gas g (kg t^{-1} d.m. burnt)

GWP_g Global warming potential for gas *g (t CO_2/t gas g)*

g 1, 2, 3 ... G greenhouse gases including carbon dioxide[1], methane and nitrous oxide (unitless)

i 1, 2, 3 … *M* strata (unitless)

t 1, 2, 3, … *t** time elapsed since the start of the project activity (years)

Ex-post GHG emissions from burning in the project scenario are duly taken into account in the present Monitoring Period, as detailed in section 3.2.2. below.

The *ex-ante* estimate of emissions from burning in the project scenario are presented below, only for the sum of strata. The complete data per stratum are available in the PD, section 3.2.2.

Table 39. Projection of ex-ante estimated project GHG emissions from biomass and peat burning (E-BPB) – sum of strata.

Ex-ante Project Emissions Biomass Burning Emissions Sum of Strata				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	0.03	0.03	0.62	0.62
2023	0.00	0.03	0.00	0.62
2024	0.00	0.03	0.00	0.62
2025	0.00	0.03	0.00	0.62

2026	0.00	0.03	0.00	0.62
2027	3.30	3.32	303.62	304.24
2028	3.19	6.51	240.71	544.95
2029	0.00	6.51	0.00	544.95
2030	0.00	6.51	0.00	544.95
2031	0.00	6.51	0.00	544.95
2032	0.43	6.94	118.27	663.23
2033	0.44	7.37	109.26	772.49
2034	4.49	11.86	314.61	1,087.10
2035	0.03	11.89	10.49	1,097.59
2036	0.00	11.89	0.00	1,097.59
2037	0.34	12.24	125.25	1,222.84
2038	0.92	13.16	21.84	1,244.68
2039	1.85	15.01	146.66	1,391.34
2040	5.68	20.68	359.14	1,750.47
2041	3.25	23.93	86.41	1,836.88
2042	0.35	24.28	8.27	1,845.15
2043	0.65	24.93	15.45	1,860.60
2044	0.00	24.93	0.00	1,860.60
2045	2.68	27.62	66.35	1,926.96
2046	0.00	27.62	0.00	1,926.96
2047	0.21	27.82	6.17	1,933.13
2048	0.00	27.82	0.00	1,933.13
2049	0.00	27.82	0.00	1,933.13
2050	1.63	29.46	111.82	2,044.95
2051	0.00	29.46	0.00	2,044.95
2052	0.23	29.69	5.52	2,050.46

Baseline Leakage Calculations

As detailed in section 3.2.3 of the PD, projections of deforestation and baseline emissions were calculated in the Leakage Belt using the same criteria as in the Project Area, following the BL-UP and LK-ASU modules. The results are presented below, only for the sum of strata, with full details per stratum available in 3.2.3 of the PD.

Table 40. Summary of gross baseline emissions from deforestation within the Leakage Belt, sum of strata.

Gross baseline emissions from deforestation in Leakage Belt Sum of Strata				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	34.58	34.58	18,370.52	18,370.52

2026	0.00	34.58	533.28	18,903.80
2027	20.22	54.80	11,275.36	30,179.16
2028	49.64	104.43	22,945.83	53,124.99
2029	29.74	134.18	17,136.02	70,261.02
2030	30.87	165.05	16,953.32	87,214.33
2031	60.23	225.28	32,240.64	119,454.97
2032	31.22	256.50	19,721.27	139,176.25
2033	60.57	317.07	35,793.46	174,969.71
2034	87.77	404.84	48,137.87	223,107.58
2035	63.29	468.13	38,887.68	261,995.25
2036	60.27	528.40	37,953.01	299,948.26
2037	4.10	532.50	9,041.60	308,989.86
2038	48.24	580.75	29,228.67	338,218.54
2039	102.23	682.98	54,109.85	392,328.38
2040	66.03	749.00	41,414.44	433,742.83
2041	32.10	781.11	22,344.96	456,087.78
2042	34.74	815.85	23,503.84	479,591.62
2043	59.10	874.95	34,443.90	514,035.52
2044	57.69	932.64	37,301.71	551,337.23
2045	16.11	948.75	14,294.13	565,631.37
2046	0.00	948.75	5,993.66	571,625.03
2047	29.63	978.38	20,904.29	592,529.32
2048	10.62	989.00	11,337.83	603,867.15
2049	16.05	1,005.05	11,115.99	614,983.14
2050	2.45	1,007.50	5,179.55	620,162.69
2051	26.07	1,033.57	17,356.71	637,519.41
2052	8.94	1,042.51	8,237.76	645,757.16

Secondly, leakage displaced from the Project Area to the Leakage Belt was duly accounted for, applying (Step 3 - LK-ASU). These were set at 10% of baseline deforestation levels within the PA, as described in section 3.2.3 of the PD. The results are presented below, only for the sum of strata, with full details per stratum available in 3.2.3 of the PD.

Table 41. Estimation of Unplanned Deforestation Displaced from the Project Area to the Leakage Belt (sum of strata)

Avoided Unplanned Deforestation Annual ex ante deforestation displaced from PA to LB Sum of Strata				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	0.03	0.03	8.97	8.97
2023	0.00	0.03	0.19	9.16
2024	0.00	0.03	0.19	9.34
2025	0.00	0.03	0.19	9.53
2026	0.00	0.03	0.19	9.71
2027	9.25	9.28	4,420.34	4,430.06
2028	7.55	16.83	3,623.50	8,053.56
2029	0.00	16.83	214.88	8,268.43
2030	0.00	16.83	214.88	8,483.31
2031	0.00	16.83	214.88	8,698.19
2032	3.29	20.11	1,939.28	10,637.47

2033	3.05	23.16	1,857.45	12,494.92
2034	9.99	33.15	4,887.12	17,382.04
2035	0.29	33.44	585.03	17,967.07
2036	0.00	33.44	436.46	18,403.53
2037	3.44	36.87	2,150.25	20,553.78
2038	0.92	37.79	605.09	21,158.87
2039	4.57	42.36	2,430.09	23,588.96
2040	11.61	53.97	5,576.09	29,165.06
2041	3.49	57.46	1,741.17	30,906.22
2042	0.35	57.81	590.34	31,496.57
2043	0.65	58.46	653.35	32,149.92
2044	0.00	58.46	321.47	32,471.38
2045	2.75	61.21	1,277.07	33,748.46
2046	0.00	61.21	337.46	34,085.92
2047	0.24	61.45	368.82	34,454.74
2048	0.00	61.45	269.18	34,723.92
2049	0.00	61.45	215.61	34,939.53
2050	3.56	65.02	1,714.95	36,654.48
2051	0.00	65.02	107.48	36,761.96
2052	0.23	65.25	182.25	36,944.21

It is further noted that Market Leakage and leakage outside the Leakage Belt were duly accounted for in *ex-ante* baseline calculations, with full details being available in PD section 3.2.3. All applicable categories of leakage applied the corresponding burning emissions.

3.2.2 Project Emissions

The Figure entitled "Deforestation in PA during the monitoring period" in section 3.1.3 displays the map of the 11.61 has of deforestation detected in MR1, which constitute Project emissions. This deforestation is primarily attributable to the invasive deforestation event (approximately 8ha or 69% of total deforestation detected in the PA during MR1), involving a road being cut through the Maanaim property. The remaining deforestation is due to community activity to clear subsistence farmland. Both types of occurrence were fully evidenced at audit , as evidenced at audit[100].

Applying the CDM tool (T-SIG) for significance of GHG emissions, as indicated by the VM0007 module, these project emissions would be eligible for omission from calculations due to being insignificant, according to the following rationale:

[100] See folders "07. Remote Monitoring" and "08. Deforestation"

"The sum of decreases in carbon pools and increases in emissions that may be neglected shall be less than 5% of the total decreases in carbon pools and increases in emissions".

This is because the project emissions totalize (see tables below) 5,956.87 tCO_2e from unavoided deforestation in the PA and 411.91 tCO_2e from associated burning emissions: their summed percentage of Net GHG emission reductions or removals (237,355 tCO2e) is 2.7%. However, the Ipoá project chose to deduct these emissions from net GHG emission reductions for conservativeness and integrity purposes.

The calculation of net change in carbon stock because of with-project deforestation in the Project Area was made according to the formula below, from the VMD0015 MREDD module. The following tables summarize the results obtained for the total net emissions in the carbon stock in the project area.

$$\Delta C_{P,DefPA,i,t} = \sum_{u=1}^{U} \left(A_{DefPA,u,i,t} * \Delta C_{pools,P,Def,u,i,t} \right)$$

$$\Delta C_{P,DefPA,i,t} = \sum_{u=1}^{U} \left(A_{DefPA,u,i,t} * \Delta C_{pools,P,Def,u,i,t} \right) \quad (3)$$

$$\Delta C_{P,DefLB,i,t} = \sum_{u=1}^{U} \left(A_{DefLB,u,i,t} * \Delta C_{pools,P,Def,u,i,t} \right) \quad (4)$$

The tables below show deforestation detected within the project area over the present monitoring period.

Table 42. Ex-post emissions from unplanned deforestation in PA per stratum (1)

Ex-post Project Emissions Annual ex post emissions from deforestation in PA Dense Tropical Submontane Rainforest				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	3.05	3.05	1,932.82	1,932.82
2023	3.93	6.98	2,490.48	4,423.30
Total	**6.98**	**6.98**	**4,423.30**	**4,423.30**

Table 43. Ex-post emissions from unplanned deforestation in PA per stratum (2)

Ex-post Project Emissions Annual ex post emissions from deforestation in PA Dense Tropical Lowland Rainforest				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	1.96	1.96	1,242.07	1,242.07
2023	2.51	4.47	1,590.61	2,832.68
Total	**4.47**	**4.47**	**2,832.68**	**2,832.68**

Table 44. Ex-post emissions from unplanned deforestation in PA per stratum (3)

Ex-post Project Emissions Annual ex post emissions from deforestation in PA Dense Tropical Alluvial Rainforest				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	0.05	0.05	31.69	31.69
2023	0.06	0.11	38.02	69.71
Total	**0.11**	**0.11**	**69.71**	**69.71**

Table 45. Ex-post emissions from unplanned deforestation in PA per stratum (3)

Ex-post Project Emissions Annual ex post emissions from deforestation in PA Pioneer Formation with fluvial influence without palm trees				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	0.02	0.02	12.67	12.67
2023	0.03	0.05	19.01	31.69
Total	**0.05**	**0.05**	**31.69**	**31.69**

Table 46. Ex-post emissions from unplanned deforestation in PA - sum of strata

Ex-post Project Emissions Annual ex post emissions from deforestation in PA Sum of Strata				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	5.08	5.08	3,219.25	3,219.25
2023	6.53	11.61	4,138.13	7,357.37

Total	**11.61**	**11.61**	**7,357.37**	**7,357.37**

As stated in the PD, no sustainable forest management activities are predicted in the project generally, or during the present monitoring period. Therefore, no logging emissions are to be reported using the VMD0005 module.

Regarding other types of greenhouse gasses in the project scenario, emissions of both CH_4 and N_2O were duly considered, in accordance with the methodology's VMD0013 module, "5 Procedures" section: "Where used in the baseline, accounting must occur under both the baseline and project scenarios in both the Project Area and in the Leakage Belt. The results are detailed in the tables below.

In the present project, such GHG emissions from biomass burning are considered to originate from the following category, provided by the methodology: "2. Periodical burning of grassland or agricultural land after deforestation". The applicable equation is as follows.

$$E_{biomassburn,i,t} = \sum_{g=1}^{G} \left(\left(\left(A_{burn,i,t} \times B_{i,t} \times COMF_i \times G_{g,i} \right) \times 10^{-3} \right) \times GWP_g \right) \quad (1)$$

Where:

$E_{biomassburn,i,t}$ Greenhouse gas emissions due to biomass burning in stratum *i* in year *t* of each GHG (CO_2, CH_4, N_2O) (t CO_{2e})

$A_{burn,i,t}$ Area burnt for stratum *i* in year *t* (ha)

$B_{i,t}$ Average aboveground biomass stock before burning stratum *i,* year (t d.m. ha^{-1})

$COMF_i$ Combustion factor for stratum *i* (unitless)

$G_{g,i}$ Emission factor for stratum *i* for gas g (kg t^{-1} d.m. burnt)

GWP_g Global warming potential for gas *g* (t CO_2/t gas g)

g (unitless) 1, 2, 3 ... G greenhouse gases including carbon dioxide[1], methane and nitrous oxide

i 1, 2, 3 … *M* strata (unitless)

t 1, 2, 3, … *t** time elapsed since the start of the project activity (years)

Ex-post GHG project emissions from biomass and peat burning (E–BPB) for MR1 are presented below, by gas: CH_4, followed by N_2O, and finally the sum of all emissions from biomass burning.

Table 47. Ex-post biomass burning emissions (CH_4) per stratum (1)

Ex-post Project Emissions Biomass Burning Emissions - CH_4 Dense Tropical Submontane Rainforest				
Year	**ha/year**	**ha (cumulative)**	**tCO2/year**	**tCO2**
2022	3.05	3.05	83.81	83.81
2023	3.93	6.98	107.99	191.81

Table 48. Ex-post biomass burning emissions (CH_4) per stratum (2)

Ex-post Project Emissions Biomass Burning Emissions - CH_4 Dense Tropical Lowland Rainforest				
Year	**ha/year**	**ha (cumulative)**	**tCO2/year**	**tCO2**
2022	1.96	1.96	54.02	54.02
2023	4.47	6.43	123.19	177.21

Table 49. Ex-post biomass burning emissions (CH_4) per stratum (3)

Ex-post Project Emissions Biomass Burning Emissions - CH_4 Dense Tropical Alluvial Rainforest				
Year	**ha/year**	**ha (cumulative)**	**tCO2/year**	**tCO2**
2022	0.05	0.05	1.56	1.56
2023	0.06	0.11	1.88	3.44

Table 50. Ex-post biomass burning emissions (CH_4) per stratum (4)

Ex-post Project Emissions Biomass Burning Emissions - CH_4 Pioneer Formation with fluvial influence without palm trees				
Year	**ha/year**	**ha (cumulative)**	**tCO2/year**	**tCO2**
2022	0.02	0.02	0.36	0.36

2023	0.03	0.05	0.54	0.90

Table 51. Ex-post biomass burning emissions (CH_4) – sum of strata

Ex-post Project Emissions Biomass Burning Emissions - CH_4 Sum of Strata				
Year	**ha/year**	**ha (cumulative)**	**tCO2/year**	**tCO2**
2022	5.08	5.08	139.75	139.75
2023	8.49	13.57	233.60	373.35

Table 52. Ex-post biomass burning emissions (N_20) per stratum (1)

Ex-post Project Emissions Biomass Burning Emissions - N_2O Dense Tropical Submontane Rainforest				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	3.05	3.05	24.92	24.92
2023	3.93	6.98	32.12	57.04

Table 53. Ex-post biomass burning emissions (N_20) per stratum (2)

Ex-post Project Emissions Biomass Burning Emissions - N_2O Dense Tropical Lowland Rainforest				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	1.96	1.96	16.06	16.06
2023	2.51	4.47	20.57	36.64

Table 54. Ex-post biomass burning emissions (N_20) per stratum (3)

Ex-post Project Emissions Biomass Burning Emissions - N_2O Dense Tropical Alluvial Rainforest				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	0.05	0.05	0.47	0.47
2023	0.11	0.16	1.02	1.49

Table 55. Ex-post biomass burning emissions (N_2O) per stratum (4)

Ex-post Project Emissions Biomass Burning Emissions - N_2O Pioneer Formation with fluvial influence without palm trees				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	0.02	0.02	0.11	0.11
2023	0.03	0.05	0.16	0.27

Table 56. Ex-post biomass burning emissions (N_2O) – sum of strata

Ex-post Project Emissions Biomass Burning Emissions - N_2O Sum of Strata				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	5.08	5.08	41.56	41.56
2023	6.58	11.66	53.87	95.43

Table 57. Total ex-post emissions from biomass burning per stratum (E–BPB) (1)

Ex-post Project Emissions Biomass Burning Emissions Dense Tropical Submontane Rainforest				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	3.05	3.05	108.74	108.74
2023	3.93	6.98	140.11	248.85

Table 58. Total ex-post emissions from biomass burning per stratum (2)

Ex-post Project Emissions Biomass Burning Emissions Dense Tropical Lowland Rainforest				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	1.96	1.96	70.08	70.08
2023	2.51	4.47	143.76	213.84

Table 59. Total ex-post emissions from biomass burning per stratum (3)

Ex-post Project Emissions Biomass Burning Emissions

Dense Tropical Alluvial Rainforest				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	0.05	0.05	2.03	2.03
2023	0.06	0.11	2.90	4.93

Table 60. Total ex-post emissions from biomass burning per stratum (4)

Ex-post Project Emissions Biomass Burning Emissions Pioneer Formation with fluvial influence without palm trees				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	0.02	0.02	0.46	0.46
2023	0.03	0.05	0.70	1.16

Table 61. Total ex-post emissions from biomass burning per stratum – sum of strata

Ex-post Project Emissions Biomass Burning Emissions Sum of Strata				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	5.08	5.08	181.31	181.31
2023	6.53	11.61	287.47	468.78

3.2.3 Leakage

As shown in Figure 14, 17 ha of deforestation were detected in the Leakage Belt during the present monitoring period.

The VMD0010 Module (LK-ASU) was applied to calculate the ex-post emissions from leakage displaced from the Project Area to the Leakage Belt (CLK-ASU-LB). As directed by equation 1, below, baseline emissions from unplanned deforestation in the LB were subtracted from the emissions in the project case in the LB.

$$\Delta C_{LK-ASU-LB} = C_{P,LB} - \Delta C_{BSL,LK,unplanned}$$

Where:

$\Delta C_{LK-ASU-LB}$	Net CO_2 emissions due to unplanned deforestation displaced from the project area to the leakage belt up to year $t*$ (t CO_{2e})
$\Delta C_{BSL,LK,unplanned}$	Net CO_2 emissions in the baseline from unplanned deforestation in the leakage belt up to year $t*$ (t CO_{2e})
$\Delta C_{P,LB}$	Net greenhouse gas emissions within the leakage belt in the project case up to year t* (t CO_{2e})

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Subsequently, the VMD0010 Module (LK-ASU) was applied to estimate the ex-post emissions due to unplanned deforestation outside the Leakage Belt ($\Delta C_{LK-ASU,OLB}$), as directed by equations 7-10 below

Applying equation 9 below, the area deforested by immigrants outside the leakage belt and project area under the project scenario ($A_{LK-OLB,t}$) was determined to be ≤ 0 in all cases, in other words the area deforested by immigrants in the Project Area and Leakage Belt under the project scenario was higher than total area deforested by immigrant agents in the baseline and project scenario, in both cases. This leads to the conclusion that leakage outside the Leakage Belt has not occurred, and therefore the emissions due to unplanned deforestation displaced outside the Leakage Belt ($\Delta C_{LK-ASU-OLB}$) are set to zero in all years and strata below.

5.1.5.6 *Ex post*, as deforestation in the project area and leakage belt will be measured, $\Delta C_{LK-ASU,OLB}$ is estimated as follows:

1. *Ex post*, the proportion of the total area deforested by immigrant agents in the project scenario must be determined from the same proportion calculated in the baseline data. The proportional area deforested by immigrant agents in the baseline and project scenarios is assumed to remain the same.

$$A_{LK\text{-}IMM,t} = PROP_{IMM} \times A_{BSL,PA,unplanned,t} \quad (7)$$

$$A_{LK-ACT-IMM,t} = PROP_{IMM} \times \left(\sum_{i=1}^{M} A_{DefPA,i,t} + A_{DefLB,i,t} \right) \quad (8)$$

Where:

$A_{LK\text{-}ACT\text{-}IMM,t}$	Area deforested by immigrants in the project area and leakage belt under the project scenario in year *t* (ha)
$PROP_{IMM}$	Proportion of area deforested by immigrant agents in the leakage belt and project area[6] (proportion)
$A_{DefPA,i,t}$	Area of recorded deforestation in the project area in the project case in stratum i in year t (ha)
$A_{DefLB,i,t}$	Area of recorded deforestation in the leakage belt in the project case in stratum i in year t (ha)
i	1, 2, 3 …*M* strata
t	*1, 2, 3 …t** time elapsed since the start of the project activity (year)

$$A_{LK\text{-}OLB,t} = A_{LK\text{-}IMM,t} - A_{LK\text{-}ACT\text{-}IMM,t} \quad (9)$$

Where:

$A_{LK\text{-}OLB,t}$	Area deforested by immigrants outside the leakage belt and project area under the project scenario in year *t* (ha)
$A_{LK\text{-}IMM,t}$	Total area deforested by immigrant agents in the baseline and project scenario in year *t* (ha)
$A_{LK\text{-}ACT\text{-}IMM,t}$	Area deforested by immigrants in the project area and leakage belt under the project scenario in year *t* (ha)
t	*1, 2, 3 …t** time elapsed since the start of the project activity (year)

4. Determine whether leakage outside the leakage belt has occurred:

   If: $A_{LK\text{-}OLB,t} \leq 0$ → Leakage outside the leakage belt has not occurred.

   If: $A_{LK\text{-}OLB,t} > 0$ → Leakage outside the leakage belt has occurred.

5. If leakage outside the leakage belt has not occurred:

$$\Delta C_{LK-ASU,OLB} = 0 \quad (10)$$

Where:

$\Delta C_{LK\text{-}ASU,OLB}$	Sum of carbon stock changes and greenhouse gas emissions due to unplanned deforestation displaced outside the leakage belt (t CO_2e)

The tables below show the ex-post sum of carbon stock changes and greenhouse gas emissions due to unplanned deforestation displaced outside the leakage belt ($\Delta C_{LK-ASU-OLB}$), by stratum.

Table 62. Ex-post carbon stock changes and GHG emissions due to unplanned deforestation displaced outside the Leakage Belt per stratum (1).

Ex-post Leakage Outside Annual Leakage Outside of Leakage Belt Dense Tropical Submontane Rainforest				
Year	$A_{LK-IMM,t}$	$A_{LK-ACT-IMM,t}$	$A_{LK-OLB,t}$	$\Delta C_{LK-ASU,OLB}$
	ha	**ha**	**ha**	**tCO2e**
2022	0.00	0.02	-0.02	0.00
2023	0.00	0.02	-0.02	0.00

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Table 63. Ex-post carbon stock changes and GHG emissions due to unplanned deforestation displaced outside the Leakage Belt per stratum (4).

Ex-post Leakage Outside Annual Leakage Outside of Leakage Belt Pioneer Formation with fluvial influence without palm trees				
Year	$A_{LK-IMM,t}$	$A_{LK-ACT-IMM,t}$	$A_{LK-OLB,t}$	$\Delta C_{LK-ASU,OLB}$
	ha	**ha**	**ha**	**tCO2e**
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00

Table 64. Ex-post carbon stock changes and GHG emissions due to unplanned deforestation displaced outside the Leakage Belt – sum of strata.

Ex-post Leakage Outside Annual Leakage Outside of Leakage Belt Sum of Strata				
Year	$A_{LK-IMM,t}$	$A_{LK-ACT-IMM,t}$	$A_{LK-OLB,t}$	$\Delta C_{LK-ASU,OLB}$
	ha	**ha**	**ha**	**tCO2e**
2022	0.00	7.48	-7.48	0.00
2023	0.00	9.60	-9.60	0.00

The VMD0011 module for estimation of emissions from market-effects (LK-ME) was used to calculate the emissions due to market leakage through decreased timber havest ($LK_{MarketEffects,timber}$), applying equations 2 – 4, below.

$$LK_{MarketEffects,Timber} = \sum_{i=1}^{M} (LF_{Me} * AL_{T,i})$$

Where:

$LK_{MarketEffects,Timber}$	Total GHG emissions due to market- effects leakage through decreased timber harvest; t $CO_{2\text{-}e}$
LF_{Me}	Leakage factor for market-effects calculations; dimensionless
$AL_{T,i}$	Summed emissions from timber harvest in stratum i in the baseline case potentially displaced through implementation of carbon project; t $CO_{2\text{-}e}$
I	*1,2, 3, …M strata*

$$AL_{T,i} = \sum_{t=1}^{t} (C_{BSL,XBT,i,t})$$

Where:

$AL_{T,i}$	Summed emissions from timber harvest in stratum i in the baseline case potentially displaced through implementation of carbon project; t $CO_{2\text{-}e}$
$C_{BSL,XBT,i,t}$	Carbon emission due to displaced timber harvests in the baseline scenario in stratum i in time t; t $CO_{2\text{-}e}$
i	*1,2, 3…M strata*
t	1, 2, 3, … t years elapsed since the projected start of the REDD project activity

$$C_{BSL,XBT,I,T} = \left(\left[V_{BSL,XE,I,t} * D_{mn} * CF\right] + \left[V_{BSL,XE,I,t} * LDF\right] + \left[V_{BSL,XE,I,t} * LIF\right]\right) * \frac{44}{12}$$

Where:

$C_{BSL,XBT,i,t}$	Carbon emission due to displaced timber harvests in the baseline scenario in stratum i in time t; t $CO_{2\text{-}e}$
$V_{BSL,XE,I,t}$	Volume of timber projected to be extracted from within the project boundary during the baseline in stratum i at time t; m^3
D_{mn}	Mean wood density of commercially harvested species; t d.m.m^{-3}
CF	Carbon fraction of biomass for commercially harvested species j; t C td.m.-1

LDF	Logging damage factor; t C m-3 (default 0.53 t C m^{-3} for broadleaf and mixed forests)
LIF	Logging infrastructure factor; t C m-3 (default 0.29 t C m^{-3})
i	*1,2, 3, …M strata*
t	1, 2, 3, … t years elapsed since the projected start of the REDD project activity

$$LK_{MarketEffects,timber} = \sum_{i=1}^{M} \left(LF_{ME} * AL_{T,i} \right) \qquad (2)$$

Where:

$LK_{MarketEffects,timber}$	Total GHG emissions due to market- effects leakage through decreased timber harvest; t CO_2-e
LF_{ME}	Leakage factor for market-effects calculations; dimensionless
$AL_{T,i}$	Summed emissions from timber harvest in stratum *i* in the baseline case potentially displaced through implementation of carbon project; t CO_2-e
i	*1,2,3,…M strata*

$$AL_{T,i} = \sum_{t=1}^{t} \left(C_{BSL,XBT,i,t} \right) \qquad (3)$$

Where:

$AL_{T,i}$	Summed emissions from timber harvest in stratum *i* in the baseline case potentially displaced through implementation of carbon project; t CO_2-e
$C_{BSL,XBT,i,t}$	Carbon emission due to displaced timber harvests in the baseline scenario in stratum *i* in time *t*; t CO_2-e
i	1, 2, 3, …*M* strata

$$C_{BSL,XBT,i,t} = ([V_{BSL,XE,i,t} * D_{mn} * CF] + [V_{BSL,XE,i,t} * LDF] + [V_{BSL,XE,i,t} * LIF]) * \frac{44}{12} \quad (4)$$

Where:

$C_{BSL,XBT,i,t}$	Carbon emission due to timber harvests in the baseline scenario in stratum *i* at time *t*; t CO_2-e
$V_{BSL,EX,i,t}$	Volume of timber projected to be extracted from within the project boundary during the baseline in stratum *i* at time *t*; m^3
D_{mn}	Mean wood density of commercially harvested species; t d.m.m^{-3}. The value must be the same as that used in the module CP-W if this pool is included in the baseline.
CF	Carbon fraction of biomass for commercially harvested species *j*; t C t d.m.$^{-1}$. The value must be the same as that used in the module CP-W if this pool is included in the baseline.
LDF	Logging damage factor; t C m^{-3} (default 0.53 t C m^{-3} for broadleaf and mixed forests; 0.25 t C m^{-3} for coniferous forests)
LIF	Logging infrastructure factor; t C m^{-3} (default 0.29 t C m^{-3})
i	1, 2, 3, … *M* strata
t	1, 2, 3, … *t** years elapsed since the projected start of the REDD project activity

It is noted that since the methodology for calculating *ex-post* emissions from market leakage through decreased timber havest is not specified in the VMD0011 module, the following procedure was created, as an ex-post calculation methodology. The assumptions were as follows:

4. subtract unavoided deforestation measured in the PA from APD and AUD baseline values (which are used in the baseline LK-ME calculation)
5. If hectare result is <0, consider it as zero to avoid positive leakage.
6. If hectare result is >0, proceed to calculate MR values for $LK_{MarketEffects,timber,}$ applying the rest of the procedure as described in VMD0011, however replacing the A_i in equation (6) with the new baseline – measured hectare value. We will call this variable MR_1A_i (ha).

Via the *ex-post* methodology described above, *ex-post* market leakage through decreased timber harvest ($LK_{MarketEffects,timber}$) was found to be zero in all years and strata, as detailed in the tables below.

Table 65. Ex-post Market Leakage (LK-ME) per stratum (1).

Ex-post Market Leakage Annual Leakage Market Effects - Timber Harvest Dense Tropical Submontane Rainforest					
Year	ha/year	ha (cumulative)	V_{MR1} m^3	$C_{MR1,XBT}$	$LK_{MarketEffects,timber}$
				tCO_2/year	tCO_2/year
2022	129.58	129.58	4,545.65	18,289.13	3,657.8
2023	849.22	978.80	29,790.69	119,860.85	23,972.2

Table 66. Ex-post Market Leakage (LK-ME) per stratum (2).

Ex-post Market Leakage Annual Leakage Market Effects - Timber Harvest Dense Tropical Lowland Rainforest					
Year	ha/year	ha (cumulative)	V_{MR1} m^3	$C_{MR1,XBT}$	$LK_{MarketEffects,timber}$
				tCO_2/year	tCO_2/year
2022	367.11	367.11	12,878.08	51,814.08	10,362.8
2023	74.31	441.41	2,606.70	10,487.90	2,097.6

Table 67. Ex-post Market Leakage (LK-ME) per stratum (4).

Ex-post Market Leakage Annual Leakage Market Effects - Timber Harvest Pioneer Formation with fluvial influence without palm trees					
Year	ha/year	ha (cumulative)	VMR1 m3	CMR1,XBT	$LKMarketEffects_{,timber}$
				tCO2/year	tCO2/year
2022	0.24	0.24	8.46	34.06	6.8
2023	0.00	0.24	0.00	0.00	0.0

Table 68. Ex-post Market Leakage (LK-ME) sum of strata.

Ex-post Market Leakage Annual Leakage Market Effects - Timber Harvest Sum of strata					
Year	**ha/year**	**ha (cumulative)**	**V_{MR1} m^3**	**$C_{MR1,XBT}$**	**AL_{MR1}*LF_{ME}**
				tCO_2/year	**tCO_2/year**
2022	5.08	5.08	17,432.19	70,137.27	14,027.5
2023	6.53	11.61	32,397.39	130,348.75	26,069.7

In accordance with requirements in the VMD0013 E-BPB module, section 5, accounting for N_20 and CH_4 emissions were applied to leakage displaced from the PA to the LB in MR1 (LK-ASU-LB), because they were accounted for in the baseline. The tables below shows the ex-post calculated emissions from biomass burning for the sum of gasses (CH_4 + N_2O) related to Unplanned Deforestation Displaced from the Project Area to the Leakage Belt (LK-ASU), while the calculation spreadsheet shows the breakdown of CH_4 and N_2O separately.

$$E_{biomassburn,i,t} = \sum_{g=1}^{G} \left(\left(\left(A_{burn,i,t} \times B_{i,t} \times COMF_i \times G_{g,i} \right) \times 10^{-3} \right) \times GWP_g \right) \qquad (1)$$

Table 69. Ex-post summed burning emissions (CH4 + N2O) associated with Leakage (LK-ASU) per stratum (1)

Ex-Post Leakage Biomass Burning Emissions Dense Tropical Submontane Rainforest				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	6.52	6.52	232.45	232.45
2023	8.38	14.90	298.76	531.21

Table 70. Ex-post summed burning emissions (CH4 + N2O) associated with Leakage (LK-ASU) per stratum (2)

Ex-Post Leakage Biomass Burning Emissions Dense Tropical Lowland Rainforest				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	0.00	0.00	0.00	0.00

2023	0.00	0.00	0.00	0.00

Table 71. Ex-post summed burning emissions (CH4 + N2O) associated with Leakage (LK-ASU) per stratum (3)

Ex-Post Leakage Biomass Burning Emissions Dense Tropical Alluvial Rainforest				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00

Table 72. Ex-post summed burning emissions (CH4 + N2O) associated with Leakage (LK-ASU) per stratum (4)

Ex-Post Leakage Biomass Burning Emissions Pioneer Formation with fluvial influence without palm trees				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	0.95	0.95	22.08	22.08
2023	1.21	2.16	28.12	50.19

Table 73. Ex-post summed burning emissions (CH4 + N2O) associated with Leakage (LK-ASU) – sum of strata

Ex-Post Leakage Biomass Burning Emissions Sum of Strata				
Year	**ha/year**	**ha (cumulative)**	**tCO_2/year**	**tCO_2**
2022	7.47	7.47	254.52	254.52
2023	9.59	17.06	326.88	581.40

3.2.4 Net GHG Emission Reductions and Removals

The first table below presents the quantification of the net GHG emission reductions and its corresponding inputs: baseline emissions, project emissions and leakage.

According to VM0007-REDDMF module, equation 2, net GHG Emission Reductions and Removals can be summarized as the “Estimated baseline emissions” minus the “Estimated project emissions” minus the “Estimated leakage emissions”, whose components are presented below.

Estimated baseline emissions
= Baseline emissions from unplanned deforestation
+ Emissions from biomass burning in the baseline
- Sum of wood products and pasture carbon pools in the baseline case

Estimated project emissions
= Emissions arising from logging gap in Project Area
+ Emissions from constructing logging infrastructure for removal of timber in Project Area
- Wood Products Carbon Pool in Project Area

Estimated leakage emissions
= (Sum of Market Leakage and Leakage outside the Leakage Belt)
** Leakage Factor*

The second table below presents net tradable VCUs, derived from net GHG reductions minus the buffer discount, defined as 10% (see separate AFOLU non-permanence risk report – NPRR). This calculation is in accordance with the VM0007 REDDMF module’s equation 19, as noted in PD section 3.2.4.

Table 74. Baseline emissions, project emissions, leakage, and net GHG emission reductions summary.

Year	Baseline emissions or removals (tCO_2e)	Project emissions or removals (tCO_2e)	Leakage emissions (tCO_2e)	Net GHG emission reductions or removals (tCO_2e)
2022	261,286	3,401	19,007	238,878
2023	490,930	4,426	32,474	454,031
Total	**752,216**	**7,826**	**51,481**	**692,909**

Table 75 - Net GHG reductions, buffer discount, and Net tradable VCUs.

REDD Benefits (buffer included) (tCO2)	Buffer discount (10.0%)	Annual Net tradeable VCUs (tCO2)	Cumulative Net Tradeable VCUs (tCO2)
238,878	25,327	213,551	213,551
454,031	47,614	406,416	619,968
692,909	**72,941**	**619,968**	**619,968**

3.3 Optional Criterion: Climate Change Adaptation Benefits

The present project did not include this item at validation.

3.3.1 Activities and/or processes implemented for Adaptation (GL1.3)

No activities in this scope were performed during this verification period.

4 COMMUNITY

4.1 Net Positive Community Impacts

4.1.1 Community Impacts (CM2.1)

The project activities implemented in the present verification period, which are in accordance with expectations set out in the PD's Theory of Change (see PD section 2.1.11, and Appendix 2), are summarized below:

Social pillar:

- Communication and Electrification for the community through the installation of solar power panels and antennas for satellite internet;
- Participatory Mapping Workshop, identifying areas of community land-use;

- Education activities in the areas of both teacher and student education and training; and donation of school equipment and material;
- Training and workshop on occupational health and safety, including distribution of personal protective equipment (PPE), providing the community members with knowledge and awareness about working safely;
- Training in associations and cooperative formation, contributing to community employment and civil rights capacity building;
- Women's workshop, including information gathering on their needs and challenges, laying the foundations for project actions addressing women's employment and wellbeing;

Climate pillar:

- Forest Conservation - Monitoring by CarbonEye for avoided deforestation; in accordance with the Forest Integrity Protocol: Reporting of invasion event to responsible authorities, and actions in the field including deployment of field team and putting up project signage [101]. The community impacts of this are to minimize their contact with illegal deforesters and associated risks.

Regarding community ownership, control and participation in these activities, their design from the beginning was in line with community expectations and desires for activities to be implemented, as presented in community votes table in section 2.3.3. During the installation of the communication and electrification equipment, the Boa Esperança community was fully involved in the technical process, which was led by the specialist company. This was the case, both for the equipment test in October 2022 and for the permanent equipment installed in February 2023. The participatory mapping workshop was attended by members of all the communities ("Boa Esperança": 15 members, Vila Batista: 7 members and Vale da Benção: 4 members) and mapped the community land-uses of all three PZ communities.

To provide the equipment, a participatory workshop was held where community members were informed that the equipment would be provided as a way of supporting the community's school as well as enabling community members to have access to the internet, as they live in a place considered remote and isolated. To this end, as a way of documenting and recording the activity, we invited the leadership of the Boa Esperença community, Mr. Magno Batista, to sign a form of acceptance and responsibility for the use of the installed equipment.

[101] See evidence folder "08. Deforestation".

Figure 18. Community leader signing responsibility agreement for installed communication and electrification equipment

The impacts, outcomes, and outputs of the activities, according to the Theory of Change (ToC), have been added to the tables below. However it is important to note that, given this is the project's first Monitoring Report, the impacts achieved so far are limited, as the latter are defined as the long-term results of the project.

Community Group	All members of the Boa Esperança community
Activity	Communication and electrification
Impact/outcome	The medium-term outcomes contributed to partially are: Functioning electricity, telecommunication, and internet infrastructure in PZ community residences. Population more informed on many topics from national current affairs to REDD+ and the Ipoá project itself. The long-term impacts contributed partially to are: An increase in community members with access to the internet, telecommunications, and electricity.

	Solar power will increase community members' access to electric light; electricity for refrigeration; electronic and its equipment; and ventilation, among others.
Output	5 equipment items purchased and transported. 6 equipment items installed and maintained in community buildings. 29 of people (Girls: 04 Women:08; Boys: 08 Men:09) with access to electricity and communication technology. 13 of telecommunication devices in the Project Zone. 6 of community members (Men: 3 Women 3) educated and trained in: i) communication technology use and ii) maintenance. 10 people (by gender/age) with access to electricity
Type of Benefit/Cost/Risk	Benefit: With the installation of energy in the school and the internet, members of the communities benefiting directly and indirectly were able to resolve issues remotely, do research, access social networks, find out about what is happening in their city, in their country and in the world , in addition to other benefits. The panels installed at the school will supply energy to the location, enabling the use of electronic equipment, such as computers, televisions, cell phones, etc. Risk: The solar panels and internet were installed in the "Boa Esperança" community, as the school was in operation during the period of this MR, benefiting directly from the project. Thus, an observed risk is that members of other communities may feel discredited for not having been directly served by the project activity. Cost: no community costs none identified
Change in Well-being	The impact is local and positive, involving only the PZ communities, with the community that benefited most directly being Boa Esperança. Internet services was the most-voted activity in the community participatory workshop, and solar energy was third, evidence was provided at audit. So, this project activity improves well-being according to the communities' own understanding of the term. The following indicators are those considered to relate to wellbeing according to the project's ToC:.

	5 equipment items installed in community buildings (WB). 29 people with access to communication technology (WB), specifically: Girls : 04 Women: 08 Boys: 08 Men: 09 10 people with access to electricity[102]: Boys: 08, Girls: 02. 6 community members educated and trained in: i) communication technology use and ii) maintenance (WB), specifically: Girls: 0 Women: 3 Boys: 0 Men: 3 An increase in overall quality of life and wellbeing is expected from both the internet and electricity installations described.

Images and further details of the activity described above are presented below.

Figure 19 - Antenna for satellite internet

Satellite internet antennas and solar panels were installed at the school in the project's Boa Esperança community. The provision of internet access serves to counter the scenario where community members had extremely limited contact with their families, friends, public services, and the outside world generally,

[102] See evidence folder: "05.Communication and Electrification"

as well as the improved power supply, which previously had been provided by small gas-powered generators.

Specialized professionals were contracted to install the necessary equipment for the internet and solar panels, as well providing instruction for the communities regarding the use and care of the devices.

Figure 20 - Solar panels for installation

Solar panels were transported to the communities, improving their electricity access. The area was signposted with instructional signs with Carbonext service channels for questions, complaints, and identification of the project's actions for community improvement.

Figure 21 - Information board about the installation of the solar panel and the antenna with internet access

Aiming to meet the recommendation made in the audit process, about the geographic delimitation of areas used by the project communities for subsistence agriculture, Carbonext.conducted an on-site participatory mapping workshop with the communities was carried out in February 2023. Representatives from all the Ipoá REDD+ project's communities attended: Boa Esperança, Vila Batista and Vale da Benção, with active participation by the community members. The mapping of the remaining two communities is predicted to be completed by end of MR3. The participatory mapping activity was based on three methodological stages: 1) Planning, consisting of all preparatory activities; 2) Execution, consisting of field data collection; and 3) Processing, consisting of post-field activities and analysis of results. After all, the planning is back to validation with de community.

The results indicate that it was possible, through this workshop, to geographically delimit the cultivated areas in the PZ, associated with the families' subsistence agriculture planting, in addition to the identification of areas used for fishing, and for wild-harvesting of forest resources, such as copaiba oil and brazil-nut, among others. The activity also enabled the collection of information associated with community dynamics and ways of life, leading to a better understanding of the project's social context.

A further description of this activity and its relevance to HCV 5 is found in section 4.3.1.

Community Group	All community members of the PZ, represented by members of each of the PZ communities who attended
Activity	Participatory Mapping Workshop
Output	16[103] areas that are fundamental for the livelihoods of communities mapped. Of which the following specific types were identified: Fishing area: 5 areas Copaiba extraction: 4 areas Brazilian nuts collection: 3 areas Timber harvesting: 1 area; 0.08 ha Churches: 3 areas 87.14 Hectares of deforested areas cultivated by communities mapped 76.72 hectares of geographic delimitation of the areas disturbed by human activity

[103] See evidence folder 06. HCVs

Impact	The long-term impacts contributed partially to are[104]: Community members able to visualize and plan their land-uses; Mitigation of future community deforestation of PZ areas; Continuous remote monitoring; Allowing communities to document local knowledge.
Type of Benefit/Cost/Risk	Planned Benefit: The objective of the participatory mapping activity was to consult the community about the areas of use, mapping them. In this context, we consider that the impacts of this activity are beneficial, directly related to territorial planning and monitoring. In addition, there are also indirect benefits through the documentation of local knowledge represented by the maps produced, which give the community a leading role in the use of the territory. It is also important to consider that one of the greatest benefits of participatory mapping is the opportunity to bring people to the table and share their experiences, knowledge, and visions about the land, contributing to build a sense of place and a connection to the territory. Potential Risk: Community conflicts involving land-use, the Operational Procedure for definition of alternative land uses was applied.
Change in Well-being	Considering the aims of participatory mapping, which include understanding, and identifying areas of use, the benefits associated with well-being are: - an increase in collaboration between community members, since the mapping allowed the exchange of knowledge and ideas between them regarding their land-use. - the consolidation and clarification of boundaries regarding community land uses.

A photographic record of the implementation of the community mapping activities is included below.

[104] In accordance with section 4.2.4 of the Ipoá REDD+ Project's PD, this HCV protection activity does not have associated outcomes

Figure 22 - Participatory Mapping Workshop

Support for school infrastructure and equipment in the Boa Esperança community was realized during MR1 through the donation of school equipment and supplies. The school was built by the residents themselves with the aim of resuming teaching for children in the communities, but they still needed some equipment to improve the school's infrastructure and facilitate teaching for the teacher and students. Thus, as prescribed in the Theory of Change (ToC), equipment and school materials including a television, a whiteboard, 4 tablets, a printer and ink refills, a computer, pens and pencils, and fans and sports and games equipment were delivered to the school in the Boa Esperança community (Figure 21)[105].

Figure 21 – Delivery of school equipment and materials.

Community Group	Schoolchildren and staff of the Boa Esperança community
Activity	School infrastructure and equipment

[105] See evidence folder “02. Education\Education equipment”

Impact/outcome	The medium-term outcomes contributed to partially are: Presence of electricity and telecommunication infrastructure within education facilities. Good school equipment in the PZ. The long-term impacts contributed partially to are: Contribute to quality education for communities. School facilities in PZ adequate and conducive to quality education.
Output	39 educational equipment and material items purchased or transported. [106] 19 electricity and telecommunication equipment items installed within education facilities. [107] 4 meetings were held with the Borba and Novo Aripuanã Secretaries of Education, with the Borba Town Hall Attorney, and with the Borba Town Pedagogical Coordinator.[108]
Type of Benefit/Cost/Risk	Benefit: School equipment and materials are essential for both students and teachers, as they facilitate the teaching and learning process. For students, these represent practical tools that help organize and absorb knowledge. For teachers are pedagogical resources that enable the creation of more dynamic and effective classes. Together contribute to a more stimulating educational environment conducive to students' academic and personal development. Risk: Wear and tear when using school equipment and materials can lead to shortages and damage. Theft of equipment and materials may also occur. Cost: no community costs none identified.
Change in Well-being	Contribute to quality education for communities. Medium Magnitude

[106] See evidence folder “02. Education/Education equipment”
[107] See evidence folder “02. Education/Education equipment”
[108] See evidence folder “02. Education/Meetings PPP”

Training in first aid and work safety was carried out in MR1, in July 2023, as further described in item 2.3.13, with 11 participants in attendance, and lasting for 6 hours. A seasoned professional, experienced in training residents of remote riverside communities where immediate healthcare might be lacking in case of workplace accidents, was recruited for this course.

Following the training, the participants were provided with a Personal Protective Equipment (PPE) kits, aiming to enhance safety among community members and to reduce the risk of workplace accidents.

Community Group	PZ community members of working age
Activity	Training and workshop: Occupational health and safety
Impact/outcome	The medium-term outcomes, contributed partially to in MR1, are: Increase in people following occupational health and safety guidelines. Reduction in work-related accidents. The long-term impacts, contributed partially to in MR1, are: Population aware of occupational risks related to health and safety. Contribution to decent work and economic growth (SDG 8)
Output	12 people (Men: 9; women: 3) trained in best practices for occupational health and safety.[109]
Type of Benefit/Cost/Risk	Benefit: Participants will have knowledge of first aid and in situations of low to medium severity accidents they will be able to use the knowledge acquired in the course. Risk: At times when first aid is required, training participants are unable to apply the techniques taught, or even cause increased harm by incorrect application. Cost: no community costs none identified.
Change in Well-being	Population aware of health and safety risks associated with occupations. Medium magnitude

[109] See evidence folder “04. Occupational Health and Safety/People trained”

Community Group	PZ community members of working age
Activity	Personal protective equipment (PPE) distribution for worker health and safety
Impact/outcome	The medium-term outcomes contributed to partially are: Increase in people following occupational health and safety guidelines. Reduction in work-related accidents. The long-term impacts contributed partially to are: Population aware of occupational risks related to health and safety. Contribution to decent work and economic growth (SDG 8)
Output	12 people (Women: 2, Boys: 10) given PPE equipment.[110]
Type of Benefit/Cost/Risk	Benefit: They will be more aware of the importance of working safely, avoiding exposure to the risk of workplace accidents. Risk: Training participants are not using PPE daily, increasing their exposure to work accidents. Cost: no community costs none identified.
Change in Well-being	Population aware of health and safety risks associated with occupations. Medium Magnitude

As mentioned in item 2.3.14, during the project monitoring period, training on associations and cooperatives was given to Project Zone community members.

The Ipoá REDD+ project understands that working on its strengthening is the basis for community be empowerment, enabling them to seek means to generate income and other related benefits. Thus, the project supports the creation of the residents' association and in this monitoring period the first step, of training provided by a consultant, was taken so that this can happen in the future.

[110] See evidence folder “04. Occupational Health and Safety/Given PPE equipment”

Community Group	PZ Community members of working age
Activity	Community association creation
Impact/outcome	The medium-term outcomes, contributed to partially in MR1, are: Strengthened community governance. Improved ability for communities to demand and obtain public services. Increase in community members (by gender) becoming members and occupying official positions in association. The long-term impacts, contributed partially to in MR1, are: Community entrepreneurship enabled. Improvement in income generation and the quality of life of employees. Reduction of poverty (SDG 1). SDG 8.3 Promote development-oriented policies that support productive activities, decent job creation, entrepreneurship, creativity and innovation, and encourage the formalization and growth of micro-, small- and medium-sized enterprises, including through access to financial services. Contribution to sustainable communities (SDG 11)
Output	One community association consultant hired for community association creation and sent to the field to assess the situation of the community's informal association and advise on next steps.[111] 17 (day one), and 26 (day two) participants in association training events, assemblies etc.
Type of Benefit/Cost/Risk	Benefit: As mentioned in 2.3.14, in practical terms a community association permits a rural community to defend the interests and rights of the local community. Furthermore, it is possible to achieve better productive efficiency, through professional training and technical assistance, incorporating technologies and better economic-financial management of agricultural activity. Risk: Over time, due to migration to large cities or lack of interest on the part of residents, the residents' association may become weakened. Furthermore, there is a possibility of greater control of the association by

[111] See evidence folder "19. Employment"

	a few residents, meaning that the transition of the association's leadership will not duly occur. Cost: no community costs identified.
Change in Well-being	Communities empowered and strengthened through their residents' association, achieving benefits to improve everyone's living conditions. High Magnitude

Also in July 2023, a women’s workshop took place, which was was led by a consultant with extensive experience with traditional communities, and had 18 women participants. The objective of the activity was to diagnose the main demands and specific needs of women, with the aim of proposing future actions aimed exclusively at them.

Based on the surveys carried out to prepare the Project Description, it was noticed that the population living in communities outside the project area, especially the female public, lived on the margins of any guaranteed rights, such as access to health, access to education, between others. Thus, the project identified the need to work on future actions specifically with women (Figure 22).

Figure 22 – Women´s workshop

Community Group	All women residents of the PZ
Activity	Women’s workshop
Impact/outcome	The medium-term outcomes contributed to partially are:

	Women generating income through activities encouraged in the project. The long-term impacts contributed partially to are: Developing the skills needed to generate employment and income Female gender empowered and aware of the importance of their role within the local social context
Output	18 women participants[112] 19 proposals for activities with women [113]
Type of Benefit/Cost/Risk	Benefit: Knowledge about the needs and desires of the female public in the project's external communities and the possibility of carrying out future actions to meet them with a view to their progress. It is expected to increase active participation in decisions related to communities and that they can, in addition to being able to obtain income. Risk: possibility that they are no longer interested in participating in activities due to their duties at home, among other reasons. Cost: no community costs none identified.
Change in Well-being	Facilitates the empowerment of women in participating in activities and also in income generation. High magnitude

Community Group	All community members of the PZ
Activity	Forest conservation
Impact/outcome	"The area started remote monitoring in CarbonEye in June 2022, and some events were dectected:

[112] See evidence folder “09. Other activities”
[113] See evidence folder “09. Other activities/Women’s Workshop/Relatório Carbonext Ipoá”

	1) The road in Maanaim Farm in July 2022; 2) The Community Ridging at Maanaim Farm in August 2022." "The events that were dectected were verified in the field: 1) The road at Maanaim Farm in September and October 2022; 2) The Community Ridging at Maanaim Farm in October 2022." Five (5) Notifications/Bulletins of occurrence in responsible command and control agencies. There were two (2) identification plaques installed at Manaaim Farm with the alert of illegal deforestation at Fazenda Maanaim (road), one (1) deforestation protocol was initiated.
Output	2 deforestation events visited in the field. [114] 5 meetings carried out to combat deforestation (WB).[115] Effectiveness of meeting Nº1[116]: 2 Effectiveness of meeting Nº2: 1 Effectiveness of meeting N°3: 1 Effectiveness of meeting N°4: 1 Effectiveness of meeting N°5: 1 2 infrastructure items installed for deforestation combat. [117] 2 protocol applications regarding community deforestation (WB)[118]
Type of Benefit/Cost/Risk	Planned benefits: Reduction of negative effects on communities – including ecological, environmental, and social – from invasive deforestation and fire.

[114] See evidence folder “08. Deforestation”.
[115] See evidence folder “\08. Deforestation\22.05.12 - Estrada Maanaim”, documents “22.06.30 - Reunião proprietários_Invasão” and “22.06.30 – Explanatory note”.
[116] Effectiveness is ranked according to an internal classification rating from 1 – 3, with 1 being lowest and 3 highest effectiveness.
[117] See evidence folder “08. Deforestation.”
[118] See evidence folder “12. Operational Procedures\Applied\”

	Potential risks: Risks to community from potentially aggressive deforestation agents invading the property. These risks are reduced by conservation project activities.
Change in Well-being	The project's overall objectives of forest conservation, GHG reductions, sustainable development, and biodiversity conservation are closely inter-related. The risks to project communities' personal property, surrounding ecosystems and physical wellbeing are reduced by forest conservation project activities, and therefore it is considered that PZ communities' wellbeing relating to these aspects is improved.

4.1.2 Negative Community Impact Mitigation (CM2.2)

During the public consultation processes, field visits, and interviews, the community groups consulted returned primarily positive feedback. The communities did not identify any potential risks that the project could pose to them on this occasion; rather, they perceived that the project activities would only help develop the region and its communities and protect the local ecosystems.

Nonetheless, measures are in place to mitigate any negative impacts on the wellbeing of community group, in accordance with the precautionary principle, and to maintain or enhance HCV attributes related to community wellbeing. At Carbonext these procedures are primarily constituted by Operational Procedures (OPs)[119]. Their applicability to the events in the MR1 period, risk evaluations and application during the MR1 period are described in the table below:

Table 76. Operational Procedures and their role in mitigating potential negative community impacts during MR1

Event	Applied OP	Risk	Evidence

[119] See evidence folder "12. Operational Procedures".

Deforestation invasion in May 2022	OP for Deforestation Events	Disputes and changes in land rights or access/use rights; Evaluation: medium. Physical risk to communities; Evaluation: medium. External risks regarding disputes and changes in land rights or access/use rights; evaluation: medium.	See Folder[120]
Conflict relating to installed communication and electrification equipment recorded on 16/11/22	OP for Community conflict resolution	Internal non-physical conflict risk to communities. Evaluation: medium	See Folder [121]
Conflict over the distribution of Personal Protective Equipment to health training participants. Recorded on 11/07/2023	OP for Community conflict resolution	Internal non-physical conflict risk to communities. Evaluation: low	See Folder [122]

[120] See evidence folder "\08. Deforestation\22.05.12 – Estrada Maanaim"
[121] See evidence folder "12. Operational Procedures\Applied"
[122] See evidence folder "12. Operational Procedures\Applied"

Conflict related to the dissatisfaction of a member of the Terra Preta do Capinarana community with the project. recorded on 14/07/2023	OP for Community conflict resolution	Internal non-physical conflict risk to communities. Evaluation: low	See Folder [123]
Communication OP Applied on the following Dates: June/2022, September/2022, January/2023 July/2023	OP for Stakeholder Communication. Visits to comunities	No risk Evaluation: low	See folder[124]
Responsibility agreement and workshop held in relation to installation of communication/ electrification infrastructure, on 06/02/2023	Responsibility agreement and workshop	Physical risk to communities; Evaluation: low Conflicts between communities over the direct benefits of projects; Evaluation: medium.	See folder[125]
Responsibility agreement for the delivery of school equipment and materials, on 10/07/2023	Responsibility agreement	Physical risk to communities; Evaluation: low Conflicts between communities over the direct benefits of	See folder[126]

[123] See evidence folder “12. Operational Procedures\Applied”
[124] See evidence folder “12. Operational Procedures\Applied”
[125] See evidence folder “12. Operational Procedures\Applied”
[126] See evidence folder “12. Operational Procedures\Applied”

		projects; Evaluation: medium.	
Deforestation in September/2023	OP for Deforestation Events	It was observed that it was a blow-down. The wind caused deforestation. Physical risk to communities; Evaluation: low	See folder[127]

Further considerations on the applied OPs, described above, and their related risks follow below:

- The OP for community conflict resolution is designed to mitigate inter- and intra-community conflict. Its application on 16/11/22, reported in the table above, was due to the dissatisfaction of certain members of one community when communication and solar equipment was installed in the neighboring community. It was clarified by Carbonext staff that the equipment would be installed in the remaining community at the earliest possible occasion and, indeed, this is planned for MR2 or MR3 at the latest. Thus, the principle of equally benefitting all communities and individuals is upheld, promoting peaceful coexistence between them.
- The OP – Deforestation Event Protocol defines the procedures to be implemented in case of invasive deforestation events, including procedures on the ground, and those involving the police and other organs. It aims to minimize exposure of project field agents and communities to potentially hostile deforestation agents, avoiding physical confrontation at all times, and promoting the correct application of command and control organs. The procedure was applied as of 27/05/22, as evidenced at audit.[128]
- The Responsibility Agreement and Workshop, was applied on 06/02/23 to define community responsibilities and mutual care agreements regarding communication and electrification equipment. [129]

In terms of mitigation measures, the OP for Deforestation Events is distinguished from the OP for Illegal Community Activity, as the procedures to be adopted for each are different. While the former involves activating command and control organs and measures in the field, the latter attempts to resolve situations through conversation, while mapping and planning of future expansion of community land-uses, and remote monitoring aim to avoid conflicts in the first place.

[127] See evidence folder “12. Operational Procedures\Applied”
[128] See evidence folder “08. Deforestation”.
[129] See evidence folder “05. Communication and Electrification”.

In this way, the OP for Illegal Community Activity promotes avoidance of negative impacts on community members, such as legal sanctions and degradation of relations between project management and proprietors.

Regarding all identified risk types, regular communication between the project team and community members was put in place in April 2022[130], with the introduction of Carbonext's means of communication, facilitating the identification of negative impacts on communities, allowing their mitigation. No criticism, observation or suggestion was reported for the project in the period of its first monitoring report period.

4.1.3 Net Positive Community Well-Being (CM2.3, GL1.4)

The table below aims to attend to criterion CM2.3[131] – which requires demonstrating that the project improves the without-project scenario – by presenting wellbeing outcomes and impacts relative to the without-project scenario. The latter scenario is described in the table below in accordance with PD section 4.1 (and its various sub-sections), which describe criterion CM1.

Regarding criterion GL1.4, the Ipoá REDD+ project is not claiming for Gold Level criterion GL1.

Table 77. Project activity improvements relative the without-project scenario.

Activity	Without-project scenario	Wellbeing Outcomes/ Impacts achieved
Communication and electrification	The communities had no electricity from the municipal grid, however, in the homes of community leaders there are solar panels providing a weak power flow, sufficient only for basic use such as night lighting, or charging a cell phone, etc. The communities did not have easy and practical access to	In MR1, the installation of communication equipment enabled communities to communicate with friends, family, public services, and the outside world more broadly, for the first time. Wide-ranging medium-term benefits ("outcomes" in CCB terminology) have already been registered in MR1, relating to communication with family

[130] See evidence folder "05. Communication and Electrification"
[131] CCB Standards v3.1: "Demonstrate that the net well-being impacts of the project are positive for all identified community groups compared with their anticipated well-being conditions under the without-project land use scenario (described in CM1)"

Activity	Without-project scenario	Wellbeing Outcomes/ Impacts achieved
	communication. There was no telephone or internet signal in the region, making it impossible for the communities to communicate with the outside world.	and friends who had long been out of contact, even presumed dead, and the areas of education, work, environmental preservation, health and safety, leisure, and entertainment, specifically. These aspects are considered to be significant wellbeing gains compared to the without-project scenario. In future, it is expected that electricity, telecommunication and internet facilities will be extended to the remaining two PZ communities, and that electrification will be reflected in quantifiable outputs including Nº of domestic appliances per household, etc. Nonetheless, it is considered that predicted outcome in section 6.2. of the PD of “functioning electricity, telecommunication, and internet infrastructure in PZ community residences” has already been largely achieved in one of the communities: Boa Esperança. .
Participatory Mapping Workshop	Lack of clear and consolidated knowledge regarding community land-uses, resulting in an inability to study and	The geographic delimitation of various categories of community land-use areas, and their documentation using Geographic Information Systems

Activity	Without-project scenario	Wellbeing Outcomes/ Impacts achieved
	monitor community land-use areas and prioritize those intended for forest conservation.	(GIS) allows continuous remote monitoring by Carbonext. As mentioned in 4.1.2., this activity plays an important role in negative community impact mitigation, as current land-uses, as well as areas for future expansion, are identified, providing a framework for regulation of potential illegal community activity. Furthermore, it promotes the protection of areas important to community ways of life, territorial planning and understanding of community dynamics. There is an important social factor to be considered: mapping permits the documentation of community reality, translating knowledge and local culture into a material, cartographic language, which promotes visibility, legitimacy and continuity for community land-uses. Significant net-positive outcomes and impacts on livelihoods, cultural values, and forest conservation are therefore considered to have been contributed to by this activity, which would not have been possible in the without-project scenario.
School infrastructure and equipment	The community school would not have the equipment and	As mentioned in item 4.1.1, both students and teachers rely on school

Activity	Without-project scenario	Wellbeing Outcomes/ Impacts achieved
	teaching materials necessary for students to learn better. The teacher would depend on her few resources provided by the city hall to teach the students.	equipment and materials, which play a crucial role in enhancing the teaching and learning experience. For students, these resources serve as invaluable aids for organization and knowledge acquisition. Meanwhile, teachers utilize them as pedagogical tools to craft engaging and impactful lessons. Collectively, these resources foster a dynamic and enriching educational setting, fostering students' academic growth and personal development. With the delivery of school equipment and materials, the teacher and her students will be able to employ more resources with the aim of providing a better lessons and promoting better learning for students. These aspects are considered to be significant wellbeing gains compared to the without-project scenario. In the future, it is expected that the school equipment and materials distributed may have effectively contributed to improving student teaching. It is also hoped that, if the school serves other levels of education, the project can also contribute to structuring the school, if necessary.

Activity	Without-project scenario	Wellbeing Outcomes/ Impacts achieved
Training and workshop: Occupational health and safety and Personal protective equipment (PPE) distribution for worker health and safety	Continued occurrences of accidents at work and a lack of awareness on work safety. Community members would continue to work, exposing themselves to the risk of accidents, without using PPE.	As mentioned in 4.1.1, the participating community members gained understanding of the importance of preserving life through the adoption of techniques that aim to protect from and prevent accidents at work. It is hoped that in the with-project scenario, the training will make community members more aware of the importance of working safely, aiming to guarantee their physical wellbeing. The expectation is that throughout the life of the project, other community members will be able to participate in training aimed at occupational safety and using PPEs.
Community association creation	The community would face difficulties in its social organization in resolving local issues. The lack of a legally representative entity could result in less effective communication between residents and local authorities. Furthermore, the absence of leadership could lead to a decreased sense of belonging and cooperation among residents. This could negatively affect quality of life. Ultimately, the lack of a neighborhood	The community will be much more engaged in defending its interests and solving its problems. The Ipoá REDD+ project wants to make it easier for external beneficiary communities to organize themselves and form their residents' association. To this end, he will spare no effort to make this happen soon. Compared to the scenario without the project, these factors are deemed as substantial improvements in well-being.

Activity	Without-project scenario	Wellbeing Outcomes/ Impacts achieved
	association could limit the community's ability to achieve its goals and resolve issues such as the need to generate income.	
Women's workshop	There would be no clear understanding of the real needs, challenges and desires that women in external communities have and face. Women would continue to live in the communities without any prospect of exclusive benefits for them.	As mentioned in 4.1.1, the project will have a Understanding the needs and aspirations of women within the project's external communities is crucial, along with the potential for future initiatives to address them and promote their advancement. This is anticipated to enhance active involvement in community-related decisions, empowering women not only to generate income but also to contribute meaningfully to their communities' development.

4.1.4 Protection of High Conservation Values (CM2.4)[132]

The diagnosis of community High Conservation Values (HCVs) was conducted in February 2023, and the research quantified the following HCVs:

Table 78. HCVs benefitted by the project and key parameters for their monitoring.

Relevant HCV	Key parameters	Monitoring Means
HCV 4	17,320.33 hectares of remaining forest conserved in the PA	Remote GIS monitoring Field mapping activities

[132] These high conservation value criteria are based on those defined by the High Conservation Value (HCV) Resource Network (see: http://hcvnetwork.org/). Practical help is available for using HCVs in each region, including generic guidance documents (Toolkits) and Country Pages.

	87.14 Hectares of deforested areas cultivated by communities in the PZ mapped 76.72 hectares areas disturbed by human activity in the PZ mapped Activity not carried out in this MR: N° of initiatives promoting community forest preservation, or other aspects of ecosystem services, in the PZ.	
HCV 5 HCV 6	16[133] areas that are fundamental for the livelihoods of communities mapped. Of which the following specific types were identified: Fishing area: 5 areas Copaiba extraction: 4 areas Brazil nut collection: 3 areas Timber harvesting: 1 area; 0.08 ha Churches: 3 areas	Participatory community mapping workshops Remote GIS monitoring

Regarding HCV 4 in the table above, the following indicators were mapped:

I. 17,320.33 hectares of remaining forest conserved in the PA;
II. 87.14 Hectares of deforested areas cultivated by communities in the PZ mapped;
III. 76.72 hectares of geographic delimitation of the areas disturbed by human activity in the PZ ;
IV. While the key parameter "n° of initiatives" has been left as is because the activities have not yet been carried out, being that the present period is MR1, and they will be conducted as of MR2.

Regarding parameter I, above, although some impact has been detected on forest in the PA and the PZ, the project's Forest Integrity Plan and Operational Procedures, which comprise the core of the present REDD project, intend to continuously improve the conservation of forest and related ecosystem services comprised within HCV 4.

Regarding parameters II and III above, it is considered that these values should be considered as baseline levels of the parameters, as they were collected in February 2023, without time for project activities to address community-driven deforestation have taken significant effect. They will be measured again at each MR to compare their performance.

[133] See evidence folder 06. HCVs

Still regarding these two parameters, it is key to note that controlled expansion of community alternative land-uses in the PZ is predicted and mapped by the participatory mapping activity. Therefore it will be key to observe in future MRs whether community deforestation has occurred within or outside areas foreseen for expansion.

Meanwhile the participatory mapping activity will be repeated in future MRs, to continue to map the community impacts on forests quantified by HCV 4 (points II and III, above), while respecting and allowing their traditional means of food production and subsistence.

Regarding HCVs 5 and 6, the participatory mapping activity carried out in MR1 attended to the indicator foreseen in the PD, "n° or size of areas of importance for livelihoods or sociocultural value of communities mapped", providing important cultural and environmental knowledge of the region.

No potential risks to HCVs 4, 5 and 6 in the areas mapped by the participatory mapping workshop were identified, since the aim of the activity to consult the community, and resultingly map, and monitor HCV areas.

4.2 Other Stakeholder Impacts

4.2.1 Mitigation of Negative Impacts on Other Stakeholders (CM3.2)

Regarding the institutional actors outlined in PD section 2.1.9, it is considered that the actions foreseen under the IPOÁ REDD+ project have the potential to generate positive impacts and synergize with their objectives. Such positive impacts are expected to be achieved through the establishment of cooperative relationships between Carbonext and the institutional stakeholders, supported by laws and ethics.

Concerning the community stakeholders, during the initial surveys of the Project Zone and surrounding communities, no stakeholders were identified other than those described in the Project Zone that had a direct relationship with the Project Zone and used or depended on the resources within it.

4.2.2 Net Impacts on Other Stakeholders (CM3.3)

The project's actions taken regarding the deforestation events in MR1, and throughout its lifetime, aim to generate net positive impacts not only for the external stakeholders involved, as well as contributing to changing negative deforestation scenarios on a broader scale.

4.3 Community Impact Monitoring

4.3.1 Community Monitoring Plan (CM4.1, CM4.2, GL1.4, GL2.2, GL2.3, GL2.5)

This topic presents the activities carried out within the period of this Monitoring Report, dates, sampling methods used, and other information regarding the monitoring process. In addition, the results of the monitoring are presented.

The table below is a summary of the activities carried out, and the evidence will be duly presented during the verification audit, as some may contain sensitive data that will not be presented in this table.

Table 79. Activities carried out during the MR1 period.

	Scope	Activity	Date	Indicator/Output from the PD	Data Collection Method/ Tool	Baseline	Results/Outputs MR 1	Risks
Community	**Health care**	Basic Health Facility (BHF)	November 2022 - December 2023	Electricity and telecommunication infrastructure installed and functioning in the BHF. Nº people (by gender/age) receiving ongoing tele-medicine medical care. Nº of medicines and medical supplies procured/transported for the BHF. Nº and effectiveness of meetings carried out to facilitate community-government PPP	Meeting notes	Meetings: 0	Activity not conducted during this MR Activity not conducted during this MR Activity not conducted during this MR 6 meetings carried out to facilitate community-government PPP with respective effectiveness ratings: Nº1:2, Nº2:2, N°3:1 N°4:2, N°5: 2, Nº6:1.[134]	Internal community conflicts of interest relating to access to healthcare initiatives Risks inherent to various healthcare procedures, including tele-medicine or emergency transport. Potential conflicts between stakeholders, service

[134] See evidence folder “01. Healthcare\Basic Health Facility”

		Support for health transport		Nº equipment items and operational supplies purchased/ transported for water ambulance. Nº of community training events relating to medical transport. Nº people transported for medical using project supported water ambulance. Nº and effectiveness of meetings carried out to facilitate community-government PPP	Meeting notes		Activity not conducted during this MR Activity not conducted during this MR Activity not conducted during this MR 4 meetings carried out to facilitate community-government PPP. Effectiveness of meetings, respectively: Nº1:2 Nº2:2 N°3:1 N°4:2 [135]	providers and the law.
	Education	School infrastructure and equipment	February 2023	Nº of educational equipment and material purchased or transported. (WB)	Statement of responsibility and receipt of materials signed by the community.	Gasoline: 0 L Oil: 0 L	39 educational equipment and material purchased or transported. [136]	Community conflicts of interest relating to access to education. Damage, loss, or misuse of educational equipment. Any situation resulting from the project's education initiatives creating discomfort or distress for students and resulting in complaints to the project's contact line
				Nº of electricity and telecommunication equipment installed and maintained within education facilities.	Contracts with suppliers and invoices.	Solar panels: 0 Antenna: 0	19 electricity and telecommunication equipment installed within education facilities. [137]	
			2022 onwards	Nº and effectiveness of meetings carried out to facilitate community-government PPP.	Minutes and notes from meetings; and screenshots of conversations in chat	Meetings: 0	[138]4 meetings held with municipal Secretaries of Education, Town Hall Attorney, and with Town Pedagogical Coordinator.	

[135] See evidence folder "01. Healthcare"
[136] See evidence folder "02. Education\Education equipment"
[137] See evidence folder "02. Education\Education equipment"
[138] See evidence folder "02. Education\Meetings PPP"

					applications.			
		Teacher support and training	February 2023	0 enrolments in training and educational courses. 0 Sum spent (R$) on teacher and student training and courses. Progress in key subject areas (by gender/age) of students and teachers Student satisfaction survey	Prints of the EAD course registration, Whatsapp application conversations and meeting minutes. Contract with the specialist company hired to tutor the teacher and students, payment invoices.	Enrollments: 0	2enrollments in training and education courses. [139] BRL 20,900 spent on teacher and student training and courses. [140] 10 students (Girls: 2, Boys: 8) students studying at PZ school [141]	
	Occupational Health and Safety	Training and workshop: Occupational health and safety	July. 2023	Nº people (by gender/age) trained and educated in best practices for occupational health and safety.	Questionnaires	Training: 0	12 people (Men: 9; women:3) trained on best practices for occupational health and safety[142]	Damage, loss, or misuse of health and safety equipment Unavoided risks relating to occupational hazards
		Personal protective equipment (PPE) distribution for worker health and safety.	July. 2023	Nº people (by gender/age) given PPE equipment.	Questionnaires	Equipment: 0	12 people (Women: 2; men: 10) given PPE equipment.[143]	

[139] See evidence folder "02. Education\Enrolments in training"
[140] See evidence folder "02. Education\Alicerce\Pagamentos"
[141] See evidence folder "02. Education\Students in PZ"
[142] See evidence folder "04. Occupational Health and Safety\People trained"
[143] See evidence folder "04. Occupational Health and Safety\Given PPE equipment"

	Communication and Electrification	Installation and maintenance in communities of infrastructure and equipment for electricity, telecommunication, and internet access	February 2023	Nº equipment items purchased and transported.	Contracts with suppliers, and invoices.	Equipment: 0	Hybrid Off Grid Solar Kit with four (4) panels and one (1) communication antenna. [144]	Internal community conflicts of interest relating to access to communication and electrification equipment Damage, loss, or misuse of communication or electrical equipment Potential accidents involving electrical or solar equipment
				Nº equipment items installed and maintained in community buildings.	Contracts with suppliers, and invoices.	Equipment: 0	Hybrid Off Grid Solar Kit with four (4) panels and one (1) communication antenna[145]	
				10 people (by gender/age) with access to electricity	Questionnaires.	People with acceess to: 0	The ten (10) students who will attend the school have access to electricity,	
				Nº of people (by gender/age) with access to electricity and communication technology.	List of enrolled, list of approved students and Social diagnosis of the region.	People with access: 0	The ten (10) students who will attend the school have access to electricity, and the ninety-four (94) residents of the Sucunduri River communities have access to the benefit of the internet*.[146]	
				Number of devices in the Project Zone	Questionnaires.	13 telecommunication devices in the Project Zone.	Thirteen (13) cellular devices were accounted for.[147]	
				Nº of community members (by gender/age) educated and trained in: i) communication technology use and ii) maintenance.	Video of the training moment and list of participants.	Member trained: 0	Seven (6) (Men: 3 Women: 3) people were trained on the day the equipment was installed to perform minor configuration changes and	

[144] See evidence folder “05.Communication and Electrification”
[145] See evidence folder “05.Communication and Electrification”
[146] See evidence folder “05.Communication and Electrification/”Diagnostico Social” and folder “Students.”
[147] See evidence folder “05.Communication and Electrification “Diagnostico_Celulares”

						simple maintenance. [148]	
			4 domestic appliances per household or community building	Questionnaires.	Domestic applances : 4	4 domestic appliances per household or community building[149]	
Employment	Community association creation	July 2023	Consultant identified/ hired for community association creation. Association constitution initiated, finalized, or maintained. Nº of members and official positions in association. Nº of women promoted to become members and official positions in association.	Questionnaires.	Association: 0	One (1) community association consultant in the field to assess the situation of the community's informal association and advise on next steps.[150]	Any situation resulting from the project initiatives creating discomfort or distress for participants and resulting in complaints to the project's contact line
High Conservation Values	Survey of High Conservation Value areas and species	February 2023	Nº of HCVs of economic and sociocultural importance to the communities identified.	Attendance List, Photos and Videos, Participatory Mapping Report, HCV List and Maps.	Identified Areas: 0	Five (5) Fishing areas, Four (4) areas of Copaíba extraction, Three (3) areas of Brazil nut collection, One (1) area of timber harvest, and Three (3) churches or areas of cultural value.[151]	Any situation resulting from the project initiatives creating discomfort or distress for participants and resulting in complaints to the project's contact line
			Community participation in identification and conservation of HCVs.	Attendance List, Photos and Videos, Participatory Mapping Report.	Identified Areas: 0	The three (3) communities participated in the Participatory Mapping Workshop.[152]	

[148] See evidence folder "05.Communication and Electrification"\Member_trained
[149] See evidence folder "05.Communication and Electrification "Diagnostico_Social_22"
[150] See evidence folder "19.Employment"
[151] See evidence folder "11. Participatory Mapping Activity"
[152] See evidence folder "11. Participatory Mapping Activity"

Climate	**Forest Conservation**	Remote Monitoring	July and August 2022	Nº of deforestation events detected remotely.	Notification Emails and Maps	Events: 0	The area started remote monitoring in CarbonEye in June 2022, and some events were dectected:[153] 1) The road in Maanaim Farm in July 2022; 2) The Community deforestation in Maanaim property, August 2022.[154]	
		Deforestation prevention and mitigation actions	September and October 2022	Nº of deforestation events visited in the field.	Photographs	Events: 0	The events that were dectected were verified in the field: 1) The road at Maanaim Farm in September and October 2022; 2) The Community Ridging at Maanaim Farm in October 2022.[155]	
			July 2022	Nº and effectiveness of meetings with **command and control organs** carried out to combat deforestation.	Notifications/Bulletins of occurrence	Meetings: 0	Five (5) Notifications/Bulletins of occurrence in responsible command and control agencies.[156]	
			September 2022	Nº of infrastructure items constructed for deforestation combat.	Photographs and invoices.	Infrestructures: 0	There were two (2) identification plaques installed at Manaaim Farm	
			June 2022	Nº of protocol applications (deforestation event protocol; forest integrity protocol, etc.)	Maps, meeting minutes, forms, etc.	Protocols: 0	With the alert of illegal deforestation at Fazenda Maanaim (road) , one (1) deforestation protocol was initiated.	

[153] See evidence folder “07. Remote Monitoring”
[154] See evidence folder “07. Remote Monitoring”
[155] See evidence folder “07. Remote Monitoring”
[156] See evidence folder “07. Remote Monitoring”

	Further Leisure, Cultural and Sporting Activities	Relationship-building actions with communities such as festive, cultural, and sports activities.	April 2022	Nº of actions carried out with communities.	Figures, photos of the posters, spreadsheets, invoices, and technical report.	Events: o	There was the distribution of 100 (un) of oral hygiene kits to all three communities involved in the project, and the distribution of 150 (kits with 3 units) ecological sanitary pads to the female public in the three communities involved in the project.[157] 9 women involved in an activity about their plans and opportunities for the community[158]	Any situation resulting from the project initiatives creating discomfort or distress for participants and resulting in complaints to the project's contact line
			April 2022	Nº of participants (by gender/age) in such activities	Attendance List	Participants: 0	33 participants (Women: 19 Men:14)[159]	

The implementation of activities on the ground is governed by a number of protocols and plans, as fully presented in PD section *2.1.11 Project Activities and Theory of Change.* The complete list of Protocols applicable to the project listed in the PD follows below, however it is important to note that during the present monitoring period, only protocols number 1, 2, 5, 6, 7 and 8 were applied:

1. OP for community conflict resolution
2. OP for stakeholder communication
3. OP for irregular or illegal community activity (Annex III of FIP)
4. OP for community health and safety at work
5. OP for definition of alternative land-uses (Annex IV of FIP)
6. FIP – Forest Integrity Plan
7. Responsibility Agreement and Workshop
8. OP for deforestation events (Annex II of FIP)

[157] See evidence folder "09. Other activities"
[158] See evidence folder "09. Other activities\Women's Workshop"
[159] See evidence folder "09. Other activities"

The application of Protocols 6 and 8 is detailed in MR section 2.5.4, the remaining protocols' relevance to the present project follows below:

- The PO for Conflict Resolution was applied on 16/11/22 regarding a situation of certain members of one community expressing dissatisfaction due to installation of communication and electrification equipment in the neighbouring community. Full details are provided in the evidence folder[160]. The forms in which this mitigates negative impacts to the community are reported in section 4.1.2.
- The PO for Stakeholder Communication was applied in June 2022, September 2022, and January 2023, to inform community members of the upcoming site visit by the Ipoá project team. Full details are provided in the evidence folder[161].
- The OP for defining alternative use areas was applied in February 2023 to carry out participatory mapping of community land-use areas, in addition to allowing the identification of HCVs. Full details are provided in the evidence folder[162].
- The Responsibility Agreement and Workshop was applied in February/2023, to establish the community's responsibility for the best use of the solar panels and internet antenna. The agreement was signed by the community leadership. Full details are provided in the evidence folder[163]

[160] See evidence folder "\12. Operational Procedures\Applied"
[161] See evidence folder "\12. Operational Procedures\Applied\OP Stakeholder Communication"
[162] See evidence folder "\12. Operational Procedures\Applied\OP for definition of Alternative Land-Uses"
[163] See evidence folder "\12. Operational Procedures\Applied"

4.3.2 Monitoring Plan Dissemination (CM4.3)

The monitoring plan results will be made available online to the voluntary carbon market audience through the VCS Monitoring Reports, which are published after verification on the IPOÁ REDD+ Project page of the VCS website[164].

Regarding the institutional stakeholders identified in PD section 2.1.9, key MR1 monitoring results, including the link to the aforementioned VCS page, including the finalized documents, will be sent on the occasion of the MR2 stakeholder consultation.

For the project's communities, presentation workshops regarding monitoring period results will be conducted in the field by a Carbonext Social Analyst after each monitoring period. They will be delivered in Portuguese, in accessible language, and with visual prompts.

4.4 Optional Criterion: Exceptional Community Benefits

4.4.1 Short-term and Long-term Community Benefits (GL2.2)

The wellbeing benefits of the Communication and Electrification activities in this remote and disconnected region should not be underestimated. In terms of personal finances, thanks to the installation[165], the community can now stay up-to-date with vital information about their federal government benefits. The benefits provided by the Brazilian State sometimes send notifications and notices about registrations and payment schedules that the community could not access, requiring 7-10 hours of boat travel, considerable fuel costs, use of paid internet, or going directly to the specialized service stations located in the nearest cities. Cases of people who have lost their right to benefits because they did not receive the necessary information to update their registration have been related to Carbonext field staff.

In the area of health, the local hospital can be informed to prepare transportation for the population, and family members can be informed and updated on health cases. Before the satellite installation, the distances and costs involved to get to hospital meant that community members had no information on

[164] https://registry.verra.org/app/projectDetail/VCS/3324
[165] See evidence folder: "5.Communication and Electrification\Community Members' Report"

relatives that were hospitalized. There have been reports of family members who died in the hospital without the people of the community being informed.

The employability actions undertaken in the present MR aim to lay the foundation for establishment of a community association. A specialized consultant was hired to provide training to the communities on associations and cooperatives, fully described under section 2.3.14 [166]. An association had existed previously, and despite being respected by the community members, it lacked legal formalization, which hampered it in terms of both production and sales. Legal formalization and strengthening of the association will help the community develop new opportunities for generating income, significantly.

According to the anthropologist hired by project developer, the women of the community have great potential. In her July 2023 report[167], she presents opportunities for women to generate income, as well as other development paths such as training courses, direct training in the community, and instruments that would strengthen women's empowerment and financial independence.

The table below presents the short- to long-term benefits contributed to by the Ipoá REDD+ project during the MR1 period. Specifically, the short-term outputs correspond to the performance indicators related to wellbeing listed in section 4.4.1 of the PD, while the medium-term outcomes and long-term impacts described are those, originally laid out in section 4.4.1 and appendix 2 of the PD, which are considered to have been contributed to during the MR1 period.

Table 80. Short- to long-term benefits achieved and contributed to during the MR1 period: the outcomes and the long-term benefits to the impacts.

Activity	Short-term outputs[168]	Medium-term outcomes	Long-term Impacts
Installation and maintenance of infrastructure and equipment for electricity, telecommunicati	5 equipment items purchased and transported. 5 equipment items installed and maintained in community buildings.[169]	Functioning electricity, telecommunicatio n, and internet infrastructure in community	Increase in community members with access to the internet and electricity, broadening their life opportunities. Population informed about key topics such as world

[166] See evidence folder “09. Other activities\Women's Workshop”
[167] See evidence folder “09. Other activities\Women’s Workshop”
[168] These correspond to the performance indicators related specifically to well-being in section 4.4.1 of the PD
[169] See evidence folder: 05.Communication and Electrification

on, and internet access	10 people with access to electricity[170]:Boys: 08; Girls: 02 29 people with access to communication technology: Men: 09; Women: 08; Boys: 8; Girls: 4[171] 13 Telecommunication devices in the Project Zone.[172] 6 Community members educated and trained in: i) communication technology use and ii) maintenance: Men: 3; Women: 3[173] 4 domestic appliances per household or community building[174]	residences of the PZ. Population more informed on many topics from national current affairs to REDD+ and project concepts.	and national affairs, REDD+ methodologies and the IPOÁ Project itself, increasing their influence and autonomy. Population able to communicate at the pace of the modern world. Contribution to ensuring access to affordable, reliable, sustainable, and modern energy (SDG 7). Contribution to industry, innovation, and infrastructure (SDG 9). SDG 9.C “Significantly increase access to information and communications technology and strive to provide universal and affordable access to the Internet in least developed countries (…)”.
Employment		Strengthened community governance. Improved ability for communities to	Community entrepreneurship enabled.

[170] See evidence folder: 05.Communication and Electrification
[171] Corresponding to the total number of residents in the Boa Esperança community. See evidence folder: 05.Communication and Electrification/”Diagnostico_Social_22
[172] See evidence folder: “\05.Communication and Electrification”; document: “Diagnóstico_Celulares_Kobo_Fev - Telecommunication Devices Diagnostic”
[173] See evidence folder: “\05.Communication and Electrification\Member_trained”
[174] See evidence folder: “\05.Communication and Electrification\”Questionários_jul23”

	1 Consultant identified/ hired for community association creation.	demand and obtain public services. Increase in community members (by gender) becoming members and occupying official positions in association.	Improvement in income generation and the quality of life of employees. Reduction of poverty (SDG 1). SDG 8.3 Promote development-oriented policies that support productive activities, decent job creation, entrepreneurship, creativity and innovation, and encourage the formalization and growth of micro-, small- and medium-sized enterprises, including through access to financial services. Contribution to sustainable communities (SDG 11)
Further Leisure, Cultural and Sporting Activities	2 of actions carried out with communities. 30 participants in the activity: Women: 19; Men: 14	Increase in number of community members with access to leisure and cultural activities. Improved intra-community relations. Improved inter-community relations between the PZ communities. Improved relations between	Improved quality of life for community members. Contribution to community Good Health and Wellbeing (SDG 3). Contribution to Equality Gender (SDG 5).

		communities and project proponents and developer.	
Forest Conservation	2 protocol applications regarding community deforestation (WB)[175] 5 meetings carried out to combat deforestation (WB).[176] Effectiveness of meeting N°1[177]: 2 Effectiveness of meeting N°2: 1 Effectiveness of meeting N°3: 1 Effectiveness of meeting N°4: 1 Effectiveness of meeting N°5: 1	Reduction in deforestation within the PZ Reduction in invasion and land-squatting attempts within the PZ	Conservation of 4,200.43 ha of forest, corresponding to avoided net emissions of 2,522,057 tCO2e over 30 years. Contribution to Life on Land (SDG 15). Contribution to Peace, Justice, and Strong Institution (SDG 16).

Given the evidence above relating to the activities carried out, and the significant impacts they have had on the residents' wellbeing, it is considered that the project is on track to generate short-term and long-term net positive well-being benefits for smallholders/community members as predicted in the PD.

4.4.2 Marginalized and/or Vulnerable Community Groups (GL2.4)

[175] See evidence folder "12. Operational Procedures."
[176] See evidence folder "\08. Deforestation\22.05.12 - Estrada Maanaim" and "22.06.30 – Explanatory note".
[177] The effectiveness is based on internal classification where the meetings are ranked in 1, 2 or 3 being 1 lower effectiveness and 3 higher.

As cited in the PD, according to the Socio-Economic Diagnosis, all the communities visited in the Reference Region are considered marginalized and vulnerable due to the difficulty of access to public policies that guarantee basic rights such as infrastructure in health, education, transportation, communication. The communities involved in the IPOÁ REDD+ Project have a low level of social organization, which weakens the local conditions to seek for these fundamental rights.

In this sense, an important measure for the access to public policies and the guarantee of exceptional rights in the communities is the empowerment of the community, based on the strengthening and consolidation of social organizations, aiming at the effective participation of community members in decision making, contributing to the improvement of socio-economic aspects by the community members themselves.

At the same time, the Vila Batista community may be considered as further marginalized in relation to Boa Esperança, due to the existing feud between the two villages. Currently, solar panels and internet access antennae are only installed in Boa Esperança, but this action is planned to be extended to Vila Batista in the next MR period. Therefore, the table below only refers to the other marginalized population identified by the project: that of women.

Still in this context, it is also important to emphasize that the approximately 12 million women who inhabit the Brazilian Amazon are the most subordinate class, within a subordinate region, within a subordinate country of Latin America (CHAVES & CESAR, 2019). The working conditions of women in the Amazon remain inferior to other states in Brazil. Women in the Amazon earn approximately 80% of the average income of Brazilian women, and 70% of men in the region. Furthermore, studies suggest that they hardly benefit at all from public social security policies (BRASIL, 2008).

Community Group	Women of all the communities in the Project Zone: Vila Batista, Boa Esperança, and Vale da Benção.
Net positive impacts	Action: Installation and maintenance of electricity, telecommunications and internet access infrastructure and equipment. Specific net positive impacts on women are: internet access permits critics knowledge regarding government benefits, and enables access to medical facilities, such as scheduling appointments in advance of visits to the city. Furthermore, the improvements that this activity brings to all community members equally apply to women. These include:

	benefits in health, education, information, communication, leisure, as well as access to the enormous amount of resources on the world wide web.
Benefit access	The installation of electrification and communication reaches all members of the community, including women. Thus, to date, no barriers have been identified that would prevent women from accessing this benefit of the project. To date, no indications or reports of any impediments to women accessing the benefits of the project have been collected or received.
Negative impacts	Repeated visits to the project area, and the always-open communication channels permit the continued identification of marginalized or vulnerable community members. For instance, the installation of solar panels and an internet antenna in the Boa Esperança community could potentially intensify the existing dispute between communities, which was identified as of February 2023. With this in mind, the Project has the Protocol "OP for community conflict resolution" in place, and during the present monitoring report, only one incident was recorded, which was not directly related womens' issues[178]. The communication channel is in place to monitor and address any increase in such negative impacts. Furthermore, for the next MR, the same technologies are planned to be installed in the Vila Batista community.

[178] See evidence folder: "12. Operational Procedures\Applied"

Community Group	Women of all the communities in the Project Zone: Vila Batista, Boa Esperança, and Vale da Benção.
Net positive impacts	Action: Women's Workshop Specific net positive impacts on women are: Women generating income through activities encouraged in the project. Developing the skills needed to generate employment and income Female gender empowered and aware of the importance of their role within the local social context
Benefit access	Research on the needs and desires of the female members of the PZ communities, enabling the possibility of carrying out future actions to meet them. The activity is expected to result in opportunities for income generation, and to increase women's active participation in community decisions.
Negative impacts	Potential conflict between women's traditional roles in the home and family and the new activities that the workshops enable.

4.4.3 Net Impacts on Women (GL2.5)

It is observed that the project activities implemented in MR1 reach all community members, including women, in an equitable fashion. For example, the communication and electrification indicator shows that all the women and girls from the Boa Esperança community were benefitted by the activity conducted in that community (8 women, 4 girls, 9 men, and 8 boys).

The actions judged as having the largest impacts on women are related here first. As of August 2023, the female community teacher benefitted from specialized training from a leading education institute in Brazil, Alicerce Educação[179]. This activity aimed to improve the teacher's professional training, including her own mastery of school fundamentals and her teaching skills themselves, so

[179] See evidence folder: 02. Education\Alicerce

that the quality of lessons greatly improves. The expected impacts of this action, in accordance with the Theory of Change (Appendix 2 of the PD) are that the community schoolgirls[180], as well as the boys, and the teacher herself, experience a dramatic improvement in quality of their education and their workplace, surpassing the regional and even national average education levels. Women are central to these transformative actions.

Secondly, as described in section 2.1.1, a women's workshop was held in July 2023 with 9 community women in attendence, led by a specialized consultant, with the objective of activity identifying their main desires and needs[181]. The aim for the coming monitoring periods is that this results in the implementation of project actions aimed exclusively at women.

Other various activities and project development actions that involved women are as follows: The women of all the PZ communities played an equitable role in the project decision making, being that of 24 participants in the activity design workshop, 10 were women[182], and many of those drawing attention to the need for electricity and internet signal, for instance, were women. Furthermore, a specific meeting just for women was held during the community consultation, in which they were encouraged to discuss what they like, what they would like to change, and what they see for the future for their community, which was called the "dreams phase"[183]. In February 2023, women were involved in the implementation of the project, notably the installation of solar panels and the Starlink antenna.

The summary of the impacts of the various activities implemented in MR1 are summarized in the table below.

Community Group	Women of the Boa Esperança community	
Impact/outcome	Communication and Electrification	The expected wide-ranging impacts are: - an increase in female community members with access to the internet, telecommunications, and electricity; - solar power will increase female community members' access to electric light; electricity for refrigeration; electronic equipment; and ventilation, among others.

[180] See evidence folder: 03. Teacher and student education\Enrolled students\2023
[181] See evidence folder: 09. Other activities\Women's Workshop
[182] See evidence folder: "\09. Other activities", file: Lista de presença_genero_idade
[183] See evidence folder: 09. Other activities\Women's Workshop

	Education	Outcomes: Improved additional training for female teachers Improved quality of lessons for female students. Impacts (among others): Contribution to quality education within the PZ (SDG 4)
	Distribution of oral hygiene kits and ecological women's sanitary pads[184] Women's Workshop	This is a one-off action to improve women's health and well-being with sanitary products, therefore the benefit is short-term rather than profound impact. Consider women's main demands when planning the project's actions; Women developing the skills needed to generate employment and income; Female gender empowered and aware of the importance of their role within the local social context.
Output	Communication and Electrification	08 Women and 04 girls with access to communication technology.[185] 02 girls with access to electricity. [186]
	Distribution of oral hygiene kits and ecological women's sanitary pads	10 women received sanitary products 33 participants (Women: 19; Men: 14) in the activities
	Women's Workshop	18 women participants 19 proposals for activities with women

[184] This activity is under the "Further Leisure, Cultural and Sporting Activities" scope of the theory of change, involving one-off, or occasional short-term actions
[185] Corresponding to the total number of women and girls in the Boa Esperança Community, see evidence folder: "\05.Communication and Electrification", document: "Diagnostico_Social_22"
[186] Corresponding to the total number of girls benefitted by solar panels installed in the school. See evidence folder: \05.Communication and Electrification\Students

	Education	2 Girl students studying at PZ school 1 female teacher at PZ school
Type of Benefit/Cost/Risk	Communication and Electrification	Benefit type: actual and direct. With the installation of solar energy and starlink internet in the Boa Esperança community, female community members were enabled to communicate with transformative ease, resolve issues relating to transport, health, education, civic duties, or any other subject, remotely. The panels installed at the school will supply energy to the location, enabling the use of electronic equipment, such as computers, televisions, cell phones, etc. Risk type: predicted and indirect. The solar panels and internet were installed in the Boa Esperança community, benefiting directly from the project. An observed risk is that female members of other communities may feel discredited for not having been directly served by the project activity. Cost: no community costs none identified
	Further Leisure, Cultural and Sporting Activities: Distribution of oral hygiene kits and ecological women's sanitary pads	Benefit type: actual and direct. The delivery of sanitary products to women contributed to the short-term improvement of female health and hygiene. Risk type: predicted and indirect. Women may interpret the supply of sanitary products negatively, thinking that the person who provided the products thinks that they do not use sanitary products.

		Cost: no community costs none identified.
	Women's Workshop	Benefit type: Knowledge about the needs and desires of the female public in the project's external communities and the possibility of carrying out future actions to meet them with a view to their progress. It is expected to increase active participation in decisions related to communities and that they can, in addition to being able to obtain income. Risk type: possibility that they are no longer interested in participating in activities due to their duties at home, among other reasons. Cost: no community costs none identified
	Education	Benefit type: actual and direct. The two girls who regularly attend the school in 2023 are enabled to to improve their reading, writing, mathematics and general knowledge skills. Risk type: predicted and indirect. Risk: The school depends on just one teacher who, for various reasons, may no longer be able to teach at the school in future years. Furthermore, it may be that the municipal education department is unable to offer school meals to students, resulting in a reduction in class hours or even the interruption of classes. Cost: no community costs none identified.
Change in Well-being	From the information above, we can observe that no direct, actual negative impacts, risks or costs to women, were identified as a result of the activities implemented in MR1. Meanwhile the benefits identified are	

	direct and actual. Some of the benefits have direct effects that could be described as transformative, notably the access to internet in the region that had hitherto been isolated, with all the deep-reaching essential capabilities, advantages, and conveniences that modern communication provides. The impact on local women is therefore evaluated as positive, involving all the PZ communities, with the community most directly benefited being Boa Esperança.

4.4.4 Benefit Sharing Mechanisms (GL2.6)

The means through which project benefits are distributed to communities is through its socio-economic, biodiversity, climate, and other initiatives, which were designed in a participatory manner, to meet the communities' needs as they see them. A particularly notable aspect of the Ipoá REDD+ Project is the up-front commitment of a specific percentage of VCU sales to CCB actions throughout the lifetime of the project[187], which forms the backbone of the present CCB project.

During the present monitoring period, the sum invested into project actions equates to R$ 9,668.56 per person benefitted, equivalent to 13.8 extra months' earnings per person, taking the regional average income as a benchmark[188].

During the present monitoring period, the primary investments were in (in order of highest to lowest cost): solar panels, PPE's distribution, school infrastructure and equipments, teacher and student education and training, Starlink internet installation, Activities for Well Being promotion and hygiene, including distribuition of Women's Sanitary products, community association creation, women workshop, training and workshop: occupational health and safety, antenna test, telephone frequency test and finally sign installations . The table below displays, in the proportion column, the percentage of total project budget during this monitoring period that was allocated to each of these activities[189].

[187] See evidence folder "13. Project Budget"
[188] In accordance with PD section 4.1.1., the PZ's population make under half minimum wage (R$1.212)
[189] See evidence folder "13. Project Budget"

Table 81. Benefit sharing: Proportion of activities in total MR1 budget.

Scope		Period	Activity	Proportion (%)
Social	**Communication and Electrification**	**out/22**	Telephone frequency test	**6.2%**
			Antenna Test	**6.3%**
		fev/23	Solar panel installation	**13.9%**
			Starlink internet installation	**8.4%**
	Occupational health and Safety	**jul/23**	Training and workshop: Occupational health and safety	**6.4%**
			Personal protective equipment (PPE) distribution for worker health and safety.	**11.6%**
	Employment	**jul/23**	Community association creation	**7.5%**
	Education	**jul/23**	School infrastructure and equipment	**10.2%**
			Teacher and student education and training	**9.8%**
Climate	**Forest Conservation**	**jul/22**	Sign Installation - Maanaim Farm	**2.3%**
			Sign Installation - Campos Farm	**2.3%**
Other	**Further Leisure, Cultural and Sporting Activities**	**abr/22**	Activities for Well Being promotion and hygiene, including distribuition of Women's Sanitary products	**8.1%**
		jul/23	Women Workshop	**6.9%**

4.4.5 Governance and Implementation Structures (GL2.8)

A particularly notable aspect of this project's Governance and Implementation Structure is the financial commitment, which makes all the CCB socio-environmental activities possible. The project owner committed, at the suggestion of Carbonext, a predetermined proportion of the total credits sold to be reinvested into the social and environmental activities over the lifetime of the project.

The project's actions are designed to enable the full and effective participation the community members from design to implementation, starting with the social diagnosis activities in March and April 2022, and again in October 2022, enabling communities to direct the actions of the project which will benefit them.

In April 2022, an initial diagnosis was carried out with women from the community, aiming to better understand their social context[190], which was followed up in July 2023 with a more robust women's workshop led by an experienced consultant (see sections 4.4.1 and 4.4.3). In this activity the women were invited to present their dreams through drawings that showed what they want for the community.

The project's community communication channel was established during the first site visit in March 2022, when Carbonext's social professional left his direct contact and the number of the company for notifications, facilitating full and effective participation of community members in project decision-making and implementation. Subsequently, in October 2022, the direct contact number for complaints, suggestions was implemented[191].

The table below shows the activities carried out during MR1, and the parties responsible for their execution.

Table 82. Responsible parties for the execution of each activity in the MR1 period.

<table>
<tr><th colspan="2">Scope</th><th>Activity</th><th>Responsible Party</th></tr>
<tr><td rowspan="8">Social</td><td rowspan="2">Education</td><td>Hiring Alicerce Educação</td><td>Carbonext</td></tr>
<tr><td>Delivery of school equipment and materials</td><td>Carbonext</td></tr>
<tr><td rowspan="2">Occupational Health and Safety</td><td>Hiring a consultant to carry out first aid training</td><td>Carbonext</td></tr>
<tr><td>Purchase and delivery of PPE</td><td>Carbonext</td></tr>
<tr><td rowspan="4">Communication and Electrification</td><td>Telephone frequency test</td><td>Proponent's Field Team</td></tr>
<tr><td>Antenna test</td><td>Proponent's Field Team</td></tr>
<tr><td>Solar panel installation</td><td>Carbonext</td></tr>
<tr><td>Starlink internet installation</td><td>Carbonext</td></tr>
</table>

[190] See evidence folder "09. Other Activities"
[191] See evidence folder "15. Communication Channel"

	Employment - Community association creation	Hiring a consultant to carry out training in associations and cooperatives	Carbonext
		Training in associations and cooperatives	Carbonext
Climate	**Forest Conservation**	Sign Installation - Maanaim Farm	Proponent's Field Team
		Sign Installation - Campos Farm	Proponent's Field Team
Other	**Further Leisure, Cultural and Sporting Activities**	Activities for Well Being promotion and hygiene, including distribution of hygiene products and Women's workshop.	Carbonext

4.4.6 Smallholders/Community Members Capacity Development (GL2.9)

As predicted in item 2.3.10 of the PD, the communities of the Ipoá project have had an important role in decision-making regarding the progress of the project in MR1, together with the other parties involved. Initially, the benefiting communities provided the information about the local context, their desires, needs and vulnerabilities, in April 2022, thus participating in the project design. This was fundamental to in the formation of the project's Theory of Change.

In terms of participating in the project implementation, in October 2022, the community assisted in the attempt to install a community internet, as well as assisting the field agent in installing identification signs in the property. In February 2023, the community helped the technician install solar panels and the school's entire electrification system.

In July 2023, the community received training in associations and cooperatives, provided by a specialized third-party consultant, who was hired by the project developer, and in which 26 people participated[192]. As mentioned in section 2.3.14, the wide-ranging benefits of an association or cooperative are, first and foremost, that it provides a legal entity for the production and trade of goods. Secondly, it enables the

[192] See evidence folder "19. Employment"

community to advocate for the interests and rights of its members, which may be wide-ranging such as civil rights, voting, education, and health. It furthermore enhances productivity through training, technical support, and implementing technologies for improved economic and financial management in agricultural endeavors. .

In the education scope, during the present MR, two activities for community capacity development occurred. The community teacher enrolled in an online course for continuous education on 13 March 2023[193], with the objective of providing improved teacher training. It consists of three modules focusing on the teacher's skills in: literacy; historical literacy and teacher training; digital literacy. The impacts here aim to improve quality education for the Boa Esperança community's children.

Secondly, instruction in use and maintenance of solar panels and antenna of internet was provided to the community by the specialized service provider, during the February 2023 site visit[194]. This aims at capacity-building enabling the Boa Esperança community to maintain and manage their own communication equipment.

These capacity-building activities permitted the project to effectively fulfil the expectations laid out in the project's education, and communication scopes of activity, for the MR 1 period.

[193] See evidence folder "03. Education"
[194] See evidence folder "04. Communication and Electrification."

5 BIODIVERSITY

5.1 Net Positive Biodiversity Impacts

5.1.1 Biodiversity Changes (B2.1)

During the present monitoring period, activities to prevent encroachment and illegal deforestation contributed to the protection of the fauna and flora present in the Project Zone.

However, inventory, monitoring, and of fauna and flora indicators will only occur after the inventories and actions indicated by specialists, as defined in the project's PD monitoring plan, changes in these parameters cannot yet be measured or reported.

Change in Biodiversity	Flora
Monitored Change	Actual positive and direct changes that resulted in quantitative changes
Justification of Change	Conducting the initial description of biodiversity in the project area. Bibliographic survey in the project area, which may be home to more than 358 tree species and with 18 species threatened at some level according to the international (IUCN) and national (CNCFlora) Red List of species. This initial description of the area is relevant for describing local biodiversity, setting the directions for conservation activities.

The research considering secondary bibliographic data of articles and studies in the region, the JUMA RDS protected area's flora inventory was conducted in October 2019 in the region of the AM 174 highway (Higuchi et al., 2010). 358 tree species, with the taxon *Protium sp.* (red rosewood) - Burseraceae being the most frequently recorded, followed by *Eschweilera* sp. (yellow matamata) - Lecythidaceae, a pattern commonly found throughout the Amazon region (Higuchi et al., 2010). The family with the highest number of species was Mimosoideae (29), followed by Sapotaceae (21), and Lauraceae and Papillionoidaeae (17).

From the list of inventoried species, available in the JUMA RDS Management Plan, we classified the taxa according to the level of extinction, based on the IUCN RED LIST (IUCN, 2018)[195] and the National Center

[195] https://www.iucnredlist.org/

for the Conservation of Flora (CNCFlora)[196]. According to the flora inventoried for the JUMA RDS, below are the species classified at some level of threat of extinction.

Change in Biodiversity	Fauna
Monitored Change	Actual positive and direct. That resulted in quantitative changes
Justification of Change	Conducting the initial description of biodiversity in the project area. Bibliographic survey in the project area, which may be home to more than 70 species of non-flying mammals, 612 avifauna species, 43 herpetofauna species and with 14 species threatened at some level according to the international (IUCN) and national (CNCFlora) Red List of species. This initial description of the area is relevant for describing local biodiversity, setting the directions for conservation activities.

The research considering secondary bibliographic data of articles and studies in the region for fauna, identified a total of 70 species of non-flying mammals (55 genera, 28 families, and 10 orders) were recorded. As for avifauna, 398 were recorded within the UC boundaries and 214 were recorded in the Reserve's surroundings, a total value (612), which represents half of the avifauna of the Brazilian Amazon, one third of the avifauna of Brazil as a whole, and among the richest avifauna of any protected area in the world. The herpetofauna survey resulted in 43 species of reptiles and 27 of amphibians.

The JUMA RDS contains 14 species classified at some level of threat of extinction ("VU", "EN", etc.), of which 8 are mammal species and three are species of the genus Chelonia, as indicated by the following table. There are two bird species listed as vulnerable recorded in the RDS. Of the mammals, four are large species with a wide distribution in Brazil: *Tapirus bairdii, Trichechus inunguis*, Priodontes maximus (giant armadillo), Pteronura brasiliensis (giant river otter). The largest monkey species (*Chiropotes albinasus*, *Ateles* sp., *Lagothrix* sp.) are classified as "endangered" and the tiny *Callibella humilis* is "Vulnerable." and the others are primates with restricted distributions. Furthermore, the fauna survey for the Maanaim and Campos Farms recorded, in addition to the *Tapirus bairdii* species other threatened species namely *Alouatta guariba clamitans*.

5.1.2 Mitigation Actions (B2.3)

[196] http://cncflora.jbrj.gov.br/portal/

The negative impacts on biodiversity in the Project Area are fragmentation of vegetation, encroachment on properties, illegal activities (timber extraction), road opening and hunting. To mitigate the negative impacts of possible activities, during the monitoring period the farm areas were monitored through CarbonEye and the Zoning workshop was conducted by the technical team in order to map the areas of use by the communities in the project area.

So, in this first monitoring period, no negative impacts on biodiversity resulting from the project were identified. Therefore, the measures taken for the maintenance or improvement of HCV attributes are associated with reduced deforestation, which means no habitat loss, fragmentation, or loss of biodiversity in the project area.

Table 83. Biodiversity HCVs and related mitigation measures.

HCV Related	Mitigation Measure
HCV 1: Species diversity **HCV 2:** Ecosystems and mosaics at landscape level, and Intact Forest Landscapes - proximity of PA to protected areas. **HCV 5:** Community needs - Brazil nut and copaíba extraction in the project zone and fishing in the Sucunduri River.	Constant surveillance and monitoring (CarbonEye System). Encouraging farmers in the region to start REDD+ projects. Identify and conserve HCV species and areas, as well as those that have economic and sociocultural importance to the communities: research considering secondary bibliographic data of articles and studies in the region. Identify species as those that have economic and sociocultural importance to the communities: Zoning Workshop (mapping of areas of use (fishing, location of Copaibas, planting areas...).

5.1.3 Net Positive Biodiversity Impacts (B2.2, GL1.4)

The IPOÁ REDD+ project has a positive impact on biodiversity in the Project Zone, as monitoring and surveillance of the area is carried out through Carbonext's CarbonEye system, protecting a total forest are of: baseline deforestation area 1,431.93 hectares, minus 11.61 hectares of unavoided unplanned deforestation in this monitoring report, totaling 1,420.32 hectares. In this way, deforestation has been avoided, along with associated habitat loss, fragmentation, and hunting. Therefore, the net impact on biodiversity is positive in this MR1.

Without the project, it is very likely that biodiversity would be negatively affected by land use transformation and degradation with the expansion of illegal activities and encroachments into the Project Zone.

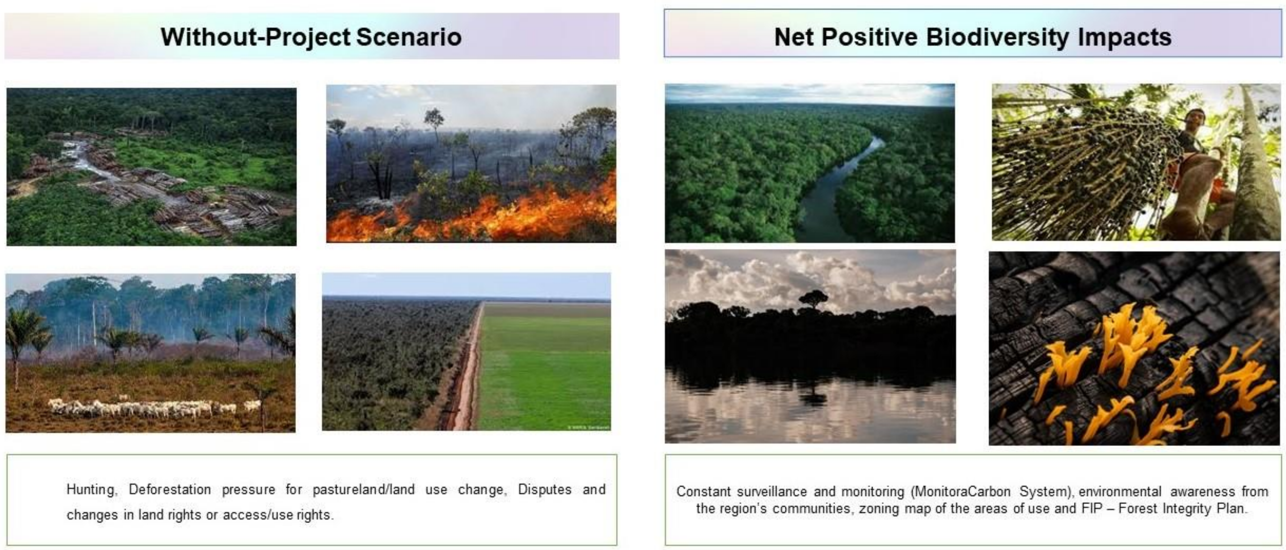

Figure 23 - Net positive Biodiversity Impacts.

5.1.4 High Conservation Values Protected (B2.4)

Project activities aim to avoid deforestation and fragmentation and maintain forest cover. No environmental impact activities will be carried out, therefore no negative effects on the HCV are expected.

Table 84. Biodiversity HCVs protected.

HCV	Activity to avoid Negative Impact	Actual Results
HCV 1: Species diversity - possible occurrence of threatened and endemic species in the Project zone	Monitoring and surveillance of the project area preventing deforestation of 1,420.32 ha of deforestation projected in the baseline (minus MR1 unavoided	Deforestation inside the PA was lower than that projected in the without-project scenario, thus it can be stated that the monitoring practices

HCV 2: Ecosystems and mosaics at landscape level, and Intact Forest Landscapes - proximity of PA to protected areas. HCV 5: Community needs - Brazil nut and copaíba extraction in the project zone and fishing in the Sucunduri River	deforestation), and associated habitat loss, disturbance and hunting. Zoning Workshop with the community.	were effective in this monitoring period. With the deforestation decrease, the negative impacts on biodiversity also decrease, because habitat loss and fragmentation of the area is minimized. Furthermore the proximity of the PA to protected areas has synergistic effects for conservation (see PD section 2.1.1).

5.1.5 Invasive Species (B2.5)

This item is not applicable as no restoration activities are planned in the area.

5.1.6 Impacts of Non-native Species (B2.6)

This item is not applicable as no restoration activities are planned in the area.

5.1.7 GMO Exclusion (B2.7)

This item is not applicable as no restoration activities are planned in the area.

5.1.8 Inputs Justification (B2.8)

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This item is not applicable as no restoration activities are planned in the area.

5.2 Offsite Biodiversity Impacts

5.2.1 Negative Offsite Biodiversity Impacts (B3.1) and Mitigation Actions (B3.2)

In this first monitoring period, no negative impacts on off-site biodiversity resulting from the implementation of project activities were identified.

5.2.2 Net Offsite Biodiversity Benefits (B3.3)

The net offsite biodiversity impacts of the Project are positive, since there are no mapped negative offsite biodiversity impacts resulting from the project's activities, and substantial benefits regarding fauna and flora conservation in the PA are foreseen. These biodiversity benefits will extend outside the project's boundaries due to factors including ecological effects, animal ranges, and dissemination of knowledge and awareness raised in local communities.

As reported in items 5.1.3, there were no negative impacts outside the project zone in this first monitoring period, and the project activities positively impact biodiversity in the project zone, since deforestation was reduced and, consequently, there was no loss of habitat in the area and fragmentation of vegetation.

5.3 Biodiversity Impact Monitoring

5.3.1 Biodiversity Monitoring Plan (B4.1, B4.2, GL1.4, GL3.4)

The project intends to develop the official monitoring plan together with specialized fauna and flora monitoring company. Thus, the objectives, sampling methods, indicators and parameters will be defined in the official biodiversity monitoring plan for the target species that will be identified *in locus* in the project area.

The monitoring plan for biodiversity is focused on the assessment of indicators listed in the Theory of Change of the Project Design (table 122), among them are: Improvement in knowledge of fauna within the

PZ or greater region; Nº of taxonomic groups studied: mammals, herpetofauna, insects; Intensity, or area of fauna sampling (Nº of plots or geographic area (ha), among others. In this first monitoring period, however, the focus of the project was on improving the surveillance and monitoring of forest areas in the project area. The direct benefit for biodiversity was the prevention of deforestation, the prevention of invasions and, consequently, the reduction of habitat loss, fragmentation, and hunting.

Table 85. Biodiversity activity monitoring results.

Activity Monitoring Results			
Scope	**Activity**	**Results**	**Period**
Surveillance	Through CarbonEye, Increase the effectiveness against invasions and illegal activities: Prevention of deforestation and degradation. Maintenance of vegetal cover.	Reducing emissions from deforestation and forest degradation;	2022-2023
HCV	Identification of the community's HCV	Start of the HCV monitoring	2022-2023
Workshops	Identify species as those that have economic and sociocultural importance to the communities: Zoning Workshop	Mapping of areas of use (fishing, location of Copaibas, planting areas).	February 2023

5.3.2 Biodiversity Monitoring Plan Dissemination (B4.3)

The results of the monitoring carried out will be made public on the Internet and through the Carbonext website. All documents and information regarding the monitoring and verification results of this project will be published on the Verra platforms.

Regarding the local stakeholders, the results obtained from the monitored will be shared during the periodic site visits, which will be arranged for events such as training or verification periods.

The biodiversity monitoring plan will be made public on Carbonext's website and Verra's website once the project is validated. Summaries of the monitoring results will be presented for the communities during the next onsite visit. For other stakeholders, the results will be sent by email.

5.4 Optional Criterion: Exceptional Biodiversity Benefits

Not applicable for the present monitoring period, as Gold Level for exceptional biodiversity benefits are not being claimed in MR1.

6 ADDITIONAL PROJECT IMPLEMENTATION INFORMATION

Below we present additional information showing how the project has been implemented in accordance with the validated Project Description, specifically the sections that discuss the Theory of Change (2.1.11 and Appendix 2), for all indicators that require implementation of an activity or process.

The baseline levels of relevant indicators, and their progress due to activities implemented during MR1, classified by pillar (social, biodiversity, and climate), are detailed in the table below.

Table 86. Baseline levels and MR1 Outputs by Scope - Social Pillar.

Scope	Activity	Baseline Outputs	Outputs MR1
Healthcare	Basic Health Facility (BHF)	0 meetings carried out to facilitate community-government PPP.	6 meetings carried out to facilitate community-government PPP.[197] Effectiveness of meetings carried out to facilitate community- government PPP[198]: Nº1:2, Nº2:2, N°3:1 N°4:2, N°5: 2, Nº6:1.
		0 people (by gender/age) receiving ongoing tele-medicine medical care.	Activity not conducted during this MR (people receiving ongoing tele-medicine medical care)
		0 of medicines and medical supplies procured for the BHF	Activity not conducted during this MR (medicines and medical supplies procured for the BHF)
		0 Electricity and telecommunication infrastructure installed and functioning	Activity not conducted during this MR (Electricity and telecommunication infrastructure installed and functioning)
	Support for health transport	0 equipment items and operational supplies purchased/ transported for water ambulance.	Activity not conducted during this MR (equipment items and operational supplies purchased/ transported for water ambulance)

[197] See evidence folder "01. Healthcare/Basic Health Facility"
[198] See evidence folder "01. Healthcare"

		0 community training events relating to water transport. 0 people transported using project supported water ambulance. 0 meetings carried out to facilitate community-government PPP	Activity not conducted during this MR (community training events relating to water transport) Activity not conducted during this MR (people transported using project supported water ambulance). 4 meetings carried out to facilitate community-government PPP[199], with respective effectiveness ratings[200]: N°1:2 N°2:2 N°3:1 N°4:2
Education	School infrastructure and equipment	0 electricity and telecommunication equipment installed and maintained within education facilities. 0 educational equipment and material purchased or transported. 0 meetings carried out to facilitate community-government PPP.	19 electricity and telecommunication equipment installed within education facilities.[201] 38 educational equipment and material purchased or transported.[202] 4 meetings carried out to facilitate community-government PPP[203], with respective effectiveness ratings: N°1:2 N°2:1 N°3:1 N°4:1
	Teacher and student education and training	0 enrolments in training and educational courses.	2 enrolments in training and educational courses[206]

[199] See evidence folder “01. Healthcare”
[200] See evidence folder “05. Communication and Electrification\Solar Energy and Internet Antenna”
[201] See evidence folder “02. Education/Education Equipment”
[202] See evidence folder “02. Education/Education Equipment”
[203] See evidence folder “02. Education/Meetings PPP”
[206] See evidence folder “02. Education/Enrolments in training”

		0 Sum spent (R$) on teacher and student training and courses. 08 students studying at PZ school [204]: Girls: 04, Boys: 04 Progress in key subject areas (by gender/age) of students and teachers Student satisfaction survey[205]	BRL 20,900 Sum spent (on teacher and student training and courses.[207] 10 students (Girls: 2, Boys: 8) students studying at PZ school [208]. Activity not monitored during this MR (Progress in key subject areas) Activity not monitored during this MR (Student satisfaction survey)
Occupational Health and Safety	Training and workshop: Occupational health and safety	0 people (by gender/age) trained in best practices on occupational health and safety.	12 people (Men: 9; Women: 3) trained in best practices for occupational health and safety.[209]
	Personal protective equipment (PPE) distribution for worker health and safety.	0 people (by gender/age) given PPE equipment.	12 people (Men: 9; Women: 3) given PPE equipment.[210]
Communication and Electrification	Installation and maintenance of infrastructure and equipment for electricity, telecommunication, and internet access	0 equipment items purchased and transported. 0 equipment items installed and maintained in community buildings.	5 equipment items purchased and transported. 5 equipment items installed and maintained in community buildings.[212]

[204] See evidence folder “03. Teacher and student education\Enrolled students\2022”
[205] Specific output suggested by specialist education service provider
[207] See evidence folder “02. Education/Alicerce/Pagamentos”
[208] See evidence folder “03. Teacher and student education\Enrolled students\2023”
[209] See evidence folder “04. Occupational Health and Safety/People trained”
[210] See evidence folder “04. Occupational Health and Safety/Given PPE equipment””

[212] See evidence folder: 05.Communication and Electrification

		0 people (by gender/age) with access to electricity 0 people (by gender/age) with access to communication technology. 7 telecommunication devices in the Project Zone. 0 community members (by gender/age) trained in: i) communication technology use and ii) maintenance. 4 domestic appliances in community households or buildings.[211]	10 people with access to electricity[213]:Boys: 08; Girls: 02 29 people with access to communication technology: Men: 09; Women: 08; Boys: 8; Girls: 4[214] 13 Telecommunication devices in the Project Zone.[215] 6 Community members trained in: i) communication technology use and ii) maintenance: Men: 3; Women: 3[216] 4 domestic appliances in community households or building[217]
Employment	Community association creation	0 participants in association training events, assemblies etc. 0 Consultants identified/ hired for community association creation. 0[218] Association constitution initiated, finalized, or maintained.	17 (day one) and 26 (day 2) participants in association training events[221] 1 Consultants identified/ hired for community association creation.[222] Activity not carried out during this MR

[211] See evidence folder: 05.Communication and Electrification
[213] See evidence folder: 05.Communication and Electrification
[214] Corresponding to the total number of residents in the Boa Esperança community. See evidence folder: 05.Communication and Electrification/"Diagnostico_Social_22
[215] See evidence folder: 05.Communication and Electrification"; document: "Diagnóstico_Celulares_Kobo_Fev - Telecommunication Devices Diagnostic"
[216] See evidence folder: 05.Communication and Electrification\Member_trained
[217] See evidence folder: 05.Communication and Electrification\"Questionários_jul23"
[218] An association has existed in the past however it was informal.
[221] See evidence folder: 19. Employment
[222] See evidence folder "19. Employment"

		0[219] Members and official positions in association. 0[220] Women promoted to become members and official positions in association.	Activity not carried out during this MR Activity not carried out during this MR

[219] An association has existed in the past however it was informal.
[220] An association has existed in the past however it was informal.

	Hiring and training monitors	0 community members identified (by gender/age) for forest monitoring work. 0 people (by gender/age) trained in forest monitoring practices. 0 monthly surveillance rounds conducted. 0 reports, protocols, and alerts sent from monitors to PP. 0 spent (R$) on monitor salaries.	Activity not carried out during this MR Activity not carried out during this MR Activity not carried out during this MR Activity not carried out during this MR Activity not carried out during this MR
Training	Sustainable and Conservation Courses	0 people (by gender/age) educated in Sustainability and Conservation practices.	Activity not carried out during this MR
High Conservation Values	Survey of High Conservation Value areas and species	**HCV 4:** 17,329.72 hectares of remaining forest in the PA at the beginning of the project 87.14 hectares deforested areas cultivated by communities in the PZ[223] 76.72 hectares disturbed by human activity in the PZ[224] **HCV 5 and 6:**	**HCV 4:** 17,318.11 hectares of remaining forest in the PA at the end of MR1 [225] Data not collected during this MR (Deforested areas cultivated in PZ) Data not collected during this MR (Areas disturbed by human activity in the PZ)

[223] See evidence folder 06. HCVs
[224] See evidence folder 06. HCVs
[225] See evidence document: IPOA_SPREADSHEET_IPOÁ_MR

		0 initiatives promoting community forest preservation, or other aspects of ecosystem services, in the PZ. 0 areas of importance for livelihoods or sociocultural value of communities mapped.	**HCV 5 and 6:** Activity not carried out during this monitoring period (initiatives promoting community forest preservation) Areas of importance for livelihoods or sociocultural values mapped: 16[226] areas that are fundamental for the livelihoods of communities mapped. Of which the following specific types were identified: Fishing area: 5 areas Copaiba extraction: 4 areas Brazil nut collection: 3 areas Timber harvesting: 1 area; 0.08 ha Churches: 3 areas

[226] See evidence folder 06. HCVs

Table 87. Baseline levels and MR1 Outputs by Scope - Biodiversity Pillar.

Scope	Activity	Baseline Outputs	Outputs MR1
Fauna and flora study initiatives	Fauna Inventory	0 taxonomic groups studied: mammals, herpetofauna, insects, etc. Intensity or area of fauna sampling (Nº of plots or geographic area (ha))	Activity not carried out during this monitoring period
		0 key indicator species identified: endemic, flagship, threatened, or other	Activity not carried out during this monitoring period
	Flora Inventory	Intensity or area of flora sampling (0 plots or geographic area (ha) (0) 0 taxonomic groups studied: Order, family, genus, etc. 0 of key indicator species identified: endemic, flagship, threatened, or other	Activity not carried out during this monitoring period
Biodiversity Research, Monitoring, or conservation activities	Environmental education	0 environmental education events conducted 0 people educated in environmental awareness	Activity not carried out during this monitoring period

	Biodiversity Research, Monitoring, or conservation activities	0 fauna research, monitoring, or conservation initiatives implemented Further biodiversity indicators to be determined by consultants during the inventory stage	Activity not carried out during this monitoring period
Biodiversity Research, Monitoring, or conservation activities	Survey of High Conservation Value areas and species	0 taxonomic groups studied: Order, family, genus, etc. 0 Intensity area of HCV's sampling (nº of plots or geographic area – ha)	Activity not carried out during this monitoring period

Table 88. Baseline levels and MR1 Outputs by Scope - Climate Pillar.

Scope	Activity	Baseline Outputs	Outputs MR1
Forest Conservation	Remote Monitoring	0 deforestation events detected remotely.	5 deforestation events detected remotely.[227]

[227] See evidence folder: "07. Remote Monitoring"

	Fire brigade training (according to PIF)	0 community firefighters identified and trained. 0 fire-fighting training events held.	Activity not carried out during this monitoring period Activity not carried out during this monitoring period
	Deforestation prevention and mitigation actions	0 deforestation events visited in the field. 0 meetings carried out to combat deforestation. 0 infrastructure items constructed for deforestation combat.	3 deforestation events visited in the field. 8 meetings carried out to combat deforestation.[228] Effectiveness of meeting Nº1[229]: 2; Effectiveness of meeting Nº2: 1; Effectiveness of meeting N°3: 1; Effectiveness of meeting N°4: 1 Effectiveness of meeting N°5: 1 1 Effectiveness of meeting N°6:1 1 Effectiveness of meeting N°7:1 1 Effectiveness of meeting N°8:1 2 infrastructure items constructed to combat deforestation. [230]

[228] See evidence folder "\08. Deforestation\22.05.12 - Estrada Maanaim", documents "22.06.30 - Reunião proprietários_Invasão" and "22.06.30 – Explanatory note".
[229] The effectiveness is based on internal classification where the meetings are ranked in 1, 2 or 3 being 1 lower effectiveness and 3 higher.
[230] See evidence folder "08. Deforestation./"Expedição Campo"

		0 equipment items and vehicles purchased for deforestation combat. 0 protocol applications (deforestation event protocol; forest integrity protocol, etc.)	Activity not carried out during this monitoring period (equipment items and vehicles) 3 protocol applications regarding community deforestation[231]

Table 89. Baseline levels and MR1 Outputs by Scope – "Other" Pillar.

Scope	Activity	Baseline Outputs	Outputs MR1
Further Leisure, Cultural and Sporting Activities	Relationship-building actions with communities	0 actions carried out with communities. 0 participants (by gender/age) in activities 0 meetings carried out to promote activities	2 actions carried out with communities: women's workshop and donation of hygiene kits[232] 30 participants in the two activities: Women: 16; Men: 14[233] Activity not carried out during this monitoring period (meetings carried out to promote activities).

[231] See evidence folder "12. Operational Procedures/Applied"
[232] See evidence folder: "09. Other activities".
[233] See evidence folder: "09. Other activities" file: "Lista de presença_Higiene_e_Oficina mulheres"

7 ADDITONAL PROJECT IMPACT INFORMATION

No additional project impact information showing how the project meets all indicators that require demonstration of impacts identified, beyond those presented in section 6 above.

APPENDICIES

Appendix 2: Project Risks Table

Appendix not used in this MR. ,

Appendix 3: Bibliographic References

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